Level-3 large deviations for the white-forced 2D Navier–Stokes system in a bounded domain

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August 4, 2026

Abstract

We study the large deviations principle (LDP) of Donsker–Varadhan type for the white-forced Navier–Stokes system in a bounded domain. Under the assumption that the noise is non-degenerate, we establish level-2 and level-3 LDPs with rate functions given by the Donsker–Varadhan formulas. The proof relies on an improved version of Kifer's criterion, a lift argument inspired by [DV83], an improved abstract result on the large-time asymptotics of generalized Markov semigroups, and a delicate approximation scheme utilizing the resolvent operators of the Markov semigroup.

AMS subject classifications: 60F10, 35Q30, 35R60, 60J35

Keywords: stochastic Navier–Stokes system, large deviations, occupation measures

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1 Introduction

In this paper, we establish both level-2 and level-3 large deviations principles (LDP) of Donsker–Varadhan type for the 2D white-forced Navier–Stokes (NS, for short) system in a smooth bounded domain D:

$$\begin{cases} \partial_t u + (u\cdot\nabla)u - \nu\Delta u + \nabla p = h + \eta, \\ \operatorname{div} u = 0, \\ u(0) = u_0, \qquad u|_{\partial D} = 0, \end{cases} \tag{1.1}$$

where u is the velocity field, p is the pressure, and $\nu > 0$ is the kinematic viscosity. Applying the Leray projection Π on both sides of $(1.1)_1$, setting $\nu = 1$ for simplicity, and writing

$$Lu := -\Pi\Delta u, \qquad B(u) := \Pi(u\cdot\nabla)u,$$

we eliminate the pressure and consider the system as an abstract evolutionary equation

$$\partial_t u + B(u) + Lu = h + \eta$$

in the space of divergence-free vector fields

$$H := \{u \in L^2 | \operatorname{div} u = 0, (u\cdot n)|_{\partial D} = 0\}. \tag{1.2}$$

The space H is endowed with the inner product $\langle u, v\rangle$ and the norm $\|u\|^2 := \langle u, u\rangle$ inherited from L^2. The driving force consists of two parts: $h \in H$ is a given deterministic function and η is a white-in-time noise of the form

$$\eta(t,x) := \partial_t \sum_{j\geq 1} b_j e_j(x)\beta_j(t), \tag{1.3}$$

where $\{b_j\}_{j\geq 1}$ is a sequence of real numbers satisfying

$$\mathcal{B}_1 := \sum_{j\geq 1} \alpha_j b_j^2 < \infty, \tag{1.4}$$

$\{e_j\}_{j\geq 1}$ is an orthonormal basis in H formed by eigenfunctions of the Stokes operator L associated with the eigenvalues $\{\alpha_j\}_{j\geq 1}$, and $\{\beta_j\}_{j\geq 1}$ are independent standard Brownian motions defined on a probability space $(\Omega, \mathscr{F}, \mathscr{F}_t, \mathbb{P})$ satisfying the usual conditions (e.g., see Definition 2.25 in [KS14]).

1.1 Main results

Under the above assumptions, the system (1.1) defines a Markov family $(u_t, \mathbb{P}_u)$ parameterized by the initial condition $u \in H$. The large-time asymptotic behaviour of this family is now well understood. In particular, it is known that there is a unique invariant measure which attracts exponentially the distribution of all solutions, provided that the noise is sufficiently non-degenerate. We refer the reader to the papers [FM95, EMS01, KS02, BKL02, HM06] for the first results, and the book [KS12] and reviews [Deb13, KS17] for the subsequent developments. Besides, the central limit theorem (CLT) holds, which describes the fluctuations of the time average of a functional around its mean value, cf. [Shi06]. In the present paper,

our main goal is to quantify the probabilities of large deviations from the mean value, which is a natural extension of the CLT. To formulate the main results of the paper, we denote the Markov operators associated with $(u_t, \mathbb{P}_u)$ by

$$\mathfrak{P}_t f(u) := \int_H f(v) P_t(u, \mathrm{d}v), \qquad \mathfrak{P}_t : C_b(H) \to C_b(H), \tag{1.5}$$

$$\mathfrak{P}_t^* \lambda(\Gamma) := \int_H P_t(u, \Gamma) \lambda(\mathrm{d}u), \qquad \mathfrak{P}_t^* : \mathcal{P}(H) \to \mathcal{P}(H), \tag{1.6}$$

where

$$P_t(u, \Gamma) := \mathbb{P}_u \{u_t \in \Gamma\} \tag{1.7}$$

is the transition function and $\mathcal{P}(H)$ denotes the space of all Borel probability measures on H endowed with the weak topology. For $\sigma \in \mathcal{P}(H)$, we write

$$\mathbb{P}_\sigma(\Gamma) := \int_H \mathbb{P}_u(\Gamma) \sigma(\mathrm{d}u), \qquad \Gamma \in \mathscr{F},$$

and consider the occupation measures

$$\zeta_t := \frac{1}{t} \int_0^t \delta_{u_s} \mathrm{d}s, \qquad t > 0 \tag{1.8}$$

defined on $(\Omega, \mathscr{F}, \mathbb{P}_\sigma)$, where δ_u is the Dirac measure centered at u. Note that $\{\zeta_t\}_{t>0}$ is a family of random probability measures on H. Recall that a function $I : \mathcal{P}(H) \to [0, \infty]$ is said to be a good rate function, if the level set $\{\lambda \in \mathcal{P}(H) | I(\lambda) \le \alpha\}$ is compact for any $\alpha \ge 0$.

Our first result establishes a level-2 LDP for $\{\zeta_t\}_{t>0}$ with a rate function I given by the Donsker–Varadhan variational formula.

Theorem 1.1. *Suppose that $h \in U := H_0^1 \bigcap H$, $b_j > 0$ for all $j \ge 1$, and (1.4) holds. Then, for any $\gamma, M > 0$, the family $\{\zeta_t\}_{t>0}$ satisfies an LDP uniformly in*

$$\sigma \in \Lambda(\gamma, M) := \left\{ \sigma \in \mathcal{P}(H) \middle| \int_H e^{\gamma \|u\|^2} \sigma(\mathrm{d}u) \le M \right\}, \tag{1.9}$$

with a good rate function $I : \mathcal{P}(H) \to [0, \infty]$ that is independent of γ, M. More precisely, the following upper and lower bounds hold:

1. for any closed set $F \subset \mathcal{P}(H)$,

$$\limsup_{t \to +\infty} \frac{1}{t} \log \left(\sup_{\sigma \in \Lambda(\gamma, M)} \mathbb{P}_\sigma \{\zeta_t \in F\} \right) \le - \inf_{\lambda \in F} I(\lambda); \tag{1.10}$$

2. for any open set $G \subset \mathcal{P}(H)$,

$$\liminf_{t \to +\infty} \frac{1}{t} \log \left(\inf_{\sigma \in \Lambda(\gamma, M)} \mathbb{P}_\sigma \{\zeta_t \in G\} \right) \ge - \inf_{\lambda \in G} I(\lambda). \tag{1.11}$$

Moreover, the rate function I is given by the Donsker–Varadhan variational formula

$$I(\lambda) = \sup_{\substack{f \in \mathcal{D}(\mathfrak{L}) \\ \inf_H f > 0}} \int_H \frac{-\mathfrak{L} f}{f} \mathrm{d}\lambda, \tag{1.12}$$

where $\mathfrak{L}$ is the generator of the Markov semigroup $\{\mathfrak{P}_t\}_{t \ge 0}$ defined on the domain

$$\mathcal{D}(\mathfrak{L}) := \left\{ f \in C_b(H) \middle| \exists g \in C_b(H),\ \mathfrak{P}_t f - f = \int_0^t \mathfrak{P}_s g \mathrm{d}s,\ \forall t \ge 0 \right\}, \tag{1.13}$$

and $\mathfrak{L} f := g$ for $f \in \mathcal{D}(\mathfrak{L})$.

Remark 1. Note that $\{\mathfrak{P}_t\}_{t\geq 0}$ may fail to be a C_0-semigroup, due to the non-compactness of the phase space H. However, thanks to the Feller property of $\mathfrak{P}_t$ and the continuity of the sample paths, the map $(t,u)\mapsto \mathfrak{P}_t f(u)$ is continuous. By the Fubini theorem, this continuity ensures that the function $s\mapsto \mathfrak{P}_s g(u)$ is integrable for fixed $u\in H$. Therefore, the integral term on the right-hand side of (1.13) is well-defined.

To present the next result, we fix any $T>0$ and set $X^T:=C([0,T];H)$. Define the empirical measures of finite-length trajectories by

$$\boldsymbol{\zeta}_t^T:=\frac{1}{t}\int_0^t \delta_{u_{[s,s+T]}}\mathrm{d}s, \tag{1.14}$$

where $u_{[t,t+T]}$ denotes the trajectory $\{u_r\}_{r\in[t,t+T]}$. Then, $\{\boldsymbol{\zeta}_t^T\}_{t>0}$ is a family of random probability measures on X^T. The following result establishes an LDP for $\{\boldsymbol{\zeta}_t^T\}_{t>0}$.

Theorem 1.2. *Under the assumptions of Theorem 1.1, for any $\gamma, M>0$, the family $\{\boldsymbol{\zeta}_t^T\}_{t>0}$ satisfies an LDP, uniformly in $\sigma\in\Lambda(\gamma,M)$, with a good rate function $I_T:\mathcal{P}(X^T)\to[0,\infty]$ that is independent of γ, M.*

Finally, to formulate the result on level-3 LDP, we recall some basic definitions. Consider the empirical measures of full trajectories

$$\boldsymbol{\zeta}_t:=\frac{1}{t}\int_0^t \delta_{u_{[s,\infty)}}\mathrm{d}s, \tag{1.15}$$

which are random probability measures on the space $X:=C([0,\infty);H)$ endowed with the topology of uniform convergence on bounded intervals. A measure $\boldsymbol{\lambda}\in\mathcal{P}(X)$ is said to be shift-invariant, if it is invariant under the shift operator θ_s, $s>0$ defined by

$$\theta_s u(\cdot):=u(s+\cdot),\qquad \theta_s: X\to X. \tag{1.16}$$

Let $\mathcal{P}_s(X)$ be the space of all shift-invariant measures on X. By the Kolmogorov theorem, each shift-invariant measure $\boldsymbol{\lambda}\in\mathcal{P}_s(X)$ can be uniquely extended to a shift-invariant measure on $X^{\mathbb{R}}:=C(\mathbb{R};H)$. With a slight abuse of notation, we still denote this extension by $\boldsymbol{\lambda}$. Let $\boldsymbol{\lambda}(u_{(-\infty,0]},\cdot)$ denote the regular conditional probability of $\boldsymbol{\lambda}$ given $\mathscr{F}_0^{-\infty}$, where for any $-\infty\leq s\leq t\leq\infty$, $\mathscr{F}_t^s$ is the σ-algebra generated by the family of projections $\{\pi_r\}_{r\in[s,t]}$ defined by

$$\pi_r u_{(-\infty,\infty)}:=u(r),\qquad \pi_r: X^{\mathbb{R}}\to H.$$

For any interval $\mathcal{J}\subset\mathbb{R}$, we define the projection $\pi_{\mathcal{J}}$ by

$$\pi_{\mathcal{J}} u_{(-\infty,\infty)}:=u_{\mathcal{J}},\qquad \pi_{\mathcal{J}}: X^{\mathbb{R}}\to C(\mathcal{J};H).$$

Theorem 1.3. *Under the assumptions of Theorem 1.1, for any $\gamma, M>0$, the family $\{\boldsymbol{\zeta}_t\}_{t>0}$ satisfies an LDP, uniformly in $\sigma\in\Lambda(\gamma,M)$, with a good rate function $\boldsymbol{I}:\mathcal{P}(X)\to[0,\infty]$ given by the Donsker–Varadhan entropy formula*

$$\boldsymbol{I}(\boldsymbol{\lambda})=\begin{cases}\int_{X^{\mathbb{R}}}\mathrm{Ent}\left(\pi^*_{[0,1]}\boldsymbol{\lambda}(u_{(-\infty,0]},\cdot)\middle|\pi^*_{[0,1]}\mathbb{P}_{u(0)}\right)\boldsymbol{\lambda}(\mathrm{d}u_{(-\infty,\infty)}) & \text{if } \boldsymbol{\lambda}\in\mathcal{P}_s(X),\\ +\infty & \text{otherwise},\end{cases} \tag{1.17}$$

where $\pi^*_{[0,1]}$ is the push-forward operator associated with $\pi_{[0,1]}$ and $\mathrm{Ent}(\mu_1|\mu_2)$ is the relative entropy of μ_1 with respect to μ_2 defined by (1.23).

1.2 Literature review and scheme of the proof

These types of LDPs were first established by Donsker and Varadhan [DV75a, DV75b, DV76, DV83] for general Markov processes under various assumptions. Later, this topic was studied by many authors, and we refer to the book [DZ10] for a detailed account of the main achievements in this field.

In stochastic PDEs, only a few results have been obtained regarding the Donsker–Varadhan type LDP. The first results are due to Gourcy [Gou07a, Gou07b], who established a level-2 LDP for the stochastic NS and Burgers equations on the torus driven by white-in-time and irregular-in-space noise. In these papers, the spatial irregularity of the noise is utilized to ensure the strong Feller property of the corresponding Markov family. However, from a physical point of view, this spatial irregularity assumption is rather restrictive. This assumption was later relaxed by Jakšić et al. in [JNPS15, JNPS18, JNPS21], where they established a level-3 LDP for a class of dissipative PDEs perturbed by random kick forces and bounded noise. Their proof combines a non-compact version of Kifer's criterion with a study of the large-time behaviour of generalized Markov semigroups. This approach was further extended to white-forced PDEs, leading to a level-2 LDP. For the case of non-degenerate noises, see [MN18, Ner19]; for the degenerate case, we refer to [NPX23]. Recently, by combining the asymptotic compactness property of the dynamics and the Kifer-type framework, Chen et al. [CLXZ24] proved a level-2 LDP for a nonlinear wave equation with localized damping and bounded noise. We also mention [LL26], where the authors generalize the results in [NPX23] to the quasi-periodic case.

To the best of our knowledge, Theorem 1.3 is the first result that establishes a level-3 LDP of the Donsker–Varadhan type for white-forced stochastic PDEs. The existing results [JNPS15, JNPS18, JNPS21] address the case of random kick forces and bounded noise, which correspond to discrete-time random dynamical systems (RDS). In the case of white-forced stochastic PDEs, the associated RDS is continuous in time. One of the main difficulties in this setting is the verification of the exponential growth condition required in the generalized Kifer criterion; see condition (3.4) in [JNPS18]. As shown in the proof of Theorem 3.3 therein, this condition is not only required for proving exponential tightness, but also plays a pivotal role in ensuring an important covering property for the level sets of the rate function. However, such a condition is not necessarily satisfied for white-forced stochastic PDEs. To be precise, one has to find a functional $\mathbf{\Phi} : C([0,T];H) \to [0,\infty)$ with compact level sets such that

$$\mathbb{E}_u \exp\left(\int_0^t \mathbf{\Phi}(u_{[s,s+T]})\mathrm{d}s\right) \lesssim e^{ct}, \qquad \forall t \geq 0. \tag{1.18}$$

In contrast, due to a priori estimates (A.9) and (A.24), only doubly-logarithmic moment estimates can be expected for the H^1-norm in the case of a bounded domain. Even in the torus case, where exponential moment estimates are available for all H^s-norms, this condition might not hold; see Remark D.2 for a detailed explanation.

To address this challenge, our approach combines two ingredients. First, inspired by [CLXZ24], see (3.6) therein, we improve the generalized Kifer criterion by replacing the exponential growth condition with a weaker condition, that is, exponential tightness along any subsequence, see Proposition D.1. Second, in order to verify this weaker condition, we develop a lift argument by utilizing some ideas from [DV83], where the authors established a level-3 LDP for general Markov processes whose transition function and generator satisfy seven specific conditions. In our current setting, it seems to be impossible to verify these conditions, since, for example, one of these conditions requires the existence of a density of the transition function with respect to a reference measure. Nevertheless, we observe that by lifting the level-2 exponential growth condition (A.3) to the trajectory level, see Lemma 3.2,

a level-3 LD upper bound holds for the empirical measures of periodized trajectories

$$\boldsymbol{\zeta}_t^{\mathrm{Per}} := \frac{1}{t}\int_0^t \delta_{\theta_s u_{\mathrm{Per},t}}\mathrm{d}s, \tag{1.19}$$

where the associated rate function is given by the Donsker–Varadhan entropy $\mathbf{H}$, see (3.1), and $u_{\mathrm{Per},t}$ is the periodized trajectory defined by

$$u_{\mathrm{Per},t}(r) = u(r), \qquad 0 \le r < t,$$
$$u_{\mathrm{Per},t}(r+t) = u_{\mathrm{Per},t}(r), \qquad r \in \mathbb{R}.$$

By using the Donsker–Varadhan variational formula (1.12) for the level-2 rate function I in Theorem 1.1, we establish a contraction inequality (3.5), which, combined with the goodness of I, leads to the compactness of the level sets of $\mathbf{H}$. Consequently, $\{\boldsymbol{\zeta}_t^{\mathrm{Per}}\}_{t>0}$ satisfies an LD upper bound with a good rate function $\mathbf{H}$. Finally, since $\{\boldsymbol{\zeta}_t^{\mathrm{Per}}\}_{t>0}$ and $\{\boldsymbol{\zeta}_t\}_{t>0}$ are exponentially equivalent, $\{\boldsymbol{\zeta}_{t_n}\}_{n\ge1}$ is exponentially tight for any sequence $\{t_n\}_{n\ge1}$ satisfying $\lim_{n\to\infty} t_n = \infty$.

In addition, we mention that with the help of the doubly logarithmic moment estimates (A.9) and (A.24), the exponential tightness holds for the whole family $\{\zeta_t\}_{t>0}$. As discussed previously, this property is unlikely to be attainable via purely PDE-based methods.

The verification of the remaining two conditions in Proposition D.1 relies on an analysis of large-time asymptotics of the Feynman–Kac semigroup. However, due to the presence of a physical boundary, the system (1.1) exhibits weaker dissipation: the H^1-norm cannot be fully dissipated, posing a significant challenge. Specifically, the weak dissipativity leads to the existence of eigenvectors satisfying only a doubly-logarithmic moment estimate for the H^1-norm, while the uniqueness guaranteed by the abstract criteria in [JNPS18, MN18] holds in a class requiring higher-order polynomial moment estimates. Consequently, the existence is established in a broader class than the uniqueness. To resolve this discrepancy, we expand the class in which the uniqueness holds by utilizing a stronger uniform irreducibility property ensured by the parabolic smoothing effect. This leads to an improved abstract criterion for large-time asymptotics of generalized Markov semigroups, see Proposition E.2, which is well-suited for the Feynman–Kac semigroup with weak dissipation.

We emphasize that Theorems 1.1 and 1.3 are the first results establishing the Donsker–Varadhan formulas (1.12) and (1.17) in the context of stochastic PDEs with non-compact phase spaces. These results can be viewed as a generalization of the compact case studied in [JNPS21]. In our setting, the main difficulty arises from the potential growth of the eigenfunctions of the Feynman–Kac semigroup. More precisely, the growth of the eigenfunctions leads to a variational formula similar to (1.12), but with the supremum taken over a class of unbounded continuous test functions, as detailed in (2.21). To address this issue, we approximate the unbounded test function through carefully designed cut-off and regularization techniques, utilizing the resolvent operators of the Markov semigroup. This approach enables the justification of the Donsker–Varadhan variational formula (1.12), see Section 2 for further details. The Donsker–Varadhan entropy formula (1.17) is proved by generalizing the approach of [JNPS21] to the continuous-time setting, combined with the variational formula and a contradiction argument. Further details can be found in Section 6.

Finally, we clarify the role of the non-degeneracy assumption. The main purpose of the present paper is not to pursue the weakest possible noise condition, but rather to isolate and resolve the new difficulties, namely, the exponential tightness at the trajectorial level, the Donsker–Varadhan formulas in non-compact phase spaces, and the presence of physical boundaries. In the periodic setting, one can exploit the algebraic structure of the Fourier basis and the associated saturating mechanism, which makes it possible to prove ergodicity and level-2 LDP for the NS system under very degenerate noise. No comparable structure is

available in a general bounded domain. The Stokes eigenmodes do not satisfy an analogous saturating condition, and even ergodicity for the NS system with very degenerate noise in bounded domains is presently out of reach, let alone a Donsker–Varadhan LDP theory.

This should not be interpreted as a limitation of the level-3 mechanism developed in this paper. On the contrary, in the periodic case, by combining the Malliavin-calculus and controllability arguments of [NPX23] with our framework, one can obtain a parallel level-3 Donsker–Varadhan LDP for the NS system on $\mathbb{T}^2$ driven by very degenerate noise. Indeed, the hypoelliptic estimates and the saturating mechanism in [NPX23] provide the uniform Feller and irreducibility properties required for the Feynman–Kac semigroup, while the lift argument developed in this paper remains essentially unchanged. Consequently, the path-space exponential tightness, the Donsker–Varadhan formulas, and the level-3 LDP can be obtained by the same mechanism as in the present work.

Structure of the paper

The paper is organized as follows. Section 2 is devoted to the proof of Theorem 1.1. Sections 3 and 4 focus on the verification of the conditions in the improved version of Kifer's criterion, cf. Proposition D.1. This verification relies on Theorem 1.1, a lift argument inspired by [DV83], and an improved abstract criterion for large-time asymptotics of generalized Markov semigroups, cf. Proposition E.2. The proofs of Theorem 1.2 and 1.3 are presented in Sections 5 and 6, respectively. Finally, the Appendix gathers a priori estimates for the solutions of the system (1.1), some properties of Markov semigroups on weighted phase spaces, basic properties of the Donsker–Varadhan entropy, and the proofs of the abstract criteria stated in Propositions D.1 and E.2.

Acknowledgements

The author was supported by the ANR project DYNACQUS through the grant ANR-24-CE40-5714-01, and by the National Natural Science Foundation of China under Grant Nos. 12171317, 12331008, 12250710674, and 12161141004. This paper was completed during the author's visit to Université Paris Cité. He would like to express sincere gratitude to the China Scholarship Council for its financial support, as well as to Professors Sergei Kuksin, Armen Shirikyan, and Vahagn Nersesyan, for the invitation to Paris, and for the valuable discussions and throughout the visit.

Notation

We shall use the following standard notations.

- For any Polish space Y, let $C_b(Y)$ be the space of all bounded continuous functions $V : Y \to \mathbb{R}$ endowed with the norm

$$\|V\|_\infty := \sup_{y\in Y} |V(y)|,$$

and let $L_b(Y)$ be the space of bounded Lipschitz functions endowed with the norm

$$\|V\|_{L_b} := \|V\|_\infty + \sup_{\substack{y_1,y_2\in Y\\ y_1\neq y_2}} \frac{|V(y_1)-V(y_2)|}{\operatorname{dist}(y_1,y_2)}. \tag{1.20}$$

- By $\mathscr{B}(Y)$, we denote the Borel σ-algebra of Y. For $A \in \mathscr{B}(Y)$, $\mathbb{I}_A$ is the indicator function of the set A. For $R > 0$ and $y \in Y$, let $B_Y(y, R)$ be the open ball centered at y of radius R, and we write $B_Y(R) := B_Y(0, R)$, if Y is a separable Banach space.
- Let $\mathcal{M}^+(Y)$ be the set of non-negative finite Borel measures on Y endowed with the topology of weak convergence, and let $\mathcal{P}(Y)$ denote the space of all Borel probability measures

on Y. For a measurable function $f : Y \to \mathbb{R}$ and a measure $\mu \in \mathcal{M}^+(Y)$, we write

$$\langle f, \mu \rangle := \int_Y f(y) \mathrm{d}\mu.$$

The weak convergence in $\mathcal{P}(Y)$ is equivalent to the convergence in the following dual-Lipschitz metric:

$$\|\mu_1 - \mu_2\|^*_{L(Y)} := \sup_{\substack{f \in L_b(Y) \\ \|f\|_{L_b} \le 1}} |\langle f, \mu_1 \rangle - \langle f, \mu_2 \rangle|, \qquad \mu_1, \mu_2 \in \mathcal{P}(Y).$$

- Define

$$H := \{u \in L^2 | \operatorname{div} u = 0, (u \cdot n)|_{\partial D} = 0\}$$

with the norm $\|\cdot\|$ and inner product $\langle \cdot, \cdot \rangle$ inherited from L^2. For any integer $j \ge 1$, we write $U^j := H^j \bigcap H^1_0 \bigcap H$ with the norm $\|\cdot\|_j$ and inner product $\langle \cdot, \cdot \rangle_j$ inherited from the usual Sobolev space H^j. Let U^* be the dual space of $U := U^1$ and denote by $\|\cdot\|_{-1}$ its norm. Let P_N be the orthogonal projection in H onto the space

$$H_N := \operatorname{span}\{e_1, e_2, \dots, e_N\},$$

where $\{e_j\}_{j \ge 1}$ is the orthonormal basis entering (1.3). Let $\mathsf{Q}_N := I - \mathsf{P}_N$.
- For any interval $\mathcal{J} \subset \mathbb{R}$, we define the space

$$X^{\mathcal{J}} := C(\mathcal{J}; H)$$

endowed with the topology of uniform convergence on any bounded intervals. Furthermore, by $\mathfrak{D}^{\mathcal{J}}$ we denote the space of functions $f : \mathcal{J} \to H$ with discontinuities only of the first kind and normalized to be right-continuous. The convergence in $\mathfrak{D}^{\mathcal{J}}$ is induced by the Skorokhod topology, cf. (16.4) in [Bil99]. For simplicity, we write

$$\mathfrak{D} := \mathfrak{D}^{[0,\infty)}, \qquad \mathfrak{D}^T := \mathfrak{D}^{[0,T]},$$

and

$$X := X^{[0,\infty)}, \qquad X^T := X^{[0,T]}.$$

- For any $r \in \mathbb{R}$, we introduce the projection from $\mathfrak{D}^{\mathbb{R}}$ to H:

$$\pi_r u_{(-\infty,\infty)} = u(r), \qquad u_{(-\infty,\infty)} \in \mathfrak{D}^{\mathbb{R}}.$$

The associated push-forward operator $\pi^*_r : \mathcal{P}(\mathfrak{D}^{\mathbb{R}}) \to \mathcal{P}(H)$ is defined as

$$\pi^*_r \boldsymbol{\lambda}(A) = \boldsymbol{\lambda}(\pi_r^{-1}(A)), \qquad \forall A \in \mathscr{B}(H). \tag{1.21}$$

For any interval $\mathcal{J} \subset \mathbb{R}$, we introduce the projection from $\mathfrak{D}^{\mathbb{R}}$ to $\mathfrak{D}^{\mathcal{J}}$:

$$\pi_{\mathcal{J}} u_{(-\infty,\infty)} = u_{\mathcal{J}}, \qquad u_{(-\infty,\infty)} \in \mathfrak{D}^{\mathbb{R}}.$$

The associated push-forward operator $\pi^*_{\mathcal{J}} : \mathcal{P}(\mathfrak{D}^{\mathbb{R}}) \to \mathcal{P}(\mathfrak{D}^{\mathcal{J}})$ is defined as

$$\pi^*_{\mathcal{J}} \boldsymbol{\lambda}(\mathbf{A}) = \boldsymbol{\lambda}(\pi_{\mathcal{J}}^{-1}(\mathbf{A})), \qquad \forall \mathbf{A} \in \mathscr{B}(\mathfrak{D}^{\mathcal{J}}). \tag{1.22}$$

For any $-\infty \le s \le t \le \infty$, $\mathscr{F}^s_t$ is the σ-algebra generated by the family of projections $\{\pi_r, r \in [s,t]\}$.
- For Banach spaces X, Y, we denote by $\mathcal{L}(X \to Y)$ the space of bounded linear operators from X to Y and write $\mathcal{L}(X) := \mathcal{L}(X \to X)$.

- Given two measures μ_1, μ_2 defined on a measurable space $(\Omega, \mathscr{F})$, the relative entropy of μ_1 with respect to μ_2 is defined by

$$\mathrm{Ent}(\mu_1|\mu_2) := \sup_{F\in\mathcal{B}(\mathscr{F})} \left(\langle F, \mu_1\rangle - \log\langle e^F, \mu_2\rangle\right), \tag{1.23}$$

  where $\mathcal{B}(\mathscr{F})$ denotes the space of all bounded $\mathscr{F}$-measurable functions $F : \Omega \to \mathbb{R}$.
- For $a, b \in \mathbb{R}$, we write

$$a \wedge b := \min\{a, b\}, \qquad a \vee b := \max\{a, b\}.$$

  By $[a]$, we denote the largest integer that does not exceed a.
- By C, C_a, C_T, and so on, we denote positive constants that are not important to the analysis, with subscripts indicating dependence on specific parameters. For simplicity, we shall frequently use the notation $\lesssim$, $\lesssim_a$, $\lesssim_T$, etc., to indicate inequalities that hold up to an unessential multiplicative constant, such as C, C_a, C_T, and so on.

2 Proof of Theorem 1.1

The LDP with a good rate function I is obtained by adapting the Kifer-type argument used in [Ner19]; see also Theorem 3.3 in [JNPS18]. In the present setting, this argument is supplemented by the H^1-doubly logarithmic moment estimate (A.9). We therefore indicate only the necessary modifications and focus on the proof of the Donsker–Varadhan variational formula (1.12).

The exponential growth condition (3.4) in [JNPS18] follows from (A.3). The other two conditions required by the Kifer-type criterion, that is, existence of pressure and uniqueness of equilibrium, are verified through a multiplicative ergodic theorem (MET), see Theorem 2.1 in [Ner19]. However, as discussed earlier, the weak dissipativity of the system (1.1), caused by the presence of a physical boundary, prevents the direct application of the proof in [Ner19] to establish the MET. To resolve this issue, we adapt and refine Theorem 4.1 of [JNPS18] to the present setting, thereby obtaining an improved abstract criterion for the large-time asymptotics of generalized Markov semigroups; see Proposition E.2. All conditions in this criterion can be verified by arguments similar to those in Sections 3 and 4 of [Ner19]; see also Section 4 in this paper for a more general case, where this criterion is applied to establish a level-3 MET. Therefore, we omit the details here.

By applying the Kifer-type criterion, see Theorem 3.3 in [JNPS18], the LD upper and lower bounds are established, with a good rate function I given by

$$I(\lambda) = \sup_{V\in C_b(H)} \left(\int_H V \mathrm{d}\lambda - Q(V)\right), \qquad \lambda \in \mathcal{P}(H), \tag{2.1}$$

where

$$Q(V) := \lim_{t\to+\infty} \frac{1}{t} \log \mathbb{E}_\sigma \exp\left(\int_0^t V(u_s)\mathrm{d}s\right), \qquad V \in C_b(H), \tag{2.2}$$

and the limit on the right-hand side does not depend on $\sigma \in \Lambda(\gamma, M)$. Here and throughout the rest of the paper, we fix the positive parameters $\gamma, M > 0$.

Now we turn to the justification of the Donsker–Varadhan variational formula (1.12). Let $\mathcal{V}$ be the space of all functions $V : H \to \mathbb{R}$ for which there exists an integer $N \geq 1$ and a function $F \in L_b(H)$ such that

$$V(u) = F(\mathsf{P}_N u), \qquad u \in H.$$

Here, P_N is the orthogonal projection in H onto the space H_N spanned by $e_1, e_2, \ldots, e_N$. We say that $\{V_n\}_{n\geq 1} \subset C_b(H)$ buc-converges to $V \in C_b(H)$, if

$$\sup_{n\geq 1} \|V_n\|_\infty < \infty, \qquad \lim_{n\to+\infty} \|V - V_n\|_{L^\infty(K)} = 0 \tag{2.3}$$

for any compact subset K of H. It is known that any function $V \in C_b(H)$ can be buc-approximated by functions in $\mathcal{V}$; see Section 5.6 in [JNPS18].

We need the following auxiliary result, which shows that the rate function I can be recovered from the pressure Q through the Legendre transform restricted to the class $\mathcal{V}$.

Lemma 2.1. *Let I be the rate function given by (2.1). Then,*

$$I(\lambda) = \sup_{V\in\mathcal{V}} \left(\int_H V \mathrm{d}\lambda - Q(V) \right) \tag{2.4}$$

for any $\lambda \in \mathcal{P}(H)$.

Proof. By definition, one has

$$I(\lambda) \geq \sup_{V\in\mathcal{V}} \left(\int_H V \mathrm{d}\lambda - Q(V) \right). \tag{2.5}$$

To prove the inequality in the other direction, we fix any $\lambda \in \mathcal{P}(H)$ and $V \in C_b(H)$. Let $\{V_n\}_{n\geq 1} \subset \mathcal{V}$ be a sequence buc-converging to V. Then,

$$\int_H V \mathrm{d}\lambda = \lim_{n\to+\infty} \int_H V_n \mathrm{d}\lambda. \tag{2.6}$$

We claim that

$$Q(V) \geq \limsup_{n\to+\infty} Q(V_n). \tag{2.7}$$

Indeed, by combining (A.3) with Lemma 3.2 in [JNPS18], the family of empirical measures $\{\zeta_t\}_{t>0}$, defined in (1.8), is exponentially tight, that is, for any $a > 0$, there is a compact set $\mathcal{K}_a \subset \mathcal{P}(H)$ such that

$$\limsup_{t\to+\infty} \frac{1}{t} \log \mathbb{P}_\sigma\{\zeta_t \in \mathcal{K}_a^c\} \leq -a, \qquad \forall \sigma \in \Lambda(\gamma, M). \tag{2.8}$$

Moreover, by the Prokhorov theorem, for any $\epsilon > 0$, there exists a compact set $K_{a,\epsilon} \subset H$ satisfying

$$\mu(K_{a,\epsilon}) \geq 1 - \epsilon \tag{2.9}$$

for any $\mu \in \mathcal{K}_a$. Let

$$a := \sup_{n\geq 1} \|V_n\|_\infty + \|V\|_\infty + |Q(V)| + 1. \tag{2.10}$$

Then,

$$\begin{aligned} Q(V_n) &= \lim_{t\to+\infty} \frac{1}{t} \log \mathbb{E}_\sigma \exp\left(t\langle V_n, \zeta_t\rangle\right) \\ &= \lim_{t\to+\infty} \frac{1}{t} \log \mathbb{E}_\sigma \left((\mathbb{I}_{\{\zeta_t\in\mathcal{K}_a\}} + \mathbb{I}_{\{\zeta_t\in\mathcal{K}_a^c\}}) \exp\left(t\langle V_n, \zeta_t\rangle\right)\right) \\ &\leq \max\{I_1, I_2\}, \end{aligned} \tag{2.11}$$

where

$$I_1 := \limsup_{t\to+\infty} \frac{1}{t} \log \mathbb{E}_\sigma \left(\mathbb{I}_{\{\zeta_t \in \mathcal{K}_a\}} \exp\left(t\langle V_n, \zeta_t\rangle\right)\right),$$

$$I_2 := \limsup_{t\to+\infty} \frac{1}{t} \log \mathbb{E}_\sigma \left(\mathbb{I}_{\{\zeta_t \in \mathcal{K}_a^c\}} \exp\left(t\langle V_n, \zeta_t\rangle\right)\right).$$

For I_1, by applying (2.9) and (2.10), it follows

$$\begin{aligned}
&\mathbb{E}_\sigma \left(\mathbb{I}_{\{\zeta_t \in \mathcal{K}_a\}} \exp\left(t\langle V_n, \zeta_t\rangle\right)\right) \\
&\qquad \le \mathbb{E}_\sigma \left(\mathbb{I}_{\{\zeta_t \in \mathcal{K}_a\}} \exp\left(t\langle \mathbb{I}_{K_{a,\epsilon}} V_n, \zeta_t\rangle + t\langle \mathbb{I}_{K_{a,\epsilon}^c} V_n, \zeta_t\rangle\right)\right) \\
&\qquad \le \mathbb{E}_\sigma \left(\mathbb{I}_{\{\zeta_t \in \mathcal{K}_a\}} \exp\left(t\langle \mathbb{I}_{K_{a,\epsilon}} V_n, \zeta_t\rangle + a\epsilon t\right)\right) \\
&\qquad \le e^{a\epsilon t} \mathbb{E}_\sigma \left(\mathbb{I}_{\{\zeta_t \in \mathcal{K}_a\}} \exp\left(t\langle \mathbb{I}_{K_{a,\epsilon}} (V_n - V), \zeta_t\rangle + t\langle \mathbb{I}_{K_{a,\epsilon}} V, \zeta_t\rangle\right)\right) \\
&\qquad \le e^{a\epsilon t} e^{t\|V_n - V\|_{L^\infty(K_{a,\epsilon})}} \mathbb{E}_\sigma \left(\mathbb{I}_{\{\zeta_t \in \mathcal{K}_a\}} \exp\left(t\langle \mathbb{I}_{K_{a,\epsilon}} V, \zeta_t\rangle\right)\right) \\
&\qquad \le e^{a\epsilon t} e^{t\|V_n - V\|_{L^\infty(K_{a,\epsilon})}} \mathbb{E}_\sigma \left(\mathbb{I}_{\{\zeta_t \in \mathcal{K}_a\}} \exp\left(t\langle V, \zeta_t\rangle - t\langle \mathbb{I}_{K_{a,\epsilon}^c} V, \zeta_t\rangle\right)\right) \\
&\qquad \le e^{2a\epsilon t} e^{t\|V_n - V\|_{L^\infty(K_{a,\epsilon})}} \mathbb{E}_\sigma \left(\exp\left(t\langle V, \zeta_t\rangle\right)\right),
\end{aligned}$$

which implies

$$I_1 \le 2a\epsilon + \|V_n - V\|_{L^\infty(K_{a,\epsilon})} + Q(V). \tag{2.12}$$

As for I_2, by applying (2.8) and (2.10), one has

$$\begin{aligned}
I_2 &\le \sup_{n\ge1} \|V_n\|_\infty + \limsup_{t\to+\infty} \frac{1}{t} \log \mathbb{P}_\sigma \left\{\zeta_t \in \mathcal{K}_a^c\right\} \\
&\le \sup_{n\ge1} \|V_n\|_\infty - a \le -|Q(V)| \le Q(V).
\end{aligned} \tag{2.13}$$

Plugging (2.12) and (2.13) into (2.11), one obtains

$$Q(V_n) \le 2a\epsilon + \|V_n - V\|_{L^\infty(K_{a,\epsilon})} + Q(V)$$

for any $n \ge 1$ and $\epsilon > 0$. Therefore, by first taking $n \to +\infty$ and then $\epsilon \to 0^+$, (2.7) holds as claimed. This, together with (2.6), implies

$$\int_H V \mathrm{d}\lambda - Q(V) \le \liminf_{n\to+\infty} \left(\int_H V_n \mathrm{d}\lambda - Q(V_n)\right) \le \sup_{V\in\mathcal{V}} \left(\int_H V \mathrm{d}\lambda - Q(V)\right),$$

thus completing the proof. □

For $V \in C_b(H)$, we define the Feynman–Kac semigroup by

$$\mathfrak{P}_t^V f(u) := \mathbb{E}_u \left(f(u_t) \exp\left(\int_0^t V(u_s) \mathrm{d}s\right)\right), \qquad f \in C_b(H). \tag{2.14}$$

Let $\mathfrak{P}_t^{V*} : \mathcal{M}^+(H) \to \mathcal{M}^+(H)$ be its dual, and let $C_{\boldsymbol{w},\kappa}(H)$ be defined in (B.1). Here and throughout this section, we shall always take κ as in Proposition B.1. In this setting, $\{\mathfrak{P}_t^V\}_{t\ge0}$ forms a semigroup of bounded linear operators in $C_{\boldsymbol{w},\kappa}(H)$. We need the following Duhamel-type formula.

Lemma 2.2. *Let $V \in C_b(H)$ and $f \in C_{\boldsymbol{w},\kappa}(H)$. Then,*

$$\mathfrak{P}_t^V f(u) - \mathfrak{P}_t f(u) = \int_0^t \mathfrak{P}_{t-s}(V \mathfrak{P}_s^V f)(u) \mathrm{d}s$$

for any $t \ge 0$ and $u \in H$.

Proof. By the Newton–Leibniz formula, the Fubini theorem, and the Markov property, we derive

$$\begin{aligned}
\mathfrak{P}_t^V f(u) - \mathfrak{P}_t f(u) &= \mathbb{E}_u \left(f(u_t) \left(\exp\left(\int_0^t V(u_s) \mathrm{d}s \right) - 1 \right) \right) \\
&= \mathbb{E}_u \left(f(u_t) \int_0^t \left(\exp\left(\int_r^t V(u_s) \mathrm{d}s \right) V(u_r) \right) \mathrm{d}r \right) \\
&= \int_0^t \mathbb{E}_u \left(f(u_t) \exp\left(\int_r^t V(u_s) \mathrm{d}s \right) V(u_r) \right) \mathrm{d}r \\
&= \int_0^t \mathbb{E}_u \left(\mathbb{E}_{u_r} \left(f(u_{t-r}) \exp\left(\int_0^{t-r} V(u_s) \mathrm{d}s \right) \right) V(u_r) \right) \mathrm{d}r \\
&= \int_0^t \mathfrak{P}_r (V \mathfrak{P}_{t-r}^V f)(u) \mathrm{d}r = \int_0^t \mathfrak{P}_{t-s} (V \mathfrak{P}_s^V f)(u) \mathrm{d}s.
\end{aligned}$$

Thus, the proof is complete. □

We introduce the following notations. Define

$$C_+(H) := \left\{ f \in C(H) \middle| f(u) \geq 0, \ \forall u \in H \right\},$$

and

$$C_{\boldsymbol{w},m}(H) := \left\{ f \in C(H) \middle| \|f\|_{C_{\boldsymbol{w},m}} := \sup_{u \in H} \frac{|f(u)|}{\|u\|^{2m} + 1} < \infty \right\},$$

and

$$\mathcal{P}_m(H) := \{ \nu \in \mathcal{P}(H) | \langle \| \cdot \|^{2m}, \nu \rangle < \infty \}.$$

A function $h_V \in C_{\boldsymbol{w},m}(H)$ is called an eigenfunction of $\{\mathfrak{P}_t^V\}_{t \geq 0}$, if there is a number $c_V \geq 0$ (called eigenvalue) such that

$$\mathfrak{P}_t^V h_V = c_V^t h_V, \qquad \forall t > 0. \tag{2.15}$$

A measure $\mu_V \in \mathcal{M}^+(H)$ is called an eigenvector of the dual semigroup $\{\mathfrak{P}_t^{V*}\}_{t \geq 0}$, if there is an eigenvalue $\tilde{c}_V \geq 0$ such that

$$\mathfrak{P}_t^{V*} \mu_V = \tilde{c}_V^t \mu_V, \qquad \forall t > 0.$$

Applying Proposition E.2, with its hypotheses verified by adapting the arguments of [Ner19] (see Theorem 2.1) as explained above, and using the doubly logarithmic moment estimate (A.9), we obtain the following result. For any $V \in \mathcal{V}$, there exists an integer $m = m(V) \geq 1$, a number $c_V > 0$, a measure $\mu_V \in \mathcal{P}_m(H)$, and a function $h_V \in C_{\boldsymbol{w},m}(H) \cap C_+(H)$ such that

$$\mathfrak{P}_t^V h_V = c_V^t h_V, \qquad \mathfrak{P}_t^{V*} \mu_V = c_V^t \mu_V, \qquad \forall t > 0.$$

The measure μ_V is the unique eigenvector of $\{\mathfrak{P}_t^{V*}\}_{t \geq 0}$ in $\mathcal{P}_m(H)$, while h_V is the unique eigenfunction of $\{\mathfrak{P}_t^V\}_{t \geq 0}$ in $C_{\boldsymbol{w},m}(H) \cap C_+(H)$ satisfying the normalization condition

$$\langle h_V, \mu_V \rangle = 1.$$

Moreover, h_V is strictly positive, and

$$c_V = \exp(Q(V)). \tag{2.16}$$

We call c_V the principal eigenvalue of $\{\mathfrak{P}_t^V\}_{t \geq 0}$.

Now we are in a position to prove the Donsker–Varadhan variational formula (1.12).

Proof of (1.12). The proof uses several auxiliary results from Appendix B and is divided into three steps.

Step 1. First, we claim that

$$I(\lambda) \geq \sup_{\substack{f\in\mathcal{D}(\mathfrak{L})\\ \inf_H f>0}} \int_H \frac{-\mathfrak{L}f}{f}\mathrm{d}\lambda, \tag{2.17}$$

where $\mathfrak{L}$ is the generator of the Markov semigroup $\{\mathfrak{P}_t\}_{t\geq 0}$, and $\mathcal{D}(\mathfrak{L})$ is defined in (1.13). Indeed, let $f \in \mathcal{D}(\mathfrak{L})$ be such that $\inf_H f > 0$. Then, for $V := \frac{-\mathfrak{L}f}{f} \in C_b(H)$, one has

$$\int_0^t \mathfrak{P}_{t-s}(Vf)\mathrm{d}s = -\int_0^t \mathfrak{P}_{t-s}(\mathfrak{L}f)\mathrm{d}s = f - \mathfrak{P}_t f. \tag{2.18}$$

On the other hand, by Lemma 2.2, it follows

$$\mathfrak{P}_t^V f - \mathfrak{P}_t f = \int_0^t \mathfrak{P}_{t-s}(V\mathfrak{P}_s^V f)\mathrm{d}s. \tag{2.19}$$

Taking difference between (2.18) and (2.19), one gets

$$\begin{aligned}\|\mathfrak{P}_t^V f - f\|_\infty &\leq \int_0^t \|\mathfrak{P}_{t-s}(V(\mathfrak{P}_s^V f - f))\|_\infty \mathrm{d}s\\ &\lesssim_V \int_0^t \|\mathfrak{P}_s^V f - f\|_\infty \mathrm{d}s.\end{aligned}$$

An application of the Gronwall inequality yields

$$\mathfrak{P}_t^V f = f, \qquad \forall t \geq 0. \tag{2.20}$$

Together with the definition (2.2) and the fact that $f \in C_b(H)$ satisfies $\inf_H f > 0$, this implies

$$Q(V) \leq \lim_{t\to+\infty} \frac{1}{t}\log\frac{\mathfrak{P}_t^V f}{\inf_H f} = \lim_{t\to+\infty}\frac{1}{t}\log\frac{f}{\inf_H f} = 0,$$

and

$$Q(V) \geq \lim_{t\to+\infty} \frac{1}{t}\log\frac{\mathfrak{P}_t^V f}{\|f\|_\infty} = \lim_{t\to+\infty}\frac{1}{t}\log\frac{f}{\|f\|_\infty} = 0.$$

Consequently, $Q(V) = 0$ and

$$\int_H \frac{-\mathfrak{L}f}{f}\mathrm{d}\lambda = \int_H V\mathrm{d}\lambda = \int_H V\mathrm{d}\lambda - Q(V) \leq I(\lambda).$$

This leads to (2.17) as claimed.

Step 2. It remains to prove the inequality (2.17) in the other direction. To this end, we claim that

$$I(\lambda) \leq \sup\left\{\int_H \frac{-\mathfrak{L}f}{f}\mathrm{d}\lambda \middle| f \in \mathcal{D}_{\boldsymbol{w},\kappa}(\mathfrak{L}),\ f>0,\ \frac{\mathfrak{L}f}{f}\in C_b(H)\right\}, \tag{2.21}$$

where $\mathcal{D}_{\boldsymbol{w},\kappa}(\mathfrak{L})$ is defined in (B.7) and $f > 0$ is understood as $f(u) > 0$ for any $u \in H$. To see this, notice that for each $V \in C_b(H)$,

$$Q(V - Q(V)) = 0.$$

Then, from Lemma 2.1, one has

$$I(\lambda) = \sup_{\substack{V \in \mathcal{V} \\ Q(V)=0}} \int_H V \mathrm{d}\lambda.$$

Let $\tilde{I}(\lambda)$ denote the right-hand side of (2.21). Let $V \in \mathcal{V}$ be such that $Q(V) = 0$. Then, by (2.16), the principal eigenvalue of $\{\mathfrak{P}_t^V\}_{t\geq 0}$ is given by $c_V = 1$. Let h_V be the positive eigenfunction associated with c_V. Combining Lemma 2.2, $c_V = 1$, and (2.15), it follows

$$\int_0^t \mathfrak{P}_s(Vh_V)\mathrm{d}s = \int_0^t \mathfrak{P}_{t-s}(V\mathfrak{P}_s^V h_V)\mathrm{d}s = \mathfrak{P}_t^V h_V - \mathfrak{P}_t h_V = h_V - \mathfrak{P}_t h_V,$$

which, by the definition (B.7), implies that $h_V \in \mathcal{D}_{\boldsymbol{w},\kappa}(\mathfrak{L})$ and

$$\mathfrak{L}h_V = -Vh_V.$$

Therefore,

$$\int_H V \mathrm{d}\lambda = \int_H \frac{-\mathfrak{L}h_V}{h_V}\mathrm{d}\lambda \leq \tilde{I}(\lambda),$$

which implies (2.21) as claimed.
Step 3. Now, it suffices to show

$$\tilde{I}(\lambda) \leq \sup_{\substack{f \in \mathcal{D}(\mathfrak{L}) \\ \inf_H f > 0}} \int_H \frac{-\mathfrak{L}f}{f}\mathrm{d}\lambda. \tag{2.22}$$

Let $\mathcal{R}_\alpha$ be the resolvent operator defined by (B.2), and let $f \in \mathcal{D}_{\boldsymbol{w},\kappa}(\mathfrak{L})$ satisfy

$$f > 0, \qquad \frac{\mathfrak{L}f}{f} \in C_b(H).$$

From Corollary B.4, it follows that

$$\int_H \frac{-\mathfrak{L}f}{f}\mathrm{d}\lambda \leq \lim_{\alpha\to+\infty} \liminf_{N\to+\infty} \int_H -\frac{\mathfrak{L}\mathcal{R}_\alpha f_N}{\mathcal{R}_\alpha f_N}\mathrm{d}\lambda,$$

where $f_N := (f \wedge N) \vee \frac{1}{N}$. Notice that for any $\alpha > 0$ and $N \geq 1$,

$$\mathcal{R}_\alpha f_N \in \mathcal{D}(\mathfrak{L}), \qquad \inf_H \mathcal{R}_\alpha f_N > 0,$$

due to Proposition B.3. Therefore, one has

$$\int_H \frac{-\mathfrak{L}f}{f}\mathrm{d}\lambda \leq \sup_{\substack{g \in \mathcal{D}(\mathfrak{L}) \\ \inf_H g > 0}} \int_H \frac{-\mathfrak{L}g}{g}\mathrm{d}\lambda.$$

This implies (2.22) and thus completes the proof. □

3 Exponential tightness

Let $\{\boldsymbol{\zeta}_t^T\}_{t>0}$ be the empirical measure defined in (1.14). In this section, our main goal is to establish Proposition 3.7, which shows the exponential tightness of $\{\boldsymbol{\zeta}_{t_n}^T\}_{n\geq 1}$ for any sequence $\{t_n\}_{n\geq 1}$ such that $\lim_{n\to+\infty} t_n = +\infty$.

As discussed in Section 1.2, to obtain this result, we consider the periodized empirical measures $\{\boldsymbol{\zeta}_t^{\mathrm{Per}}\}_{t>0}$ defined by (1.19). Let $\mathfrak{D}^{\mathbb{R}}$ be the Skorokhod space of right-continuous paths with left limits. Let $\mathcal{P}_s(\mathfrak{D}^{\mathbb{R}})$ be the space of all Borel probability measures which are

invariant under the shift operators $\{\theta_t\}_{t>0}$. Notice that for any $s>0$ and $A\in\mathscr{B}(\mathfrak{D}^{\mathbb{R}})$, by the periodicity of $u_{\mathrm{Per},t}$, one has

$$\begin{aligned}\theta_s^*\boldsymbol{\zeta}_t^{\mathrm{Per}}(A)&=\frac{1}{t}\int_0^t\mathbb{I}_{\{\theta_r u_{\mathrm{Per},t}\in\theta_s^{-1}A\}}\mathrm{d}r=\frac{1}{t}\int_0^t\mathbb{I}_{\{\theta_{r+s}u_{\mathrm{Per},t}\in A\}}\mathrm{d}r\\&=\frac{1}{t}\int_s^{t+s}\mathbb{I}_{\{\theta_r u_{\mathrm{Per},t}\in A\}}\mathrm{d}r=\frac{1}{t}\int_0^t\mathbb{I}_{\{\theta_r u_{\mathrm{Per},t}\in A\}}\mathrm{d}r=\boldsymbol{\zeta}_t^{\mathrm{Per}}(A).\end{aligned}$$

Therefore, $\{\boldsymbol{\zeta}_t^{\mathrm{Per}}\}_{t>0}$ is a family of random probability measures valued in $\mathcal{P}_s(\mathfrak{D}^{\mathbb{R}})$. Define the Donsker–Varadhan entropy

$$\mathbf{H}(\boldsymbol{\lambda}):=\begin{cases}\int_{\mathfrak{D}^{\mathbb{R}}}\mathrm{Ent}\left(\pi^*_{[0,1]}\boldsymbol{\lambda}(u_{(-\infty,0]},\cdot)\middle|\pi^*_{[0,1]}\mathbb{P}_{u(0)}\right)\boldsymbol{\lambda}(\mathrm{d}u_{(-\infty,\infty)}) & \text{if }\boldsymbol{\lambda}\in\mathcal{P}_s(\mathfrak{D}^{\mathbb{R}}),\\ +\infty & \text{otherwise.}\end{cases}\tag{3.1}$$

Here and in what follows, for any $r\in\mathbb{R}$ and $\mathcal{J}\subset\mathbb{R}$, π_r^* and $\pi_{\mathcal{J}}^*$ are the push-forward operators defined in (1.21) and (1.22), respectively. The basic properties of $\mathbf{H}$ are gathered in the Appendix C.

3.1 LD upper bound for periodized empirical measures

In this subsection, we prove the following uniform LD upper bound for $\{\boldsymbol{\zeta}_t^{\mathrm{Per}}\}_{t>0}$.

Proposition 3.1. *For any closed set $F\subset\mathcal{P}(\mathfrak{D}^{\mathbb{R}})$, it follows*

$$\limsup_{t\to+\infty}\frac{1}{t}\log\left(\sup_{\sigma\in\Lambda(\gamma,M)}\mathbb{P}_\sigma\{\boldsymbol{\zeta}_t^{\mathrm{Per}}\in F\}\right)\le-\inf_{\boldsymbol{\lambda}\in F}\mathbf{H}(\boldsymbol{\lambda}).\tag{3.2}$$

To establish this proposition, we need the following auxiliary lemma, which provides an exponential-tightness-type estimate. The proof is carried out by lifting the level-2 exponential growth estimate (A.3) to the trajectory level.

Lemma 3.2. *For any $a>0$, there is a closed set $F_a\subset\mathcal{P}_s(\mathfrak{D}^{\mathbb{R}})$ such that the family $\{\pi_0^*\boldsymbol{\lambda}|\boldsymbol{\lambda}\in F_a\}$ is tight in $\mathcal{P}(H)$ and*

$$\limsup_{t\to+\infty}\frac{1}{t}\log\left(\sup_{\sigma\in\Lambda(\gamma,M)}\mathbb{P}_\sigma\{\boldsymbol{\zeta}_t^{\mathrm{Per}}\in F_a^c\}\right)\le-a.\tag{3.3}$$

Proof. Define

$$\Phi(u):=(\kappa\wedge\gamma)\|u\|_1^2,\tag{3.4}$$

where κ is chosen in Proposition B.1 and γ is given in (1.9). By the definition (1.19), one has

$$\int_0^t\Phi(u_s)\mathrm{d}s=t\int_{\mathfrak{D}^{\mathbb{R}}}\Phi(u(0))\boldsymbol{\zeta}_t^{\mathrm{Per}}(\mathrm{d}u_{(-\infty,\infty)}).$$

Define $\mathfrak{A}_t:=(\boldsymbol{\zeta}_t^{\mathrm{Per}})^*\mathbb{P}_\sigma$, so $\mathfrak{A}_t$ is a probability measure on the space $\mathcal{P}_s(\mathfrak{D}^{\mathbb{R}})$. Then,

$$\mathbb{E}_\sigma\exp\left(\int_0^t\Phi(u_s)\mathrm{d}s\right)=\int_{\mathcal{P}_s(\mathfrak{D}^{\mathbb{R}})}\exp(t\langle\Phi,\pi_0^*\boldsymbol{\lambda}\rangle)\mathfrak{A}_t(\mathrm{d}\boldsymbol{\lambda}).$$

Together with the level-2 exponential growth estimate (A.3) and the Chebyshev inequality, this implies the existence of a constant $C>0$ depending on $M,h,\mathcal{B}_0:=\sum_{j=1}^\infty b_j^2$ such that

$$\begin{aligned}Ce^{Ct}&\ge\mathbb{E}_\sigma\exp\left(\int_0^t\Phi(u_s)\mathrm{d}s\right)\ge\int_{\mathcal{P}_s(\mathfrak{D}^{\mathbb{R}})}\exp(tl\pi_0^*\boldsymbol{\lambda}(K_l^c))\mathfrak{A}_t(\mathrm{d}\boldsymbol{\lambda})\\&\ge\exp\left(\frac{tl}{n}\right)\mathfrak{A}_t\left\{\boldsymbol{\lambda}\in\mathcal{P}_s(\mathfrak{D}^{\mathbb{R}})\middle|\pi_0^*\boldsymbol{\lambda}(K_l^c)>\frac{1}{n}\right\},\end{aligned}$$

where $K_l := \{u|\Phi(u) \le l\}$. By choosing $l_{n,a} := Cn + an^2$, one has

$$\mathfrak{A}_t \left\{ \boldsymbol{\lambda} \in \mathcal{P}_s(\mathfrak{D}^{\mathbb{R}}) \Big| \pi_0^* \boldsymbol{\lambda}(K_{l_{n,a}}^c) > \frac{1}{n} \right\} \le Ce^{-ant}.$$

Introduce

$$F_a := \bigcap_{n=1}^{\infty} \left\{ \boldsymbol{\lambda} \in \mathcal{P}_s(\mathfrak{D}^{\mathbb{R}}) \Big| \pi_0^* \boldsymbol{\lambda}(K_{l_{n,a}}^c) \le \frac{1}{n} \right\}.$$

Then, it follows that

$$\mathbb{P}_\sigma\{\boldsymbol{\zeta}_t^{\mathrm{Per}} \in F_a^c\} = \mathfrak{A}_t(F_a^c) \le C \frac{e^{-at}}{1 - e^{-at}},$$

which implies (3.3). Using the Portmanteau theorem and the Prokhorov theorem, we see that F_a satisfies the desired properties. This completes the proof. □

We turn to the proof of Proposition 3.1.

Proof of Proposition 3.1. Let F be a closed subset of $\mathcal{P}(\mathfrak{D}^{\mathbb{R}})$. By applying Lemma 3.2 and Proposition C.3, for any $a > 0$, it follows

$$\begin{aligned}
&\limsup_{t\to+\infty} \frac{1}{t} \log \left(\sup_{\sigma \in \Lambda(\gamma, M)} \mathbb{P}_\sigma\{\boldsymbol{\zeta}_t^{\mathrm{Per}} \in F\} \right) \\
&\quad = \limsup_{t\to+\infty} \frac{1}{t} \log \left(\sup_{\sigma \in \Lambda(\gamma, M)} \mathbb{P}_\sigma \left\{\boldsymbol{\zeta}_t^{\mathrm{Per}} \in F \bigcap \mathcal{P}_s(\mathfrak{D}^{\mathbb{R}})\right\} \right) \\
&\quad \le \max \left\{ \limsup_{t\to+\infty} \frac{1}{t} \log \left(\sup_{\sigma \in \Lambda(\gamma, M)} \mathbb{P}_\sigma \left\{\boldsymbol{\zeta}_t^{\mathrm{Per}} \in F \bigcap F_a\right\} \right), -a \right\} \\
&\quad \le \max \left\{ - \inf_{\boldsymbol{\lambda} \in F} \mathbf{H}(\boldsymbol{\lambda}), -a \right\}.
\end{aligned}$$

By sending $a \to \infty$, one obtains (3.2). □

3.2 Contraction inequality and exponential tightness

In this subsection, we establish the exponential tightness of $\{\boldsymbol{\zeta}_t^T\}_{t>0}$. To this end, we first prove the goodness of the Donsker–Varadhan entropy $\mathbf{H}$ defined in (3.1). The proof is based on the following contraction inequality.

Lemma 3.3. *Let I be the level-2 rate function given by (1.12). Then,*

$$I(\lambda) \le \inf\{\mathbf{H}(\boldsymbol{\lambda}) | \pi_0^* \boldsymbol{\lambda} = \lambda\}, \qquad \forall \lambda \in \mathcal{P}(H). \tag{3.5}$$

Proof. By the definition (3.1), it suffices to establish

$$I(\lambda) \le \inf\{\mathbf{H}(\boldsymbol{\lambda}) | \boldsymbol{\lambda} \in \mathcal{P}_s(\mathfrak{D}^{\mathbb{R}}), \pi_0^* \boldsymbol{\lambda} = \lambda\}.$$

To see this, we take $\boldsymbol{\lambda} \in \mathcal{P}_s(\mathfrak{D}^{\mathbb{R}})$ and $f \in \mathcal{D}(\mathfrak{L})$ such that $\inf_H f > 0$. Notice that

$$\Psi_t(u_{(-\infty,\infty)}) := \log \frac{f(u(t))}{\mathfrak{P}_t f(u(0))}, \qquad \Psi_t : \mathfrak{D}^{\mathbb{R}} \to \mathbb{R}$$

is a bounded $\mathscr{F}_t^0$-measurable function satisfying

$$\log \left(\int_{\mathfrak{D}} e^{\Psi_t} \mathbb{P}_u \right) = \log \left(\frac{\int_{\mathfrak{D}} f(u(t)) \mathrm{d}\mathbb{P}_u}{\mathfrak{P}_t f(u)} \right) = 0.$$

From the definition (C.2) and the relation (C.3), one has

$$\int_{\mathfrak{D}^{\mathbb{R}}} \Psi_t \mathrm{d}\boldsymbol{\lambda} \le t\mathbf{H}(\boldsymbol{\lambda}).$$

This, together with the translation-invariance of $\boldsymbol{\lambda}$, implies

$$\begin{aligned} t\mathbf{H}(\boldsymbol{\lambda}) &\ge \int_{\mathfrak{D}^{\mathbb{R}}} \log \frac{f(u(t))}{\mathfrak{P}_t f(u(0))} \boldsymbol{\lambda}(\mathrm{d}u_{(-\infty,\infty)}) \\ &= \int_{\mathfrak{D}^{\mathbb{R}}} \log f(u(t)) \boldsymbol{\lambda}(\mathrm{d}u_{(-\infty,\infty)}) - \int_{\mathfrak{D}^{\mathbb{R}}} \log \mathfrak{P}_t f(u(0)) \boldsymbol{\lambda}(\mathrm{d}u_{(-\infty,\infty)}) \\ &= \int_H \log f(u) \lambda(\mathrm{d}u) - \int_H \log \mathfrak{P}_t f(u) \lambda(\mathrm{d}u), \end{aligned}$$

where $\lambda := \pi_0^* \boldsymbol{\lambda}$. Therefore, it follows

$$\begin{aligned} \mathbf{H}(\boldsymbol{\lambda}) &\ge -\lim_{t\to 0^+} \frac{1}{t} \int_H \int_0^t \frac{\mathfrak{P}_s \mathfrak{L} f}{\mathfrak{P}_s f} \mathrm{d}s \mathrm{d}\lambda \\ &= -\lim_{t\to 0^+} \frac{1}{t} \int_0^t \int_H \frac{\mathfrak{P}_s \mathfrak{L} f}{\mathfrak{P}_s f} \mathrm{d}\lambda \mathrm{d}s = \int_H \frac{-\mathfrak{L} f}{f} \mathrm{d}\lambda, \end{aligned}$$

which implies $\mathbf{H}(\boldsymbol{\lambda}) \ge I(\lambda)$ and thus (3.5). □

Now we prove the goodness of the Donsker–Varadhan entropy $\mathbf{H}$.

Proposition 3.4. *For any $\alpha \ge 0$, $A_\alpha := \{\boldsymbol{\lambda} \in \mathcal{P}(\mathfrak{D}^{\mathbb{R}}) | \mathbf{H}(\boldsymbol{\lambda}) \le \alpha\}$ is compact.*

Proof. Step 1: Tightness of marginals. We claim that the family $A_{\mathcal{M},\alpha} := \{\pi_0^* \boldsymbol{\lambda} | \boldsymbol{\lambda} \in A_\alpha\}$ is tight in $\mathcal{P}(H)$. In fact, by Lemma 3.3, for any $\boldsymbol{\lambda} \in A_\alpha$, one has $I(\lambda) \le \alpha$, where $\lambda := \pi_0^* \boldsymbol{\lambda}$. Consequently,

$$A_{\mathcal{M},\alpha} \subset \{\lambda \in \mathcal{P}(H) | I(\lambda) \le \alpha\}.$$

Together with the goodness of I, this ensures the tightness of $A_{\mathcal{M},\alpha}$ as claimed.
Step 2: Construction of an auxiliary test function. Fix any $\epsilon, \delta \in (0,1)$. By the tightness of $A_{\mathcal{M},\alpha}$, there is a compact subset K of H such that

$$\lambda(K^c) < \frac{\delta}{2}, \qquad \forall \lambda \in A_{\mathcal{M},\alpha}. \tag{3.6}$$

On the other hand, from the compactness of K and the continuity of the map $u \mapsto \mathbb{P}_u$, see Proposition 2.4.7 in [KS12], it follows that $\{\mathbb{P}_u\}_{u\in K}$ is tight in $\mathcal{P}(\mathfrak{D}^{[0,1]})$. Thus, there is a compact set $\mathcal{K} \subset \mathfrak{D}^{[0,1]}$ satisfying $\mathbb{P}_u(\mathcal{K}^c) < \epsilon$ for any $u \in K$. Introduce the following auxiliary test function

$$\Psi(u_{(-\infty,\infty)}) := a \int_0^1 \mathbb{I}_K(u(s)) \mathbb{I}_{\mathcal{K}^c}(u_{[s,s+1]}) \mathrm{d}s, \qquad \Psi : \mathfrak{D}^{\mathbb{R}} \to \mathbb{R},$$

where $a > 0$. Then, for any $u \in K$, one has

$$\begin{aligned} \mathbb{E}_u \exp\left(a \mathbb{I}_K(u(0)) \mathbb{I}_{\mathcal{K}^c}(u_{[0,1]})\right) &\le \mathbb{E}_u \exp\left(a \mathbb{I}_{\mathcal{K}^c}(u_{[0,1]})\right) \\ &= (e^a - 1)\mathbb{P}_u(\mathcal{K}^c) + 1 \le (e^a - 1)\epsilon + 1. \end{aligned} \tag{3.7}$$

For $u \notin K$, the above inequality holds trivially. This, together with the Jensen inequality and the Markov property, implies

$$\begin{aligned}\mathbb{E}_u e^{\Psi} &\le \int_0^1 \mathbb{E}_u \exp\left(a\mathbb{I}_K(u(s))\mathbb{I}_{\mathcal{K}^c}(u_{[s,s+1]})\right) \mathrm{d}s \\ &= \int_0^1 \mathbb{E}_u \left(\mathbb{E}_{u(s)} \exp\left(a\mathbb{I}_K(u(0))\mathbb{I}_{\mathcal{K}^c}(u_{[0,1]})\right)\right) \mathrm{d}s \le (e^a-1)\epsilon + 1 \end{aligned} \tag{3.8}$$

for any $u \in H$.

Step 3: Tightness of A_α. Fix any $\boldsymbol{\lambda} \in A_\alpha$. Combining (3.8) with the definition (C.2) of $\bar{\mathbf{H}}$ and the relation (C.3), one has

$$\begin{aligned}\int_{\mathfrak{D}^{\mathbb{R}}} \Psi \mathrm{d}\boldsymbol{\lambda} &\le \bar{\mathbf{H}}(2,\boldsymbol{\lambda}) + \int_{\mathfrak{D}^{\mathbb{R}}} \log\left(\int_{\mathfrak{D}} e^{\Psi} \mathrm{d}\mathbb{P}_{u(0)}\right) \boldsymbol{\lambda}(\mathrm{d}u_{(-\infty,\infty)}) \\ &\le 2\alpha + \log((e^a-1)\epsilon+1). \end{aligned} \tag{3.9}$$

On the other hand, by the Fubini theorem and the translation-invariance of $\boldsymbol{\lambda}$, it follows

$$\begin{aligned}\int_{\mathfrak{D}^{\mathbb{R}}} \Psi \mathrm{d}\boldsymbol{\lambda} &= a\int_0^1 \int_{\mathfrak{D}^{\mathbb{R}}} \mathbb{I}_K(u(s))\mathbb{I}_{\mathcal{K}^c}(u_{[s,s+1]})\mathrm{d}\boldsymbol{\lambda}\mathrm{d}s \\ &= a\int_{\mathfrak{D}^{\mathbb{R}}} \mathbb{I}_K(u(0))\mathbb{I}_{\mathcal{K}^c}(u_{[0,1]})\mathrm{d}\boldsymbol{\lambda} \ge a\pi^*_{[0,1]}\boldsymbol{\lambda}(\mathcal{K}^c) - a\lambda(K^c), \end{aligned}$$

where $\lambda = \pi_0^*\boldsymbol{\lambda}$. This, together with (3.6) and (3.9), implies

$$\pi^*_{[0,1]}\boldsymbol{\lambda}(\mathcal{K}^c) < \frac{\delta}{2} + \frac{1}{a}(2\alpha + \log((e^a-1)\epsilon+1)).$$

Choosing $\epsilon := \frac{1}{e^a-1}$ with a sufficiently large so that $\frac{2\alpha+\log 2}{a} < \frac{\delta}{2}$, one gets

$$\pi^*_{[0,1]}\boldsymbol{\lambda}(\mathcal{K}^c) < \delta.$$

Therefore, the projections of measures in A_α to $\mathfrak{D}^{[0,1]}$ form a tight family. Together with Lemma C.4, this leads to the tightness of A_α. Finally, by using the lower semi-continuity of $\mathbf{H}$, see Proposition C.2, combined with the Prokhorov theorem, A_α is compact. □

By Propositions 3.1 and 3.4, $\{\boldsymbol{\zeta}_t^{\mathrm{Per}}\}_{t>0}$ admit a uniform LD upper bound with a good rate function characterized by the Donsker–Varadhan entropy $\mathbf{H}$. This, together with the exponential equivalence of $\{\pi^*_{[0,\infty)}\boldsymbol{\zeta}_t^{\mathrm{Per}}\}_{t>0}$ and $\{\boldsymbol{\zeta}_t\}_{t>0}$, ensures the exponential tightness of $\{\boldsymbol{\zeta}_{t_n}^T\}_{n\ge1}$ for any sequence $\{t_n\}_{n\ge1}$ such that $\lim_{n\to\infty} t_n = +\infty$, as detailed below.

Lemma 3.5. *The families $\{\pi^*_{[0,\infty)}\boldsymbol{\zeta}_t^{\mathrm{Per}}\}_{t>0}$ and $\{\boldsymbol{\zeta}_t\}_{t>0}$ are uniformly exponentially equivalent, that is, for any $\delta > 0$,*

$$\lim_{t\to+\infty} \frac{1}{t} \log\left(\sup_{\sigma\in\Lambda(\gamma,M)} \mathbb{P}_\sigma\left\{\|\pi^*_{[0,\infty)}\boldsymbol{\zeta}_t^{\mathrm{Per}} - \boldsymbol{\zeta}_t\|^*_{L(\mathfrak{D})} > \delta\right\}\right) = -\infty. \tag{3.10}$$

Proof. Let $L_b(\mathfrak{D})$ denote the space of all bounded Lipschitz functions on $\mathfrak{D}$ with the norm defined by (1.20). For any $\boldsymbol{V} \in L_b(\mathfrak{D})$ such that $\|\boldsymbol{V}\|_{L_b} \le 1$,

$$\begin{aligned}|\langle \boldsymbol{V}, \boldsymbol{\zeta}_t\rangle - \langle \boldsymbol{V}, \pi^*_{[0,\infty)}\boldsymbol{\zeta}_t^{\mathrm{Per}}\rangle| &\le \frac{1}{t}\int_0^t |\boldsymbol{V}(u_{[s,\infty)}) - \boldsymbol{V}(\pi_{[s,\infty)}u_{\mathrm{Per},t})|\mathrm{d}s \\ &\le \frac{1}{t}\int_0^t \mathrm{dist}_{\mathfrak{D}}(u_{[s,\infty)}, \pi_{[s,\infty)}u_{\mathrm{Per},t})\mathrm{d}s. \end{aligned}$$

Since

$$u_{[0,\infty)}(s+r) = u_{\mathrm{Per},t}(s+r), \qquad \forall r \in [0, t-s),$$

one has

$$\mathrm{dist}_{\mathfrak{D}}(u_{[s,\infty)}, \pi_{[s,\infty)} u_{\mathrm{Per},t}) \le \sum_{m=[t-s]+1}^{\infty} \frac{1}{2^m} = \frac{1}{2^{[t-s]}},$$

where $[x]$ denotes the largest integer not exceeding x, and we refer to (16.4) in [Bil99] for the definition of $\mathrm{dist}_{\mathfrak{D}}$. Hence, it follows

$$|\langle V, \boldsymbol{\zeta}_t\rangle - \langle V, \pi^*_{[0,\infty)} \boldsymbol{\zeta}^{\mathrm{Per}}_t\rangle| \le \frac{1}{t}\int_0^t \frac{1}{2^{[s]}} \mathrm{d}s \le \frac{2}{t}.$$

This implies

$$\|\pi^*_{[0,\infty)} \boldsymbol{\zeta}^{\mathrm{Per}}_t - \boldsymbol{\zeta}_t\|^*_{L(\mathfrak{D})} < \delta, \qquad \forall t > \frac{2}{\delta},$$

and thus the uniform exponential equivalence (3.10). □

The above exponential equivalence ensures the following uniform LD upper bound for $\{\boldsymbol{\zeta}_t\}_{t>0}$ with a good rate function $\mathcal{I}$.

Proposition 3.6. *For any closed set $F \subset \mathcal{P}(X)$, it follows*

$$\limsup_{t\to+\infty} \frac{1}{t} \log \left(\sup_{\sigma\in\Lambda(\gamma,M)} \mathbb{P}_\sigma\{\boldsymbol{\zeta}_t \in F\} \right) \le - \inf_{\boldsymbol{\lambda}\in F} \mathcal{I}(\boldsymbol{\lambda}), \tag{3.11}$$

where $\mathcal{I}$ is a good rate function satisfying the Donsker–Varadhan entropy formula (1.17).

Proof. Recall that any probability measure $\boldsymbol{\lambda} \in \mathcal{P}_s(\mathfrak{D})$ can be uniquely extended to a probability measure $\boldsymbol{\lambda} \in \mathcal{P}_s(\mathfrak{D}^{\mathbb{R}})$. Together with Proposition 3.1 and the contraction principle (Theorem 4.2.1 in [DZ10]), this yields a uniform LD upper bound for $\{\pi^*_{[0,\infty)} \boldsymbol{\zeta}^{\mathrm{Per}}_t\}_{t>0}$ with a good rate function

$$\mathcal{I}(\boldsymbol{\lambda}) := \begin{cases} \int_{\mathfrak{D}^{\mathbb{R}}} \mathrm{Ent}\left(\pi^*_{[0,1]}\boldsymbol{\lambda}(u_{(-\infty,0]},\cdot)\middle|\pi^*_{[0,1]}\mathbb{P}_{u(0)}\right)\boldsymbol{\lambda}(\mathrm{d}u_{(-\infty,\infty)}) & \text{if } \boldsymbol{\lambda}\in\mathcal{P}_s(\mathfrak{D}), \\ +\infty & \text{otherwise.} \end{cases}$$

Step 1: LD upper bound on $\mathcal{P}(\mathfrak{D})$. Let $F \subset \mathcal{P}(\mathfrak{D})$ be a closed subset. For $\delta > 0$, let

$$F_\delta := \{\boldsymbol{\lambda} \in \mathcal{P}(\mathfrak{D}) | \, \mathrm{dist}_{\mathcal{P}(\mathfrak{D})}(\boldsymbol{\lambda}, F) \le \delta\}.$$

Applying Lemma 3.5, it follows

$$\begin{aligned} &\limsup_{t\to+\infty} \frac{1}{t} \log \left(\sup_{\sigma\in\Lambda(\gamma,M)} \mathbb{P}_\sigma\{\boldsymbol{\zeta}_t \in F\} \right) \\ &\qquad \le \limsup_{t\to+\infty} \frac{1}{t} \log \left(\sup_{\sigma\in\Lambda(\gamma,M)} \mathbb{P}_\sigma\{\pi^*_{[0,\infty)}\boldsymbol{\zeta}^{\mathrm{Per}}_t \in F_\delta\} \right) \le - \inf_{\boldsymbol{\lambda}\in F_\delta} \mathcal{I}(\boldsymbol{\lambda}). \end{aligned}$$

This, together with Lemma 4.1.6 in [DZ10], yields

$$\limsup_{t\to+\infty} \frac{1}{t} \log \left(\sup_{\sigma\in\Lambda(\gamma,M)} \mathbb{P}_\sigma\{\boldsymbol{\zeta}_t \in F\} \right) \le - \lim_{\delta\to 0^+} \inf_{\boldsymbol{\lambda}\in F_\delta} \mathcal{I}(\boldsymbol{\lambda}) = - \inf_{\boldsymbol{\lambda}\in F} \mathcal{I}(\boldsymbol{\lambda}). \tag{3.12}$$

Step 2: support property of $\boldsymbol{\lambda}$ *with* $\mathcal{I}(\boldsymbol{\lambda}) < +\infty$. We claim that

$$\boldsymbol{\lambda}(X) = 1 \tag{3.13}$$

for any $\boldsymbol{\lambda} \in \mathcal{P}(\mathfrak{D})$ satisfying $\mathcal{I}(\boldsymbol{\lambda}) < +\infty$. Indeed, in this case, one has

$$\mathrm{Ent}\left(\pi^*_{[0,1]}\boldsymbol{\lambda}(u_{(-\infty,0]},\cdot)\middle|\pi^*_{[0,1]}\mathbb{P}_{u(0)}\right) < +\infty, \qquad \boldsymbol{\lambda}\text{-a.s.},$$

which ensures the absolute continuity of the conditional distribution $\pi^*_{[0,1]}\boldsymbol{\lambda}(u_{(-\infty,0]},\cdot)$ with respect to $\pi^*_{[0,1]}\mathbb{P}_{u(0)}$. On the other hand, as $\mathbb{P}_{u(0)}$ is supported on X, it follows

$$\pi^*_{[0,1]}\boldsymbol{\lambda}(u_{(-\infty,0]}, X^{[0,1]}) = 1, \qquad \boldsymbol{\lambda}\text{-a.s.}$$

Combining this with the translation invariance of $\boldsymbol{\lambda}$, one obtains (3.13).

Step 3: LD upper bound on $\mathcal{P}(X)$. Let $F \subset \mathcal{P}(X)$ be a closed subset, and set $\tilde{F}$ as the closure of F in the topology of $\mathcal{P}(\mathfrak{D})$. Since $\boldsymbol{\zeta}_t$ is supported on $\mathcal{P}(X)$ for all $t > 0$, from (3.12), one has

$$\limsup_{t\to+\infty}\frac{1}{t}\log\left(\sup_{\sigma\in\Lambda(\gamma,M)}\mathbb{P}_\sigma\{\boldsymbol{\zeta}_t \in F\}\right) \le -\inf_{\boldsymbol{\lambda}\in\tilde{F}}\mathcal{I}(\boldsymbol{\lambda}).$$

We claim that

$$\inf_{\boldsymbol{\lambda}\in\tilde{F}}\mathcal{I}(\boldsymbol{\lambda}) \ge \inf_{\boldsymbol{\lambda}\in F}\mathcal{I}(\boldsymbol{\lambda}), \tag{3.14}$$

so the upper bound (3.11) holds. To see this, it suffices to show

$$\tilde{F}\bigcap\{\mathcal{I}(\boldsymbol{\lambda}) < \infty\} \subset F. \tag{3.15}$$

Indeed, if $\boldsymbol{\lambda} \in \tilde{F}\bigcap\{\mathcal{I}(\boldsymbol{\lambda}) < \infty\}$, then $\boldsymbol{\lambda}$ can be approximated by $\boldsymbol{\lambda}_n \in F \subset \mathcal{P}(X)$ under the topology of $\mathcal{P}(\mathfrak{D})$. Moreover, by applying (3.13), one has $\boldsymbol{\lambda} \in \mathcal{P}_s(X)$. Therefore, by the definition of the Skorohod metric on $\mathfrak{D}$, one gets $\boldsymbol{\lambda}_n \to \boldsymbol{\lambda}$ in the topology of $\mathcal{P}(X)$. Since F is closed, one has $\boldsymbol{\lambda} \in F$ and thus (3.15).

Step 4: Goodness of the rate function $\mathcal{I}$ *on* $\mathcal{P}(X)$. By applying (3.13), one has

$$K_a := \{\boldsymbol{\lambda} \in \mathcal{P}(X) | \mathcal{I}(\boldsymbol{\lambda}) \le a\} = \{\boldsymbol{\lambda} \in \mathcal{P}(\mathfrak{D}) | \mathcal{I}(\boldsymbol{\lambda}) \le a\},$$

which, together with the goodness of $\mathcal{I}$ on $\mathcal{P}(\mathfrak{D})$, implies that K_a is compact in the topology of $\mathcal{P}(\mathfrak{D})$. Let $\{\boldsymbol{\lambda}_n\}_{n\ge1} \subset K_a$. Since K_a is compact in $\mathcal{P}(\mathfrak{D})$, there exist a subsequence, still denoted by $\{\boldsymbol{\lambda}_n\}$, and $\boldsymbol{\lambda} \in K_a$ such that

$$\boldsymbol{\lambda}_n \to \boldsymbol{\lambda} \qquad \text{in } \mathcal{P}(\mathfrak{D}).$$

As $\mathcal{I}(\boldsymbol{\lambda}) \le a$, (3.13) gives $\boldsymbol{\lambda} \in \mathcal{P}(X)$. By the same argument used in the proof of (3.15), the convergence also holds in $\mathcal{P}(X)$. Hence, K_a is compact in $\mathcal{P}(X)$, and $\mathcal{I}$ is a good rate function on $\mathcal{P}(X)$. □

Using the contraction principle again, we see that a uniform LD upper bound holds for $\{\boldsymbol{\zeta}^T_t\}_{t>0}$ with a good rate function. It is well-known that in the discrete-time case, an LD upper bound with a good rate function implies the exponential tightness, see Lemma 2.6 in [LS87]. Therefore, we have the following result.

Proposition 3.7. *For each sequence* $\{t_n\}_{n\ge1}$ *such that* $\lim_{n\to+\infty} t_n = +\infty$, *the family* $\{\boldsymbol{\zeta}^T_{t_n}\}_{n\ge1}$ *is uniformly exponentially tight in* $\sigma \in \Lambda(\gamma, M)$, *that is, for any* $a > 0$, *there is a compact set* $\mathcal{K}_a \subset \mathcal{P}(X^T)$ *such that*

$$\limsup_{n\to\infty}\frac{1}{t_n}\log\left(\sup_{\sigma\in\Lambda(\gamma,M)}\mathbb{P}_\sigma\{\boldsymbol{\zeta}^T_{t_n} \in \mathcal{K}^c_a\}\right) \le -a.$$

4 Large-time asymptotics for the Feynman–Kac semigroup

The goal of this section is to study the large-time behavior of the Feynman–Kac semigroup. We begin with some definitions. Fix $T > 0$ and introduce a Markov T-process associated with the NS system (1.1): the phase space is $X^T := C([0,T];H)$, and the transition function is defined in the following way. Let $\boldsymbol{u} \in X^T$ and consider

$$\Gamma := \bigcap_{j=1}^n \pi_{t_j}^{-1}(A_j),$$

where $0 \le t_1 < \ldots < t_n \le T$, and $A_1, A_2, \ldots, A_n \in \mathscr{B}(H)$. Set $\mathbb{P}_{\boldsymbol{u}(T)} := \mathbb{P}_u|_{u=\boldsymbol{u}(T)}$, where $(u_t, \mathbb{P}_u)$ is the Markov family associated with (1.1). Define the transition function by

$$P_t^T(\boldsymbol{u}, \Gamma) := \prod_{j=1}^k \mathbb{I}_{A_j}(\boldsymbol{u}(t+t_j)) \mathbb{P}_{\boldsymbol{u}(T)} \left(\bigcap_{j=k+1}^n \{u_{t+t_j-T} \in A_j\} \right), \tag{4.1}$$

where the integer k is such that $t + t_k < T \le t + t_{k+1}$. If $t + t_1 \ge T$, then

$$P_t^T(\boldsymbol{u}, \Gamma) := \mathbb{P}_{\boldsymbol{u}(T)} \left(\bigcap_{j=1}^n \{u_{t+t_j-T} \in A_j\} \right). \tag{4.2}$$

If $t + t_n < T$, then

$$P_t^T(\boldsymbol{u}, \Gamma) := \prod_{j=1}^n \mathbb{I}_{A_j}(\boldsymbol{u}(t+t_j)). \tag{4.3}$$

By the Kolmogorov theorem, this transition function is well-defined. Let $(\boldsymbol{u}_t, \mathbb{P}_{\boldsymbol{u}})$ denote the Markov family corresponding to this transition function. For $\boldsymbol{V} \in C_b(X^T)$, define the Feynman–Kac semigroup by the formula

$$\mathfrak{P}_t^{T,\boldsymbol{V}} \boldsymbol{f}(\boldsymbol{u}) := \mathbb{E}_{\boldsymbol{u}} \left(\boldsymbol{f}(\boldsymbol{u}_t) \exp\left(\int_0^t \boldsymbol{V}(\boldsymbol{u}_s) \mathrm{d}s \right) \right), \qquad \boldsymbol{f} \in C_b(X^T). \tag{4.4}$$

By $\mathfrak{P}_t^{T,\boldsymbol{V}*}$, we denote the dual semigroup of $\mathfrak{P}_t^{T,\boldsymbol{V}}$. For simplicity, we write

$$\mathfrak{P}_t^T := \mathfrak{P}_t^{T,0}, \qquad \mathfrak{P}_t^{T*} := \mathfrak{P}_t^{T,0*}. \tag{4.5}$$

4.1 Hyper-exponential recurrence

Define the space

$$X_\infty^T := C([0,T];U) \bigcap C^{\frac{1}{4}}([0,T];U^*)$$

endowed with the norm

$$\|\boldsymbol{u}\|_{X_\infty^T} := \|\boldsymbol{u}\|_{C([0,T];U)} + \|\boldsymbol{u}\|_{C^{1/4}([0,T];U^*)}.$$

For any integer $R \ge 1$, let X_R^T be the closure of $B_{X_\infty^T}(R)$ under the topology of X^T. By the Aubin–Lions lemma, X_R^T is a compact subset of X^T. Moreover, X_∞^T is dense in X^T. Define the first hitting time

$$\tau(R) := \inf\left\{ t \ge 0 \middle| \boldsymbol{u}_t \in X_R^T \right\}. \tag{4.6}$$

Proposition 4.1. *For any $\kappa > 0$, there are integers $m, R \ge 1$ such that*

$$\mathbb{E}_{\boldsymbol{u}} \exp(\kappa \tau(R)) \lesssim_{\kappa,T} \|\boldsymbol{u}\|_{X^T}^{2m} + 1, \qquad \forall \boldsymbol{u} \in X^T. \tag{4.7}$$

Proof. Step 1: Hyper-exponential recurrence in X^T. For $r > 0$, define

$$\tau_0(r) := \inf\{t \geq 0 | \boldsymbol{u}_t \in \overline{B_{X^T}(r)}\}. \tag{4.8}$$

We claim that for any $\kappa > 0$, there are integers $m, r \geq 1$ such that

$$\mathbb{E}_{\boldsymbol{u}} \exp(\kappa \tau_0(r)) \lesssim_{\kappa,T} \|\boldsymbol{u}\|_{X^T}^{2m} + 1, \qquad \forall \boldsymbol{u} \in X^T. \tag{4.9}$$

Indeed, by applying (A.7), it follows

$$\mathbb{E}_{\boldsymbol{u}} \|\boldsymbol{u}_{T+1}\|_{X^T}^{2m} \leq 4e^{-m\alpha_1} \|\boldsymbol{u}\|_{X^T}^{2m} + C_{m,\mathcal{B}_0,h,T}. \tag{4.10}$$

Then, by choosing m, r sufficiently large so that

$$q := 8e^{-m\alpha_1} < 1, \qquad e^{\kappa(T+1)} q < 1, \qquad r \geq \max\left\{1, \frac{1}{4} e^{m\alpha_1} C_{m,\mathcal{B}_0,h,T}\right\}, \tag{4.11}$$

one has

$$\mathbb{E}_{\boldsymbol{u}} \|\boldsymbol{u}_{T+1}\|_{X^T}^{2m} \leq q(\|\boldsymbol{u}\|_{X^T}^{2m} \vee r). \tag{4.12}$$

Together with the Markov property, this implies

$$\begin{aligned}
p_k(\boldsymbol{u}) :&= \mathbb{E}_{\boldsymbol{u}} \left(\mathbb{I}_{\{\tau_0(r) > k(T+1)\}} \|\boldsymbol{u}_{k(T+1)}\|_{X^T}^{2m}\right) \\
&\leq \mathbb{E}_{\boldsymbol{u}} \left(\mathbb{I}_{\{\tau_0(r) > (k-1)(T+1)\}} \|\boldsymbol{u}_{k(T+1)}\|_{X^T}^{2m}\right) \\
&= \mathbb{E}_{\boldsymbol{u}} \left(\mathbb{I}_{\{\tau_0(r) > (k-1)(T+1)\}} \mathbb{E}_{\boldsymbol{u}_{(k-1)(T+1)}} \|\boldsymbol{u}_{T+1}\|_{X^T}^{2m}\right) \\
&\leq q\mathbb{E}_{\boldsymbol{u}} \left(\mathbb{I}_{\{\tau_0(r) > (k-1)(T+1)\}} \left(\|\boldsymbol{u}_{(k-1)(T+1)}\|_{X^T}^{2m} \vee r\right)\right) = q p_{k-1}(u),
\end{aligned}$$

which leads to

$$p_k(\boldsymbol{u}) \leq q^k \|\boldsymbol{u}\|_{X^T}^{2m}$$

for any integer $k \geq 1$ and $\boldsymbol{u} \in X^T$. Hence,

$$\mathbb{P}_{\boldsymbol{u}}\{\tau_0(r) > k(T+1)\} \leq r^{-2m} p_k(\boldsymbol{u}) \leq r^{-2m} q^k \|\boldsymbol{u}\|_{X^T}^{2m}.$$

Combining this with (4.11), one has

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{u}} \exp(\kappa\tau_0(r)) &\leq 1 + \mathbb{E}_{\boldsymbol{u}} \left(\exp(\kappa\tau_0(r)) \sum_{k=0}^{\infty} \mathbb{I}_{\{\tau_0(r) \in (k(T+1),(k+1)(T+1)]\}}\right) \\
&\leq 1 + \sum_{k=0}^{\infty} e^{\kappa(k+1)(T+1)} \mathbb{P}_{\boldsymbol{u}}\{\tau_0(r) > k(T+1)\} \\
&\leq 1 + \sum_{k=0}^{\infty} e^{\kappa(k+1)(T+1)} r^{-2m} q^k \|\boldsymbol{u}\|_{X^T}^{2m} \lesssim_{\kappa,T} \|\boldsymbol{u}\|_{X^T}^{2m} + 1.
\end{aligned}$$

Step 2: Auxiliary stopping times and estimates. Introduce

$$\tilde{\tau}_0(r) := \tau_0(r) + T + 1,$$

where $\tau_0(r)$ is defined in (4.8), and

$$\tau_n(r) := \inf\{t > \tilde{\tau}_{n-1}(r) | \boldsymbol{u}_t \in \overline{B_{X^T}(r)}\}, \qquad \tilde{\tau}_n(r) := \tau_n(r) + T + 1, \qquad n \geq 1.$$

Let

$$\tilde{n} := \min\{n \in \mathbb{N} | \boldsymbol{u}_{\tilde{\tau}_n(r)} \in X_R^T\}.$$

Applying (A.9) and Corollary A.2, one has

$$\mathbb{E}_{\boldsymbol{u}} \log\left(1+\log\left(1+\|\boldsymbol{u}_{T+1}\|_{X_\infty^T}\right)\right) \lesssim_{\mathcal{B}_1,h,T} \|\boldsymbol{u}\|_{X^T}^2+1. \tag{4.13}$$

This, together with the Chebyshev inequality, implies

$$\mathbb{P}_{\boldsymbol{u}}\left\{\boldsymbol{u}_{T+1}\in X_R^T\right\} \geq 1-\mathbb{P}_{\boldsymbol{u}}\left\{\boldsymbol{u}_{T+1}\notin B_{X_\infty^T}(R)\right\} \geq 1-\frac{C_{\mathcal{B}_1,h,T}(\|\boldsymbol{u}\|_{X^T}^2+1)}{\log(1+\log(1+R))}.$$

Therefore, for any $p\in(0,1)$, $r\geq 1$ and $\boldsymbol{u}\in\overline{B_{X^T}(r)}$, there is an integer $R=R(p,r)>0$ such that

$$\mathbb{P}_{\boldsymbol{u}}\{\boldsymbol{u}_{T+1}\in X_R^T\}\geq 1-p.$$

Together with the strong Markov property, this yields

$$\begin{aligned}\mathbb{P}_{\boldsymbol{u}}\{\tilde{n}>k\} &\leq \mathbb{E}_{\boldsymbol{u}}\left(\prod_{j=0}^{k}\mathbb{I}_{\left\{\boldsymbol{u}_{\tilde{\tau}_j(r)}\notin X_R^T\right\}}\right) \leq \mathbb{E}_{\boldsymbol{u}}\left(\prod_{j=0}^{k-1}\mathbb{I}_{\left\{\boldsymbol{u}_{\tilde{\tau}_j(r)}\notin X_R^T\right\}}\mathbb{P}_{u_{\tau_k(r)}}\{\boldsymbol{u}_{T+1}\notin X_R^T\}\right)\\ &\leq p\mathbb{E}_{\boldsymbol{u}}\left(\prod_{j=0}^{k-1}\mathbb{I}_{\left\{\boldsymbol{u}_{\tilde{\tau}_j(r)}\notin X_R^T\right\}}\right)\leq\ldots\leq p^k\end{aligned} \tag{4.14}$$

for any $\boldsymbol{u}\in X^T$ and integer $k\geq 1$. On the other hand, by applying (4.9) with 3κ, estimate (4.12), and the Markov property, for

$$\tau_0':=\inf\{t\geq T+1|\boldsymbol{u}_t\in\overline{B_{X^T}(r)}\},$$

it follows

$$\begin{aligned}\mathbb{E}_{\boldsymbol{u}}(\exp(3\kappa\tau_0')) &\leq e^{3\kappa(T+1)}\mathbb{E}_{\boldsymbol{u}}\left(\mathbb{E}_{\boldsymbol{u}_{T+1}}(\exp(3\kappa\tau_0))\right)\\ &\leq Ce^{3\kappa(T+1)}\mathbb{E}_{\boldsymbol{u}}(1+\|\boldsymbol{u}_{T+1}\|_{X^T}^{2m}) \lesssim_{\kappa,T}\left(1+\|\boldsymbol{u}\|_{X^T}^{2m}\vee r\right),\end{aligned}$$

which, combined with the strong Markov property and the definition of τ_{k-1}, yields

$$\begin{aligned}\mathbb{E}_{\boldsymbol{u}}\exp(3\kappa\tau_k) &= \mathbb{E}_{\boldsymbol{u}}\left(\mathbb{E}_{\boldsymbol{u}}(\exp(3\kappa\tau_k)|\mathscr{F}_{\tau_{k-1}})\right)\\ &= \mathbb{E}_{\boldsymbol{u}}\left(\exp(3\kappa\tau_{k-1})\mathbb{E}_{\boldsymbol{u}_{\tau_{k-1}}}(\exp(3\kappa\tau_0'))\right) \lesssim_{\kappa,T}\mathbb{E}_{\boldsymbol{u}}\left(\exp(3\kappa\tau_{k-1})\right).\end{aligned}$$

Combining this with (4.9), one gets

$$\mathbb{E}_{\boldsymbol{u}}\exp(3\kappa\tilde{\tau}_k)\leq C_{\kappa,T}^k\left(\|\boldsymbol{u}\|_{X^T}^{2m}+1\right),\qquad\forall\boldsymbol{u}\in X^T. \tag{4.15}$$

Step 3: Hyper-exponential recurrence in X_∞^T. Applying (4.14) and (4.15), it follows

$$\begin{aligned}\mathbb{P}_{\boldsymbol{u}}\{\tau(R)\geq l(T+1)\} &\leq \mathbb{P}_{\boldsymbol{u}}\{\tau(R)>\tilde{\tau}_k\}+\mathbb{P}_{\boldsymbol{u}}\{\tilde{\tau}_k\geq l(T+1)\}\\ &\leq \mathbb{P}_{\boldsymbol{u}}\{\tilde{n}>k\}+\mathbb{P}_{\boldsymbol{u}}\{\tilde{\tau}_k\geq l(T+1)\}\\ &\leq p^k+e^{-3\kappa l(T+1)}C_{\kappa,T}^k\left(\|\boldsymbol{u}\|_{X^T}^{2m}+1\right).\end{aligned}$$

By choosing $k=\lceil cl\rceil$, where c is such that

$$c\log(C_{\kappa,T})\leq\kappa(T+1),$$

and taking p sufficiently small so that

$$c\log p\leq -2\kappa(T+1),$$

one obtains

$$\mathbb{P}_{\boldsymbol{u}}\{\tau(R)\geq l(T+1)\}\leq(1+C_{\kappa,T})e^{-2\kappa l(T+1)}\left(\|u\|_{X^T}^{2m}+1\right).$$

As shown previously, this implies (4.7) and thus completes the proof. □

4.2 *Concentration property and growth condition*

Define the weight function

$$\mathfrak{w}_m(\boldsymbol{u}) := \|\boldsymbol{u}\|_{X^T}^{2m} + 1, \tag{4.16}$$

and denote by $C_{\mathfrak{w},m}(X^T)$ (respectively, $L^\infty_{\mathfrak{w},m}(X^T)$) the space of all continuous (measurable) functions $\boldsymbol{f} : X^T \to \mathbb{R}$ such that

$$|\boldsymbol{f}(\boldsymbol{u})| \le C\mathfrak{w}_m(\boldsymbol{u}), \qquad \forall \boldsymbol{u} \in X^T.$$

The spaces $C_{\mathfrak{w},m}(X^T)$ and $L^\infty_{\mathfrak{w},m}(X^T)$ are endowed with the norm

$$\|\boldsymbol{f}\|_{L^\infty_{\mathfrak{w},m}} := \sup_{\boldsymbol{u}\in X^T} \left|\frac{\boldsymbol{f}(\boldsymbol{u})}{\mathfrak{w}_m(\boldsymbol{u})}\right|. \tag{4.17}$$

Proposition 4.2. *For any $\boldsymbol{V} \in C_b(X^T)$, $\boldsymbol{u} \in X^T$, and $t > T$, the measure*

$$P_t^{T,\boldsymbol{V}}(\boldsymbol{u}, \cdot) := \mathfrak{P}_t^{T,\boldsymbol{V}*}\delta_{\boldsymbol{u}}$$

is concentrated on X^T_∞. Moreover, for any $\boldsymbol{V} \in C_b(X^T)$, there are integers $m, R_0 \ge 1$ such that[1]

$$\sup_{t\ge 0} \frac{\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathfrak{w}_l\|_{L^\infty_{\mathfrak{w},l}}}{\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}\|_{R_0}} < \infty, \tag{4.18}$$

$$\sup_{t\in[0,1]} \|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}\|_\infty < \infty \tag{4.19}$$

for any $l \ge m$, where $\mathbf{1}$ denotes the function on X^T identically equal to 1, and $\|\cdot\|_\infty$, $\|\cdot\|_R$ are the L^∞ norms on X^T and X^T_R, respectively.

Proof. Combining the estimate (A.9) and Corollary A.2, we see that the measure $P_t^{T,\boldsymbol{V}}(\boldsymbol{u}, \cdot)$ is concentrated on X^T_∞ for any $\boldsymbol{V} \in C_b(X^T)$, $\boldsymbol{u} \in X^T$, and $t > T$. The estimate (4.19) follows from the definition (4.4). Thus, it remains to establish (4.18), whose proof is divided into the following four steps.

Step 1. We claim that for any $\boldsymbol{V} \in C_b(X^T)$, $t \ge 0$, and integer $m \ge 1$, $\mathfrak{P}_t^{T,\boldsymbol{V}}$ is a bounded linear operator in $L^\infty_{\mathfrak{w},m}$. Indeed, from (A.7), it follows

$$\mathbb{E}_{\boldsymbol{u}}\mathfrak{w}_m(\boldsymbol{u}_t) \lesssim_{m,\mathcal{B}_0,h,T} \mathfrak{w}_m(\boldsymbol{u}) \tag{4.20}$$

for any $t \ge 0$ and $m \ge 1$. Then, for any $\boldsymbol{f} \in L^\infty_{\mathfrak{w},m}$, one has

$$\begin{aligned}\mathfrak{P}_t^{T,\boldsymbol{V}}\boldsymbol{f}(u) &= \mathbb{E}_{\boldsymbol{u}}\left(\boldsymbol{f}(\boldsymbol{u}_t)\exp\left(\int_0^t \boldsymbol{V}(\boldsymbol{u}_s)\mathrm{d}s\right)\right)\\ &\le \|\boldsymbol{f}\|_{L^\infty_{\mathfrak{w},m}} e^{t\|\boldsymbol{V}\|_\infty}\mathbb{E}_{\boldsymbol{u}}\mathfrak{w}_m(\boldsymbol{u}_t) \le C_{m,\mathcal{B}_0,h,T}e^{t\|\boldsymbol{V}\|_\infty}\|\boldsymbol{f}\|_{L^\infty_{\mathfrak{w},m}}\mathfrak{w}_m(\boldsymbol{u}),\end{aligned}$$

which implies

$$\|\mathfrak{P}_t^{T,\boldsymbol{V}}\|_{\mathcal{L}(L^\infty_{\mathfrak{w},m})} \le C_{m,\mathcal{B}_0,h,T}e^{t\|\boldsymbol{V}\|_\infty} \tag{4.21}$$

as claimed.

Step 2. Notice that by replacing $\boldsymbol{V}$ by $\boldsymbol{V} - \inf_{X^T}\boldsymbol{V}$, one may assume that $\boldsymbol{V}$ is non-negative. To prove (4.18), it suffices to establish

$$\sup_{k\in\mathbb{N}} \frac{\|\mathfrak{P}_{k(T+1)}^{T,\boldsymbol{V}}\mathfrak{w}_m\|_{L^\infty_{\mathfrak{w},m}}}{\|\mathfrak{P}_{k(T+1)}^{T,\boldsymbol{V}}\mathbf{1}\|_{R_0}} < \infty. \tag{4.22}$$

[1]The expression $\mathfrak{P}_t^{T,\boldsymbol{V}}\mathfrak{w}_m$ is understood in the sense of (4.4), though $\mathfrak{w}_m$ is not bounded.

Indeed, suppose that (4.22) holds. For any $t \geq 0$, let

$$\alpha(t) := \left[\frac{t}{(T+1)}\right](T+1).$$

Then, from (4.21) and the non-negativity of $\boldsymbol{V}$, one has

$$\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathfrak{w}_m\|_{L^\infty_{\mathfrak{w},m}} \leq \|\mathfrak{P}_{t-\alpha(t)}^{T,\boldsymbol{V}}\|_{\mathcal{L}(L^\infty_{\mathfrak{w},m})}\|\mathfrak{P}_{\alpha(t)}^{T,\boldsymbol{V}}\mathfrak{w}_m\|_{L^\infty_{\mathfrak{w},m}} \lesssim_{\boldsymbol{V},m,\mathcal{B}_0,h,T} \|\mathfrak{P}_{\alpha(t)}^{T,\boldsymbol{V}}\mathfrak{w}_m\|_{L^\infty_{\mathfrak{w},m}},$$

and

$$\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}\|_{R_0} \geq \|\mathfrak{P}_{\alpha(t)}^{T,\boldsymbol{V}}\mathbf{1}\|_{R_0}.$$

Consequently,

$$\sup_{t\geq 0}\frac{\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathfrak{w}_m\|_{L^\infty_{\mathfrak{w},m}}}{\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}\|_{R_0}} \leq C_{\boldsymbol{V},m,\mathcal{B}_0,h,T}\sup_{k\in\mathbb{N}}\frac{\|\mathfrak{P}_{k(T+1)}^{T,\boldsymbol{V}}\mathfrak{w}_m\|_{L^\infty_{\mathfrak{w},m}}}{\|\mathfrak{P}_{k(T+1)}^{T,\boldsymbol{V}}\mathbf{1}\|_{R_0}} < \infty.$$

Step 3. We claim that there are integers $m_0, R_0 \geq 1$ such that

$$\sup_{t\geq 0}\frac{\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}\|_{L^\infty_{\mathfrak{w},m}}}{\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}\|_{R_0}} < \infty \tag{4.23}$$

for any $m \geq m_0$. To see this, let $\tau(R)$ be defined in (4.6), and let the integers $m_0, R_0 \geq 1$ and the constant $C_0 > 0$ be such that (4.7) holds with $\kappa := \|\boldsymbol{V}\|_\infty$. Then,

$$\begin{aligned}\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}(\boldsymbol{u}) &= \mathbb{E}_{\boldsymbol{u}}\left(\mathbb{I}_{A_t}\exp\left(\int_0^t \boldsymbol{V}(\boldsymbol{u}_s)\mathrm{d}s\right)\right) + \mathbb{E}_{\boldsymbol{u}}\left(\mathbb{I}_{A_t^c}\exp\left(\int_0^t \boldsymbol{V}(\boldsymbol{u}_s)\mathrm{d}s\right)\right)\\ &=: I_1 + I_2,\end{aligned} \tag{4.24}$$

where $A_t := \{\tau(R_0) > t\}$. By Proposition 4.1, one has

$$I_1 \leq \mathbb{E}_{\boldsymbol{u}}\exp(\kappa\tau(R_0)) \leq C_0\mathfrak{w}_{m_0}(\boldsymbol{u}). \tag{4.25}$$

As for I_2, by using the strong Markov property, Proposition 4.1, and the non-negativity of $\boldsymbol{V}$, it follows

$$\begin{aligned}I_2 &\leq \mathbb{E}_{\boldsymbol{u}}\left(\mathbb{I}_{A_t^c}\exp\left(\int_0^{\tau(R_0)} \boldsymbol{V}(\boldsymbol{u}_s)\mathrm{d}s\right)\mathbb{E}_{\boldsymbol{u}_{\tau(R_0)}}\exp\left(\int_0^t \boldsymbol{V}(\boldsymbol{u}_s)\mathrm{d}s\right)\right)\\ &\leq C_0\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}\|_{R_0}\mathfrak{w}_{m_0}(\boldsymbol{u}).\end{aligned} \tag{4.26}$$

Since $\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}(\boldsymbol{u}) \geq 1$ for any $\boldsymbol{u} \in X^T$, by applying (4.24)–(4.26) and the fact

$$\mathfrak{w}_m(\boldsymbol{u}) \leq 2\mathfrak{w}_{m+1}(\boldsymbol{u}), \qquad \forall m \geq 1,$$

one gets (4.23).

Step 4. We turn to the proof of (4.18). Applying the Markov property, estimate (4.10), and the non-negativity of $\boldsymbol{V}$, it follows

$$\begin{aligned}&\mathfrak{P}_{k(T+1)}^{T,\boldsymbol{V}}\mathfrak{w}_m(\boldsymbol{u})\\ &\quad\leq e^{(T+1)\|\boldsymbol{V}\|_\infty}\mathbb{E}_{\boldsymbol{u}}\left(\exp\left(\int_0^{(k-1)(T+1)} \boldsymbol{V}(\boldsymbol{u}_s)\mathrm{d}s\right)\mathbb{E}_{\boldsymbol{u}_{(k-1)(T+1)}}\mathfrak{w}_m(\boldsymbol{u}_{T+1})\right)\\ &\quad\leq 4e^{(T+1)\|\boldsymbol{V}\|_\infty - m\alpha_1}\mathfrak{P}_{(k-1)(T+1)}^{T,\boldsymbol{V}}\mathfrak{w}_m(u) + C_{\boldsymbol{V},m,\mathcal{B}_0,h,T}\mathfrak{P}_{k(T+1)}^{T,\boldsymbol{V}}\mathbf{1}(\boldsymbol{u}).\end{aligned} \tag{4.27}$$

Therefore, by choosing m sufficiently large so that

$$q := 4e^{(T+1)\|\boldsymbol{V}\|_\infty - m\alpha_1} < 1$$

and iterating the inequality (4.27), one gets

$$\mathfrak{P}^{T,\boldsymbol{V}}_{k(T+1)}\mathfrak{w}_m(\boldsymbol{u}) \le q^k \mathfrak{w}_m(\boldsymbol{u}) + \frac{1}{1-q} C_{\boldsymbol{V},m,\mathcal{B}_0,h,T} \mathfrak{P}^{T,\boldsymbol{V}}_{k(T+1)} \mathbf{1}(\boldsymbol{u}).$$

This, together with (4.23), implies (4.22). □

4.3 Time-continuity and dissipativity condition

The following result verifies the time-continuity condition in Proposition E.2.

Proposition 4.3. *For any integer $m \ge 1$, $\boldsymbol{f} \in C_{\mathfrak{w},m}(X^T)$, $\boldsymbol{V} \in C_b(X^T)$, and $\boldsymbol{u} \in X^T$, the map $t \mapsto \mathfrak{P}^{T,\boldsymbol{V}}_t \boldsymbol{f}(\boldsymbol{u})$ is continuous from $[0,\infty)$ to $\mathbb{R}$.*

Proof. We claim that the family

$$\mathcal{A} := \left\{ \boldsymbol{f}(\boldsymbol{u}_t) \exp\left(\int_0^t \boldsymbol{V}(\boldsymbol{u}_s) \mathrm{d}s \right) \middle| 0 \le t \le t_0 + 1 \right\}$$

is uniformly integrable. Indeed, by applying (4.20), one has

$$\begin{aligned} \mathbb{E}_{\boldsymbol{u}} \left(\boldsymbol{f}(\boldsymbol{u}_t) \exp\left(\int_0^t \boldsymbol{V}(\boldsymbol{u}_s) \mathrm{d}s \right) \right)^2 &\le 2\|\boldsymbol{f}\|^2_{L^\infty_{\mathfrak{w},m}} e^{2(t_0+1)\|\boldsymbol{V}\|_\infty} \mathbb{E}_{\boldsymbol{u}} \mathfrak{w}_{2m}(\boldsymbol{u}_t) \\ &\lesssim_{m,\mathcal{B}_0,h,T} \|\boldsymbol{f}\|^2_{L^\infty_{\mathfrak{w},m}} e^{2(t_0+1)\|\boldsymbol{V}\|_\infty} \mathfrak{w}_{2m}(\boldsymbol{u}). \end{aligned}$$

This ensures the uniform integrability of $\mathcal{A}$ as claimed. On the other hand, by the continuity of the sample paths, it follows

$$\lim_{t \to t_0} \boldsymbol{f}(\boldsymbol{u}_t) \exp\left(\int_0^t \boldsymbol{V}(\boldsymbol{u}_s) \mathrm{d}s \right) = \boldsymbol{f}(\boldsymbol{u}_{t_0}) \exp\left(\int_0^{t_0} \boldsymbol{V}(\boldsymbol{u}_s) \mathrm{d}s \right)$$

almost surely. Consequently, an application of the Vitali convergence theorem gives the continuity of $t \mapsto \mathfrak{P}^{T,\boldsymbol{V}}_t \boldsymbol{f}(\boldsymbol{u})$. □

We turn to the dissipativity condition in Proposition E.2.

Proposition 4.4. *Let $\boldsymbol{V} \in C_b(X^T)$. Then, there is an integer $C = C_{\mathcal{B}_0,h,T} \ge 1$ such that for any $t \ge 0$, integer $R \ge 1$, and $\boldsymbol{u} \in \overline{B_{X^T}(R)}$,*

$$P^{T,\boldsymbol{V}}_t(\boldsymbol{u}, \overline{B_{X^T}(\mathcal{R})}) \ge \frac{1}{2} e^{-t\|\boldsymbol{V}\|_\infty}, \tag{4.28}$$

where $\mathcal{R} = C(R+1)$.

Proof. Step 1. First, we establish (4.28) for $P^T_t(\boldsymbol{u}, \cdot)$. Applying (4.20), it follows

$$\mathbb{E}_{\boldsymbol{u}} \|\boldsymbol{u}_t\|^2_{X^T} \lesssim_{\mathcal{B}_0,h,T} \|\boldsymbol{u}\|^2_{X^T} + 1, \qquad \forall t \ge 0.$$

In particular, for $\boldsymbol{u} \in \overline{B_{X^T}(R)}$, one has

$$\mathbb{E}_{\boldsymbol{u}} \|\boldsymbol{u}_t\|^2_{X^T} \le C_{\mathcal{B}_0,h,T}(R^2+1),$$

which, by the Chebyshev inequality, implies

$$P^T_t(\boldsymbol{u}, \overline{B_{X^T}(\mathcal{R})}) \ge 1 - \mathcal{R}^{-2} \mathbb{E}_{\boldsymbol{u}} \|\boldsymbol{u}_t\|^2_{X^T} \ge 1 - \frac{C_{\mathcal{B}_0,h,T}(R^2+1)}{\mathcal{R}^2}.$$

Therefore, by choosing $\mathcal{R} = C(R+1)$ with sufficiently large C and taking infimum on both sides of the above estimate, one gets

$$\inf_{t\geq 0}\inf_{\boldsymbol{u}\in\overline{B_{X^T}(R)}} P_t^T\left(\boldsymbol{u}, \overline{B_{X^T}(\mathcal{R})}\right) \geq \frac{1}{2}. \tag{4.29}$$

Step 2. To conclude the proof, notice that

$$P_t^{T,\boldsymbol{V}}\left(\boldsymbol{u}, \overline{B_{X^T}(\mathcal{R})}\right) \geq e^{-t\|\boldsymbol{V}\|_\infty} P_t^T\left(\boldsymbol{u}, \overline{B_{X^T}(\mathcal{R})}\right).$$

Thus, it follows

$$P_t^{T,\boldsymbol{V}}\left(\boldsymbol{u}, \overline{B_{X^T}(\mathcal{R})}\right) \geq \frac{1}{2}e^{-t\|V\|_\infty}$$

for any $t \geq 0$ and $\boldsymbol{u} \in \overline{B_{X^T}(R)}$. This completes the proof. □

4.4 Uniform irreducibility

In this subsection, our goal is to establish the uniform irreducibility condition required in Proposition E.2. Motivated by the relation

$$P_t^{T,\boldsymbol{V}}(\boldsymbol{u}, \mathrm{d}\boldsymbol{v}) \geq e^{-t\|\boldsymbol{V}\|_\infty} P_t^T(\boldsymbol{u}, \mathrm{d}\boldsymbol{v}), \tag{4.30}$$

we start with the transition function $P_t^T(\boldsymbol{u}, \cdot)$.

Lemma 4.5. *For any integers $\rho, R \geq 1$ and $r > 0$, there exist positive numbers $l = l(R)$ and $p = p(\rho, r)$ such that*

$$P_l^T\left(\boldsymbol{u}, X_\rho^T \bigcap B_{X^T}(\hat{\boldsymbol{u}}, r)\right) \geq p, \tag{4.31}$$

where $\boldsymbol{u} \in \overline{B_{X^T}(R)}$ and $\hat{\boldsymbol{u}} \in X_\rho^T$.

Proof. Step 1. First, we study the irreducibility property of the transition function $P_{T+1}^T(\boldsymbol{u}, \cdot)$. Notice that from the definition (4.2), one has

$$P_{T+1}^T(\boldsymbol{u}, \Gamma) = \mathbb{P}_{\boldsymbol{u}(T)}\{u_{[1,T+1]} \in \Gamma\}, \qquad \forall \Gamma \in \mathscr{B}(X^T). \tag{4.32}$$

Define

$$\tilde{P}_{T+1}^T(u, \Gamma) := \mathbb{P}_u\{u_{[1,T+1]} \in \Gamma\}, \qquad u \in H.$$

Under this notation, one has $P_{T+1}^T(\boldsymbol{u}, \cdot) = \tilde{P}_{T+1}^T(u, \cdot)|_{u=\boldsymbol{u}(T)}$. We claim that for any integer $\rho \geq 1$ and $r, d > 0$,

$$p_1(\rho, r, d) := \inf_{(u,\hat{\boldsymbol{u}})\in\overline{B_U(d)}\times X_\rho^T} \tilde{P}_{T+1}^T\left(u, X_\rho^T \bigcap B_{X^T}(\hat{\boldsymbol{u}}, r)\right) > 0. \tag{4.33}$$

Before proceeding into the proof of (4.33), we reduce its verification to some simplified situations. Notice that

$$\tilde{P}_{T+1}^T\left(u, X_\rho^T \bigcap B_{X^T}(\hat{\boldsymbol{u}}, r)\right) \geq \tilde{P}_{T+1}^T\left(u, B_{X_\infty^T}(\rho) \bigcap B_{X^T}(\hat{\boldsymbol{u}}, r)\right), \tag{4.34}$$

where, by Proposition A.3, the right-hand side, considered as a function of $(u, \hat{\boldsymbol{u}}) \in H \times X^T$, is lower semi-continuous. Together with the compactness of $\overline{B_U(d)} \times X_\rho^T$, this reduces the proof of (4.33) to

$$\tilde{P}_{T+1}^T\left(u, B_{X_\infty^T}(\rho) \bigcap B_{X^T}(\hat{\boldsymbol{u}}, r)\right) > 0 \tag{4.35}$$

for each fixed $(u, \hat{\boldsymbol{u}}) \in \overline{B_U(d)} \times X_\rho^T$.

Furthermore, we show that it suffices to establish (4.35) with $\hat{\boldsymbol{u}} \in C^1([0,T];U^2)$. Indeed, for any $\hat{u} \in X_\rho^T$, by definition, there is $\tilde{\boldsymbol{u}} \in B_{X_\infty^T}(\rho)$ such that

$$\|\tilde{u} - \hat{u}\|_{X^T} < \frac{r}{4}.$$

Define

$$\tilde{u}_{\epsilon,N} := \varphi_\epsilon * (\mathsf{P}_N \tilde{u}),$$

where P_N is the orthogonal projection in H onto the space H_N spanned by $e_1, e_2, \dots, e_N$[2], φ_ϵ is the standard mollifier in the time variable t, and $\mathsf{P}_N\tilde{u}(t)$ is understood as constant extension outside $[0,T]$. It is straightforward to check that

$$\tilde{u}_{\epsilon,N} \in C^1([0,T];U^2), \qquad \|\tilde{u}_{\epsilon,N}\|_{X_\infty^T} \le \|\tilde{u}\|_{X_\infty^T} < \rho,$$

and

$$\lim_{\substack{\epsilon\to 0\\ N\to\infty}} \|\tilde{u}_{\epsilon,N} - \tilde{u}\|_{X^T} = 0.$$

Consequently, by choosing sufficiently small ϵ and large N, it follows that

$$\tilde{u}_{\epsilon,N} \in C^1([0,T];U^2) \bigcap B_{X_\infty^T}(\rho), \qquad \|\tilde{u}_{\epsilon,N} - \hat{u}\|_{X^T} < \frac{r}{2}.$$

Set $\tilde{r} := \min\{\rho - \|\tilde{u}_{\epsilon,N}\|_{X_\infty^T}, \frac{r}{4}\}$. Then, one has

$$B_{X_\infty^T}(\tilde{\boldsymbol{u}}_{\epsilon,N}, \tilde{r}) \subset B_{X_\infty^T}(\rho) \bigcap B_{X^T}(\hat{\boldsymbol{u}}, r),$$

which implies

$$\tilde{P}_{T+1}^T\left(u, B_{X_\infty^T}(\rho) \bigcap B_{X^T}(\hat{\boldsymbol{u}}, r)\right) \ge \tilde{P}_{T+1}^T\left(u, B_{X_\infty^T}(\tilde{\boldsymbol{u}}_{\epsilon,N}, \tilde{r})\right).$$

This reduces the verification of (4.35) to

$$\tilde{P}_{T+1}^T\left(u, B_{X_\infty^T}(\hat{\boldsymbol{u}}, r)\right) > 0 \tag{4.36}$$

for any fixed $(u, \hat{\boldsymbol{u}}) \in \overline{B_U(d)} \times C^1([0,T];U^2)$ and $d, r > 0$.

Step 2. We prove (4.36) by using a control-type argument, see Section 3.6.1 in [KS12]. We need the following auxiliary lemmas whose proofs are postponed until the end of this subsection.

Lemma 4.6. *For any $\hat{\boldsymbol{u}} \in C^1([0,T];U^2)$, there is a local Lipschitz nonlinear operator*

$$\begin{aligned} Z_{\hat{\boldsymbol{u}}} : U &\to L^2(0,T+1;U^2) \bigcap H^1(0,T+1;H) \bigcap \mathcal{X}^1(0,T+1),\\ u &\mapsto Z_{\hat{\boldsymbol{u}}}(u;t) \end{aligned}$$

such that the solution of (1.1), with the initial data u and the profile of the noise η given by $\partial_t Z_{\hat{\boldsymbol{u}}} + L Z_{\hat{\boldsymbol{u}}}$, is equal to $\hat{\boldsymbol{u}}$ when $t \in [1, T+1]$, that is,

$$\hat{\boldsymbol{u}} = Z_{\hat{\boldsymbol{u}}}(u;t)|_{[1,T+1]} + \mathfrak{R}_1^{T+1}(u, Z_{\hat{\boldsymbol{u}}}(u)), \tag{4.37}$$

where the operator $\mathfrak{R}_1^{1+T}$ and the space $\mathcal{X}^1(0,T+1)$ are defined in (A.27) and (A.28), respectively.

[2]Recall that $\{e_j\}_{j\ge1}$ is the orthonormal basis of H formed by eigenfunctions of the Stokes operator.

Lemma 4.7. *For any $r > 0$ and*

$$\hat{z} \in L^2(0, T+1; U^2) \bigcap H^1(0, T+1; H) \bigcap \mathcal{X}^1(0, T+1),$$

it follows

$$\mathbb{P}\left\{ \|z - \hat{z}\|_{\mathcal{X}^1(0,T+1)} + \|z - \hat{z}\|_{C^{\frac{1}{4}}([0,T+1];U^*)} < r \right\} > 0, \tag{4.38}$$

where z is the solution of the stochastic Stokes system (A.14).

Let $Z_{\hat{\boldsymbol{u}}}$ be the nonlinear operator given in Lemma 4.6 and choose $C_d > 0$ such that the image of $\overline{B_U(d)}$ under $Z_{\hat{\boldsymbol{u}}}$ is contained in $B_{\mathcal{X}^1(0,T+1)}(C_d)$. Let M_d be the Lipschitz constant of $\mathfrak{R}_1^{T+1}$ on the bounded set $\overline{B_U(d)} \times B_{\mathcal{X}^1(0,T+1)}(C_d)$. Then, by applying Proposition A.3, one has

$$\|\mathfrak{R}_1^{T+1}(u, Z_{\hat{\boldsymbol{u}}}(u)) - \mathfrak{R}_1^{T+1}(u, z)\|_{X_\infty^T} < \frac{r}{2},$$

provided that $\|Z_{\hat{\boldsymbol{u}}}(u) - z\|_{\mathcal{X}^1(0,T+1)} < \frac{r}{2M_d}$. Let $u(t)$ denote the solution of (1.1) issued from u. Then, by the definition of $\tilde{P}_{T+1}^T$ and the decomposition (A.12), it follows

$$\tilde{P}_{T+1}^T \left(u, B_{X_\infty^T}(\hat{\boldsymbol{u}}, r)\right) = \mathbb{P}_u\{u_{[1,T+1]} \in B_{X_\infty^T}(\hat{\boldsymbol{u}}, r)\}$$

and

$$\left\{ \|z - Z_{\hat{\boldsymbol{u}}}(u)\|_{\mathcal{X}^1(0,T+1)} + \|z - Z_{\hat{\boldsymbol{u}}}(u)\|_{C^{\frac{1}{4}}([0,T+1];U^*)} < \frac{r}{2M_d} \wedge \frac{r}{2} \right\}$$
$$\subset \{u_{[1,T+1]} \in B_{X_\infty^T}(\hat{\boldsymbol{u}}, r)\}.$$

Combining this with Lemma 4.7, one obtains (4.36) and thus (4.33).
Step 3. We turn to (4.31). To this end, we introduce the set

$$\mathbf{A}^T := \{\boldsymbol{u} \in X^T | \boldsymbol{u}(T) \in \overline{B_U(d)}\},$$

and claim that there is a universal constant $d > 0$ and $t = t(R) > 0$ such that

$$P_t^T(\boldsymbol{u}, \mathbf{A}^T) \geq \frac{1}{2}, \qquad \forall \boldsymbol{u} \in \overline{B_{X^T}(R)}. \tag{4.39}$$

Indeed, by the definition (4.2), it follows

$$P_t^T(\boldsymbol{u}, \mathbf{A}^T) = \mathbb{P}_{\boldsymbol{u}(T)}\{u_t \in \overline{B_U(d)}\}.$$

Combining this with the estimates (A.5) and (A.9), for $t > 1$, one has

$$\mathbb{E}_{\boldsymbol{u}(T)} \log(1 + \log(1 + \|u_t\|_1^2)) \lesssim_{\mathcal{B}_1,h} e^{-\alpha_1 t}\|\boldsymbol{u}(T)\|^2 + 1 \lesssim_{\mathcal{B}_1,h} e^{-\alpha_1 t} R^2 + 1.$$

Together with the Chebyshev inequality, this implies

$$P_t^T(\boldsymbol{u}, \mathbf{A}^T) = 1 - \mathbb{P}_{\boldsymbol{u}(T)}\{u_t \notin \overline{B_U(d)}\} \geq 1 - \frac{C_{\mathcal{B}_1,h}(e^{-\alpha_1 t} R^2 + 1)}{\log(1 + \log(1 + d^2))}.$$

Choosing $d > 0$ sufficiently large so that

$$\log(1 + \log(1 + d^2)) = 4C_{\mathcal{B}_1,h}$$

and

$$t = t(R) := \frac{2 \log R}{\alpha_1} \vee 1,$$

one gets (4.39) as claimed. Finally, an application of the Kolmogorov–Chapman relation combined with (4.33) and (4.39) yields (4.31) with

$$p = p(\rho, r) := \frac{1}{2} p_1(\rho, r, d) > 0$$

and

$$l = l(R) := \left(\frac{2 \log R}{\alpha_1} \vee 1 \right) + T + 1. \tag{4.40}$$

This completes the proof. □

Combining the relation (4.30), Lemma 4.5, and (4.40), we obtain the uniform irreducibility property for the Feynman–Kac semigroup.

Proposition 4.8. *For any integers $\rho, R \geq 1$ and $r > 0$, there are positive numbers $l = l(R)$ and $p = p(\rho, r)$ such that*

$$P_l^{T, \boldsymbol{V}} \left(\boldsymbol{u}, X_\rho^T \bigcap B_{X^T}(\hat{\boldsymbol{u}}, r) \right) \geq C_{\boldsymbol{V}}^{-1} R^{-C_{\boldsymbol{V}}} p, \tag{4.41}$$

where $\boldsymbol{u} \in \overline{B_{X^T}(R)}$, $\hat{\boldsymbol{u}} \in X_\rho^T$, and $C_{\boldsymbol{V}} > 0$ is a constant depending only on $\boldsymbol{V}$.

We present the proofs of Lemmas 4.6 and 4.7.

Proof of Lemma 4.6. Let $\chi : \mathbb{R} \to [0, 1]$ be a smooth cut-off function such that $\chi(t) \equiv 1$ for $t \leq \frac{1}{2}$ and $\chi(t) \equiv 0$ for $t \geq 1$. We define

$$w_{\hat{\boldsymbol{u}}}(u; t) := \begin{cases} \chi(t) e^{-Lt} u + (\partial_t \hat{\boldsymbol{u}}(0) - \hat{\boldsymbol{u}}(0)) t^2 + (2\hat{\boldsymbol{u}}(0) - \partial_t \hat{\boldsymbol{u}}(0)) t, & t \in [0, 1], \\ \hat{\boldsymbol{u}}(t - 1), & t \in [1, T + 1], \end{cases}$$

and

$$\xi_{\hat{\boldsymbol{u}}}(u) := \partial_t w + Lw + B(w) - h,$$

where $w = w_{\hat{\boldsymbol{u}}}(u)$. Let $Z_{\hat{\boldsymbol{u}}}(u)$ be the solution of the following forced Stokes system

$$\partial_t z + Lz = \xi_{\hat{\boldsymbol{u}}}(u)$$

with zero initial data $z(0) = 0$. By applying the standard regularity theory for parabolic equations, the map

$$u \mapsto Z_{\hat{\boldsymbol{u}}}(u)$$

is locally Lipschitz. Notice that $w_{\hat{\boldsymbol{u}}}(u)$ is the solution of (1.1) satisfying all aforementioned properties. This completes the proof. □

Proof of Lemma 4.7. Define

$$\tilde{\mathcal{X}}^1(0, T + 1) := L^2(0, T + 1; U^2) \bigcap H^1(0, T + 1; H) \bigcap \mathcal{X}^1(0, T + 1)$$

equipped with the equivalent norm

$$\|z\|_{\tilde{\mathcal{X}}^1(0, T+1)} := \|z\|_{L^2(0, T+1; U^2)} + \|\partial_t z\|_{L^2(0, T+1; H)}.$$

Using the Sobolev embedding

$$H^1(0, T + 1; H) \subset C^{\frac{1}{4}}([0, T + 1]; H),$$

one has

$$\|\mathsf{Q}_N \hat{z}\|_{\mathcal{X}^1(0, T+1)} + \|\mathsf{Q}_N \hat{z}\|_{C^{\frac{1}{4}}([0, T+1]; U^*)} \lesssim \|\mathsf{Q}_N \hat{z}\|_{\tilde{\mathcal{X}}^1(0, T+1)} \to 0, \tag{4.42}$$

as $N \to +\infty$. Here, $\mathsf{Q}_N := I - \mathsf{P}_N$. Notice that

$$z(t) = \sum_{j\geq 1} b_j z_j(t) e_j,$$

where

$$z_j(t) := \int_0^t e^{-\alpha_j(t-s)} \mathrm{d}\beta_j(s) = \beta_j(t) - \alpha_j \int_0^t e^{-\alpha_j(t-s)} \beta_j(s) \mathrm{d}s.$$

This gives

$$\|z_j\|_{C^{\frac{1}{4}}([0,T+1])} \lesssim \alpha_j \|\beta_j\|_{C^{\frac{1}{4}}([0,T+1])}, \qquad \forall j \geq 1.$$

Consequently, for any $s, t \in [0, T+1]$ satisfying $s \neq t$, one has

$$\begin{aligned}\frac{\|\mathsf{Q}_N z(t) - \mathsf{Q}_N z(s)\|_{U^*}^2}{\sqrt{|t-s|}} &= \sum_{j>N} b_j^2 \alpha_j^{-1} \frac{|z_j(t) - z_j(s)|^2}{\sqrt{|t-s|}} \\ &\leq \sum_{j>N} b_j^2 \alpha_j^{-1} \|z_j\|_{C^{\frac{1}{4}}([0,T+1])}^2 \leq \sum_{j>N} b_j^2 \alpha_j \|\beta_j\|_{C^{\frac{1}{4}}([0,T+1])}^2,\end{aligned}$$

which leads to

$$\mathbb{E}\|\mathsf{Q}_N z\|_{C^{\frac{1}{4}}([0,T+1];U^*)}^2 \lesssim \sum_{j>N} b_j^2 \alpha_j \to 0, \qquad \text{as } N \to +\infty, \tag{4.43}$$

Similarly,

$$\mathbb{E}\|\mathsf{Q}_N z\|_{\mathcal{X}^1(0,T+1)}^2 \lesssim \sum_{j>N} b_j^2 \alpha_j \to 0, \qquad \text{as } N \to +\infty. \tag{4.44}$$

Fix $r > 0$. By applying (4.42)–(4.44), there is an integer $N = N(r) \geq 1$ such that

$$\|\mathsf{Q}_N \hat{z}\|_{\mathcal{X}^1(0,T+1)} + \|\mathsf{Q}_N \hat{z}\|_{C^{\frac{1}{4}}([0,T+1];U^*)} < \frac{r}{3}$$

and

$$\mathbb{P}\left\{\|\mathsf{Q}_N z\|_{\mathcal{X}^1(0,T+1)} + \|\mathsf{Q}_N z\|_{C^{\frac{1}{4}}([0,T+1];U^*)} < \frac{r}{3}\right\} > 0.$$

Notice that

$$\begin{aligned}&\|\mathsf{P}_N z - \mathsf{P}_N \hat{z}\|_{\mathcal{X}^1(0,T+1)} + \|\mathsf{P}_N z - \mathsf{P}_N \hat{z}\|_{C^{\frac{1}{4}}([0,T+1];U^*)} \\ &\qquad \leq C(r) \sup_{j\leq N} \|b_j z_j - \hat{z}_j\|_{C^{\frac{1}{4}}([0,T+1])},\end{aligned}$$

where $\hat{z}_j(t) := \langle \hat{z}(t), e_j \rangle$. For $j \leq N$, define

$$A_j := \left\{\|b_j z_j - \hat{z}_j\|_{C^{\frac{1}{4}}([0,T+1])} < \frac{r}{3C(r)}\right\}$$

and

$$A_{>N} := \left\{\|\mathsf{Q}_N z\|_{\mathcal{X}^1(0,T+1)} + \|\mathsf{Q}_N z\|_{C^{\frac{1}{4}}([0,T+1];U^*)} < \frac{r}{3}\right\}.$$

As z_j has a non-degenerate Gaussian distribution and $b_j > 0$, one has $\mathbb{P}(A_j) > 0$ for all $j \leq N$. Hence, using the independence of the processes $\{z_j\}_{j\geq 1}$, it follows

$$\mathbb{P}\left(\bigcap_{j\leq N} A_j \bigcap A_{>N}\right) > 0.$$

Since

$$\bigcap_{j\leq N} A_j \bigcap A_{>N} \subset \left\{\|z - \hat{z}\|_{\mathcal{X}^1(0,T+1)} + \|z - \hat{z}\|_{C^{\frac{1}{4}}([0,T+1];U^*)} < r\right\},$$

the proof is complete. □

4.5 Uniform Feller property

Let $\mathcal{V}^T$ be the space of functions $\boldsymbol{V} : X^T \to \mathbb{R}$ for which there is an integer $N \geq 1$ and a function $\boldsymbol{F} \in L_b(X^T)$ such that

$$\boldsymbol{V}(\boldsymbol{u}) = \boldsymbol{F}(\mathsf{P}_N \boldsymbol{u}), \qquad \boldsymbol{u} \in X^T. \tag{4.45}$$

Here, P_N is the orthogonal projection in H understood as the natural extension

$$\mathsf{P}_N \boldsymbol{u}(t) := \sum_{j \leq N} \langle \boldsymbol{u}(t), e_j \rangle e_j, \qquad t \in [0,T].$$

The main result is stated as follows.

Proposition 4.9. *For any $\boldsymbol{V} \in \mathcal{V}^T$, there is an integer $R_0 \geq 1$ such that for any $\boldsymbol{f} \in \mathcal{V}^T$ and $R \geq R_0$, the family $\{\|\mathfrak{P}_t^{T,\boldsymbol{V}} \mathbf{1}\|_R^{-1} \mathfrak{P}_t^{T,\boldsymbol{V}} \boldsymbol{f}\}_{t \geq 0}$ is uniformly equicontinuous on X_R^T.*

Proof. Step 1: Construction of coupling processes. For any $u, u' \in H$, let $\{u_t\}_{t \geq 0}$ and $\{u'_t\}_{t \geq 0}$ be the solutions of (1.1) issued from u and u', respectively, and let $\{v_t\}_{t \geq 0}$ be the solution of the following auxiliary problem

$$\begin{cases} \partial_t v + B(v) + Lv + \mathsf{P}_N[a(v-u) + B(u) - B(v)] = h + \eta, \\ v|_{t=0} = u'. \end{cases}$$

Denote by $\lambda(u,u')$ and $\lambda'(u,u')$ the distributions of processes $\{v_t\}_{0 \leq t \leq 1}$ and $\{u'_t\}_{0 \leq t \leq 1}$, respectively. By Theorem 1.2.28 in [KS12], there is a maximal coupling $(\nu(u,u'), \nu'(u,u'))$ defined on some probability space $(\tilde{\Omega}, \tilde{\mathscr{F}}, \tilde{\mathbb{P}})$ for the pair $(\lambda(u,u'), \lambda'(u,u'))$. Let $\{\tilde{v}_t\}_{0 \leq t \leq 1}$ and $\{\tilde{u}'_t\}_{0 \leq t \leq 1}$ be the flows of this maximal coupling. Then, $\{\tilde{v}_t\}_{0 \leq t \leq 1}$ solves

$$\begin{cases} \partial_t \tilde{v} + B(\tilde{v}) + L\tilde{v} + \mathsf{P}_N[a\tilde{v} - B(\tilde{v})] = h + \psi, \\ \tilde{v}|_{t=0} = u', \end{cases}$$

where the process $\left\{\int_0^t \psi(s) \mathrm{d}s\right\}_{0 \leq t \leq 1}$ has the same distribution as that of

$$\left\{\int_0^t (\eta(s) + \mathsf{P}_N(au - B(u)))\, \mathrm{d}s\right\}_{0 \leq t \leq 1}.$$

Let $\{\tilde{u}_t\}_{0 \leq t \leq 1}$ be the solution of

$$\begin{cases} \partial_t \tilde{u} + B(\tilde{u}) + L\tilde{u} + \mathsf{P}_N[a\tilde{u} - B(\tilde{u})] = h + \psi, \\ \tilde{u}|_{t=0} = u. \end{cases}$$

By the uniqueness-in-law for the above equation, the process $\{\tilde{u}_t\}_{0 \leq t \leq 1}$ is distributed as that of $\{u_t\}_{0 \leq t \leq 1}$. Define

$$\mathfrak{G}_t(u,u',\omega) := \tilde{u}_t, \quad \mathfrak{G}'_t(u,u',\omega) := \tilde{u}'_t, \quad \hat{\mathfrak{G}}_t(u,u',\omega) := \tilde{v}_t,$$

where $u, u' \in H$, $\omega \in \tilde{\Omega}$, and $t \in [0,1]$. Let $\{(\Omega^k, \mathscr{F}^k, \mathbb{P}^k)\}_{k \geq 0}$ be independent copies of the probability space $(\tilde{\Omega}, \tilde{\mathscr{F}}, \tilde{\mathbb{P}})$, and denote by $(\Omega, \mathscr{F}, \mathbb{P})$ their direct product. For $\omega = (\omega^0, \omega^1, \ldots) \in \Omega$, define $\tilde{u}_0 = u$, $\tilde{u}'_0 = u'$, and

$$\tilde{u}_t(\omega) := \mathfrak{G}_s(\tilde{u}_k(\omega), \tilde{u}'_k(\omega), \omega^k), \quad \tilde{u}'_t(\omega) := \mathfrak{G}'_s(\tilde{u}_k(\omega), \tilde{u}'_k(\omega), \omega^k), \tag{4.46}$$

$$\tilde{v}_t(\omega) := \hat{\mathfrak{G}}_s(\tilde{u}_k(\omega), \tilde{u}'_k(\omega), \omega^k), \tag{4.47}$$

where $t = s + k$ with $s \in [0,1)$ and $k \geq 0$. We call the pair $(\tilde{u}_t, \tilde{u}'_t)$ a coupled trajectory at level (N,a) issued from (u,u').

Step 2: Stratification. In what follows, we shall simply denote by (u_t, u'_t) and v_t the coupled trajectory and the associated process constructed above. For any integers $r, \rho \geq 0$, define the events

$$\hat{G}_r := \bigcap_{j=0}^{r} G_j, \qquad G_j := \{v_t = u'_t,\ \forall t \in [j, j+1)\}, F_{r,0} = \emptyset,$$

and

$$F_{r,\rho} := \left\{ \sup_{t\in[0,r]} \left(\int_0^t \|u_s\|_1^2 + \|u'_s\|_1^2 \mathrm{d}s - 2Kt \right) \leq \|u\|^2 + \|u'\|^2 + \rho \right\}$$
$$\bigcap \{\|u_r\|^2 + \|u'_r\|^2 \leq \rho\}, \qquad \rho \geq 1,$$

where K is the constant in (A.2). Furthermore, we introduce the following disjoint partitions of Ω. Let $A_0 := G_0^c$, $A_\infty := \hat{G}_{+\infty}$, and

$$A_{r,\rho} := \left(\hat{G}_{r-1} \bigcap G_r^c\right) \bigcap \left(F_{r,\rho} \bigcap F_{r,\rho-1}^c\right), \qquad r, \rho \geq 1.$$

We need the following probabilistic estimates for the events A_0 and $A_{r,\rho}$, whose proof can be found in [Ner19], see Lemma 4.3 therein.

Lemma 4.10. *Let $u, u' \in \overline{B_H(R)}$ be such that $d := \|u - u'\| \leq 1$ and $R \geq 1$. For any $\alpha > 0$, there is an integer $N_0 = N_0(\alpha)$ and universal constants $c_1, c_2, c_3 > 0$ such that*

$$\mathbb{P}(A_0) \lesssim_{R,N,a} d^{\frac{c_1}{2}}, \tag{4.48}$$

$$\mathbb{P}(A_{r,\rho}) \lesssim_R e^{-c_2\rho} \wedge \left(d^{c_1} e^{-c_1\alpha r} + \left[\exp\left(C_{R,N,a} d^{c_1} e^{c_3\rho - c_1\alpha r}\right) - 1\right]^{\frac{1}{2}} \right), \tag{4.49}$$

provided that the coupling parameter (N, a) satisfies $N \geq N_0(\alpha)$ and $a \geq \frac{N^2}{2}$.

Step 3: Proof of the uniform equicontinuity. Let m, R_0 be given in Proposition 4.2, and let $\boldsymbol{V}, \boldsymbol{f} \in \mathcal{V}^T$, $R \geq R_0$. Our goal is to prove the uniform equicontinuity of

$$g_t := \|\mathfrak{P}_t^{T,\boldsymbol{V}} \mathbf{1}\|_R^{-1} \mathfrak{P}_t^{T,\boldsymbol{V}} \boldsymbol{f}, \qquad t \geq 0$$

on X_R^T. Without loss of generality, we assume that $\boldsymbol{f}$ is non-negative,

$$\|f\|_\infty \leq 1, \qquad \inf_{X^T} \boldsymbol{V} > 0,$$

and the integer N in (4.45) is the same for $\boldsymbol{V}, \boldsymbol{f}$. Let

$$\boldsymbol{u}, \boldsymbol{u}' \in X_R^T, \qquad d := \|\boldsymbol{u} - \boldsymbol{u}'\|_{X^T} \leq 1.$$

Since $X_R^T \subset \overline{B_{X^T}(R)}$, one has

$$\boldsymbol{u}(T), \boldsymbol{u}'(T) \in \overline{B_H(R)}, \qquad \|\boldsymbol{u}(T) - \boldsymbol{u}'(T)\| \leq d \leq 1.$$

Let (u_t, u'_t) be the coupled trajectory at level (N, a) issued from $(\boldsymbol{u}(T), \boldsymbol{u}'(T))$, and let v_t be the associated process. Define

$$\boldsymbol{u}_t := u_{[t-T,t]}, \qquad \boldsymbol{u}'_t := u'_{[t-T,t]}, \qquad t \geq 0,$$

where

$$u_t := \boldsymbol{u}(T + t), \qquad u'_t := \boldsymbol{u}'(T + t), \qquad t \in [-T, 0].$$

Let

$$\Xi_t^{\boldsymbol{V},\boldsymbol{u}} := \exp\left(\int_0^t \boldsymbol{V}(\boldsymbol{u}_s)\mathrm{d}s\right), \qquad \Xi_t^{\boldsymbol{V},\boldsymbol{u}'} := \exp\left(\int_0^t \boldsymbol{V}(\boldsymbol{u}'_s)\mathrm{d}s\right).$$

Under the above notations, it follows

$$\begin{aligned}\mathfrak{P}_t^{T,\boldsymbol{V}}\boldsymbol{f}(\boldsymbol{u})-\mathfrak{P}_t^{T,\boldsymbol{V}}\boldsymbol{f}(\boldsymbol{u}')&=\mathbb{E}\left(\Xi_t^{\boldsymbol{V},\boldsymbol{u}}\boldsymbol{f}(\boldsymbol{u}_t)-\Xi_t^{\boldsymbol{V},\boldsymbol{u}'}\boldsymbol{f}(\boldsymbol{u}_t')\right)\\&=\mathbb{E}\left(\mathbb{I}_{A_0}\left(\Xi_t^{\boldsymbol{V},\boldsymbol{u}}\boldsymbol{f}(\boldsymbol{u}_t)-\Xi_t^{\boldsymbol{V},\boldsymbol{u}'}\boldsymbol{f}(\boldsymbol{u}_t')\right)\right)\\&\quad+\mathbb{E}\left(\mathbb{I}_{A_\infty}\left(\Xi_t^{\boldsymbol{V},\boldsymbol{u}}\boldsymbol{f}(\boldsymbol{u}_t)-\Xi_t^{\boldsymbol{V},\boldsymbol{u}'}\boldsymbol{f}(\boldsymbol{u}_t')\right)\right)\\&\quad+\sum_{r,\rho\geq 1}\mathbb{E}\left(\mathbb{I}_{A_{r,\rho}}\left(\Xi_t^{\boldsymbol{V},\boldsymbol{u}}\boldsymbol{f}(\boldsymbol{u}_t)-\Xi_t^{\boldsymbol{V},\boldsymbol{u}'}\boldsymbol{f}(\boldsymbol{u}_t')\right)\right)\\&=:I_0^t+I_\infty^t+\sum_{r,\rho\geq 1}I_{r,\rho}^t.\end{aligned}\tag{4.50}$$

We claim that

$$|I_0^t|\lesssim_{R,\boldsymbol{V},T}\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}\|_R\mathbb{P}(A_0)^{\frac{1}{2}},\tag{4.51}$$

$$|I_{r,\rho}^t|\lesssim_{R,\boldsymbol{V},T}e^{r\|\boldsymbol{V}\|_\infty}\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}\|_R\mathbb{P}(A_{r,\rho})^{\frac{1}{2}}\tag{4.52}$$

for any integers $r,\rho\geq 1$ and $R\geq R_0$. We turn to the proof of (4.52), as the other follows similarly. Indeed, if $t\leq r+T+1$, then by the positivity and boundedness of $\boldsymbol{f}$, one has

$$I_{r,\rho}^t\leq\mathbb{E}\left(\mathbb{I}_{A_{r,\rho}}\Xi_t^{\boldsymbol{V},\boldsymbol{u}}\boldsymbol{f}(\boldsymbol{u}_t)\right)\lesssim_{\boldsymbol{V},T}e^{r\|\boldsymbol{V}\|_\infty}\mathbb{P}(A_{r,\rho}),$$

which, by symmetry, implies (4.52). On the other hand, for $t>r+T+1$, by using the Markov property, (4.20), and (4.23), it follows

$$\begin{aligned}I_{r,\rho}^t&\leq\mathbb{E}\left(\mathbb{I}_{A_{r,\rho}}\Xi_t^{\boldsymbol{V},\boldsymbol{u}}\right)\\&\lesssim_{\boldsymbol{V},T}e^{r\|\boldsymbol{V}\|_\infty}\mathbb{E}\left(\mathbb{I}_{A_{r,\rho}}\mathbb{E}\left(\exp\left(\int_{r+T+1}^t\boldsymbol{V}(\boldsymbol{u}_s)\mathrm{d}s\right)\Big|\mathscr{F}_{r+1}\right)\right)\\&\lesssim_{\boldsymbol{V},T}e^{r\|\boldsymbol{V}\|_\infty}\mathbb{E}\left(\mathbb{I}_{A_{r,\rho}}\mathfrak{P}_{t-r-1}^{T,\boldsymbol{V}}\mathbf{1}(\boldsymbol{u}_{r+1})\right)\\&\lesssim_{\boldsymbol{V},T}e^{r\|\boldsymbol{V}\|_\infty}\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}\|_{L^\infty_{\mathfrak{w},m}}\mathbb{E}\left(\mathbb{I}_{A_{r,\rho}}\mathfrak{w}_m(\boldsymbol{u}_{r+1})\right)\\&\lesssim_{\boldsymbol{V},T}e^{r\|\boldsymbol{V}\|_\infty}\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}\|_{R_0}(\mathbb{E}\mathfrak{w}_{2m}(\boldsymbol{u}_{r+1}))^{\frac{1}{2}}\mathbb{P}(A_{r,\rho})^{\frac{1}{2}}\\&\lesssim_{R,\boldsymbol{V},T}e^{r\|\boldsymbol{V}\|_\infty}\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}\|_R\mathbb{P}(A_{r,\rho})^{\frac{1}{2}}.\end{aligned}$$

This, together with symmetry again, implies (4.52).

It remains to estimate I_∞^t. To this end, we decompose

$$I_\infty^t=I_{\infty,1}^t+I_{\infty,2}^t,$$

where

$$\begin{aligned}I_{\infty,1}^t&:=\mathbb{E}\left(\mathbb{I}_{A_\infty}\Xi_t^{\boldsymbol{V},\boldsymbol{u}}\left(\boldsymbol{f}(\boldsymbol{u}_t)-\boldsymbol{f}(\boldsymbol{u}_t')\right)\right),\\I_{\infty,2}^t&:=\mathbb{E}\left(\mathbb{I}_{A_\infty}\left(\Xi_t^{\boldsymbol{V},\boldsymbol{u}}-\Xi_t^{\boldsymbol{V},\boldsymbol{u}'}\right)\boldsymbol{f}(\boldsymbol{u}_t')\right).\end{aligned}$$

Notice that on the event A_∞,

$$v_s=u_s',\qquad\forall s\geq 0.$$

Then, $w_s:=u_s-u_s'$ satisfies the equation

$$\partial_s w+Lw+a\mathsf{P}_N w+\mathsf{Q}_N(B(u)-B(u'))=0,$$

which gives

$$\|\mathsf{P}_N(u_s - u'_s)\| \le e^{-as} d, \qquad \forall s \ge 0.$$

This, together with the Lipschitz property of $\boldsymbol{f}$ and $\boldsymbol{V}$, gives

$$|I^t_{\infty,1}| \lesssim_{\boldsymbol{f}} \mathbb{E}\left(\mathbb{I}_{A_\infty} \Xi^{\boldsymbol{V},\boldsymbol{u}}_t \|\mathsf{P}_N(\boldsymbol{u}_t - \boldsymbol{u}'_t)\|_{X^T}\right) \lesssim_{\boldsymbol{f}} d\|\mathfrak{P}^{T,\boldsymbol{V}}_t \mathbf{1}\|_R,$$

and

$$\begin{aligned}|I^t_{\infty,2}| &\le \mathbb{E}\left(\mathbb{I}_{A_\infty}\left(\exp\left(\int_0^t |\boldsymbol{V}(\boldsymbol{u}_s) - \boldsymbol{V}(\boldsymbol{u}'_s)|\mathrm{d}s\right) - 1\right)\Xi^{\boldsymbol{V},\boldsymbol{u}'}_t \boldsymbol{f}(\boldsymbol{u}'_t)\right)\\ &\le \left(\exp\left(C_{\boldsymbol{V},a} d\right) - 1\right)\|\mathfrak{P}^{T,\boldsymbol{V}}_t \mathbf{1}\|_R \lesssim_{\boldsymbol{V},a} d\|\mathfrak{P}^{T,\boldsymbol{V}}_t \mathbf{1}\|_R.\end{aligned}$$

Hence, it follows

$$|I^t_\infty| \lesssim_{\boldsymbol{f},\boldsymbol{V},a} d\|\mathfrak{P}^{T,\boldsymbol{V}}_t \mathbf{1}\|_R. \tag{4.53}$$

By plugging the estimates (4.51)–(4.53) into (4.50), one gets

$$\frac{|\mathfrak{P}^{T,\boldsymbol{V}}_t \boldsymbol{f}(\boldsymbol{u}) - \mathfrak{P}^{T,\boldsymbol{V}}_t \boldsymbol{f}(\boldsymbol{u}')|}{\|\mathfrak{P}^{T,\boldsymbol{V}}_t \mathbf{1}\|_R} \lesssim_{R,\boldsymbol{f},\boldsymbol{V},T,a} d + \mathbb{P}(A_0)^{\frac{1}{2}} + \sum_{r,\rho\ge 1} e^{r\|\boldsymbol{V}\|_\infty}\mathbb{P}(A_{r,\rho})^{\frac{1}{2}}. \tag{4.54}$$

Combining this with Lemma 4.10, it remains to prove the uniform convergence in d of the series on the right-hand side of (4.54), by appropriately choosing the parameters α and (N, a). Notice that each term in the series is positive. Thus, one only needs to consider the case $d = 1$:

$$\sum_{r,\rho\ge 1} e^{r\|\boldsymbol{V}\|_\infty}\left(e^{-\frac{c_2}{2}\rho} \wedge \left(e^{-\frac{c_1}{2}\alpha r} + \left[\exp\left(C_{R,N,a} e^{c_3\rho - c_1\alpha r}\right) - 1\right]^{\frac{1}{4}}\right)\right) < \infty.$$

Define $\mathbb{S} := \{(r,\rho) | \rho \le \frac{c_1\alpha r}{2c_3}\}$. By choosing

$$\alpha = \frac{16\|\boldsymbol{V}\|_\infty}{c_1} \vee \frac{8c_3\|\boldsymbol{V}\|_\infty}{c_1 c_2}, \qquad N \ge N_0(\alpha), \qquad a = \frac{N^2}{2},$$

one obtains

$$\begin{aligned}&\sum_{(r,\rho)\in\mathbb{S}} e^{r\|\boldsymbol{V}\|_\infty}\left(e^{-\frac{c_1}{2}\alpha r} + \left[\exp\left(C_{R,N,a} e^{c_3\rho - c_1\alpha r}\right) - 1\right]^{\frac{1}{4}}\right)\\ &\qquad \lesssim_{R,N,a} \sum_{r,\rho\in\mathbb{S}} e^{r\|\boldsymbol{V}\|_\infty} e^{-\frac{c_1\alpha r}{8}} \lesssim_{R,\boldsymbol{V},N,a} \sum_{r\ge 1} r e^{-r\|\boldsymbol{V}\|_\infty} < \infty,\end{aligned}$$

and

$$\sum_{(r,\rho)\notin\mathbb{S}} e^{r\|\boldsymbol{V}\|_\infty} e^{-\frac{c_2}{2}\rho} \le \sum_{(r,\rho)\notin\mathbb{S}} e^{-\frac{c_2\rho}{4}} \lesssim_{\boldsymbol{V}} \sum_{\rho\ge 1} \rho e^{-\frac{c_2\rho}{4}} < \infty.$$

This shows the uniform equicontinuity of $\{g_t\}_{t\ge 0}$ on X^T_R. □

4.6 Uniform tail probability estimate

The goal of this subsection is to establish the uniform tail probability estimate required in Proposition E.2 for the Feynman–Kac semigroup.

Proposition 4.11. *Let $\boldsymbol{V} \in C_b(X^T)$ and $m \ge 1$. Then, for any integer $\mathcal{R} \ge 1$ and $t \ge T+1$,*

$$\sup_{\boldsymbol{u}\in\overline{B_{X^T}(\mathcal{R})}} \int_{(X^T_R)^c} \mathfrak{w}_m(\boldsymbol{v}) P^{T,\boldsymbol{V}}_t(\boldsymbol{u}, \mathrm{d}\boldsymbol{v}) \to 0,$$

as $R \to +\infty$.

Proof. Fix $\boldsymbol{V} \in C_b(X^T)$, $t \geq T+1$, and $m, \mathcal{R} \geq 1$. Let

$$g(x) := \log(1+\log(1+x)).$$

Applying the definition of $P_t^{T,\boldsymbol{V}}$ and the Cauchy–Schwarz inequality, one gets

$$\begin{aligned}\int_{(X_R^T)^c} \mathfrak{w}_m(\boldsymbol{v}) P_t^{T,\boldsymbol{V}}(\boldsymbol{u}, \mathrm{d}\boldsymbol{v}) &\leq \frac{e^{t\|\boldsymbol{V}\|_\infty}}{\sqrt{g(R)}} \int_{(X_R^T)^c} \mathfrak{w}_m(\boldsymbol{v}) \sqrt{g\left(\|\boldsymbol{v}\|_{X_\infty^T}\right)} P_t^T(\boldsymbol{u}, \mathrm{d}\boldsymbol{v}) \\ &\leq \frac{e^{t\|\boldsymbol{V}\|_\infty}}{\sqrt{g(R)}} \left(\mathbb{E}_{\boldsymbol{u}} \mathfrak{w}_{2m}(\boldsymbol{u}_t)\right)^{\frac{1}{2}} \left(\mathbb{E}_{\boldsymbol{u}} g\left(\|\boldsymbol{u}_t\|_{X_\infty^T}\right)\right)^{\frac{1}{2}}. \end{aligned} \tag{4.55}$$

Moreover, utilizing the estimates (4.13) and (4.20) combined with the Markov property, it follows

$$\begin{aligned}\mathbb{E}_{\boldsymbol{u}} \log\left(1+\log\left(1+\|\boldsymbol{u}_t\|_{X_\infty^T}\right)\right) &= \mathbb{E}_{\boldsymbol{u}} \left(\mathbb{E}_{\boldsymbol{u}_{t-T-1}} \log\left(1+\log\left(1+\|\boldsymbol{u}_{T+1}\|_{X_\infty^T}\right)\right)\right) \\ &\lesssim_{\mathcal{B}_1,h,T} \mathbb{E}_{\boldsymbol{u}} \mathfrak{w}_1(\boldsymbol{u}_{t-T-1}) \lesssim_{\mathcal{B}_1,h,T} \mathfrak{w}_1(\boldsymbol{u}), \end{aligned} \tag{4.56}$$

which implies

$$\left(\mathbb{E}_{\boldsymbol{u}} \mathfrak{w}_{2m}(\boldsymbol{u}_t)\right)^{\frac{1}{2}} \left(\mathbb{E}_{\boldsymbol{u}} g\left(\|\boldsymbol{u}_t\|_{X_\infty^T}\right)\right)^{\frac{1}{2}} \lesssim_{m,\mathcal{B}_1,h,T} \sqrt{\mathfrak{w}_{2m}(\boldsymbol{u}) \mathfrak{w}_1(\boldsymbol{u})}.$$

This, together with (4.55), yields

$$\sup_{\boldsymbol{u} \in \overline{B_{X^T}(\mathcal{R})}} \int_{(X_R^T)^c} \mathfrak{w}_m(\boldsymbol{v}) P_t^{T,\boldsymbol{V}}(\boldsymbol{u}, \mathrm{d}\boldsymbol{v}) \lesssim_{\boldsymbol{V},t,\mathcal{R},m,\mathcal{B}_1,h,T} \frac{1}{\sqrt{g(R)}} \to 0,$$

as $R \to +\infty$. □

4.7 Existence of an eigenvector

Here, we show that the dual semigroup $\mathfrak{P}_t^{T,\boldsymbol{V}*}$ has at least one eigenvector which satisfies an exponential moment estimate.

Proposition 4.12. *For any $\boldsymbol{V} \in C_b(X^T)$ and $t > 0$, there is a measure $\mu_{t,\boldsymbol{V}} \in \mathcal{P}(X^T)$ and a positive constant $c_{t,\boldsymbol{V}}$ satisfying*

$$\mathfrak{P}_t^{T,\boldsymbol{V}*} \mu_{t,\boldsymbol{V}} = c_{t,\boldsymbol{V}} \mu_{t,\boldsymbol{V}}, \tag{4.57}$$

and

$$\int_{X^T} \exp\left(\frac{\gamma_0}{2} \|\boldsymbol{u}\|_{X^T}^2\right) \mu_{t,\boldsymbol{V}}(\mathrm{d}\boldsymbol{u}) < \infty, \tag{4.58}$$

where γ_0 is the constant given in Lemma A.1.

Proof. Step 1: Existence of an eigenvector for $t \geq T+1$. Let $t \geq T+1$ and $\boldsymbol{V} \in C_b(X^T)$. For simplicity, write

$$\boldsymbol{f}(\boldsymbol{u}) := \exp\left(\frac{\gamma_0}{2} \|\boldsymbol{u}\|_{X^T}^2\right).$$

Define

$$D_l := \{\boldsymbol{\nu} \in \mathcal{P}(X^T) | \langle \boldsymbol{f}, \boldsymbol{\nu} \rangle \leq l\}, \qquad l > 0.$$

By the above definition and the Fatou lemma, D_l is a closed and convex subset of $\mathcal{P}(X^T)$. Introduce the following continuous map

$$G_{t,\boldsymbol{V}} : D_l \to \mathcal{P}(X^T), \qquad \boldsymbol{\nu} \mapsto \frac{\mathfrak{P}_t^{T,\boldsymbol{V}*} \boldsymbol{\nu}}{\mathfrak{P}_t^{T,\boldsymbol{V}*} \boldsymbol{\nu}(X^T)}.$$

We claim that there exists $l > 0$ such that $G_{t,\boldsymbol{V}}(D_l) \subset D_l$. Indeed, from (A.8), one has

$$\mathbb{E}_{\boldsymbol{u}}\left(\|\boldsymbol{u}_t\|_{X^T}^2 \boldsymbol{f}(\boldsymbol{u}_t)\right) \lesssim_{\mathcal{B}_0,h,t,T} \boldsymbol{f}(\boldsymbol{u}), \tag{4.59}$$

which implies

$$\begin{aligned}\mathfrak{P}_t^T \boldsymbol{f}(\boldsymbol{u}) = \mathbb{E}_{\boldsymbol{u}} \boldsymbol{f}(\boldsymbol{u}_t) &= \mathbb{E}_{\boldsymbol{u}}\left(\boldsymbol{f}(\boldsymbol{u}_t)\mathbb{I}_{\{\|\boldsymbol{u}_t\|_{X^T}\le R\}} + \boldsymbol{f}(\boldsymbol{u}_t)\mathbb{I}_{\{\|\boldsymbol{u}_t\|_{X^T}> R\}}\right)\\ &\le e^{\frac{\gamma_0 R^2}{2}} + R^{-2} C_{\mathcal{B}_0,h,t,T}\boldsymbol{f}(\boldsymbol{u}).\end{aligned}$$

Consequently, for $\boldsymbol{\nu} \in D_l$, it follows

$$\begin{aligned}\langle \boldsymbol{f}, G_{t,\boldsymbol{V}}\boldsymbol{\nu}\rangle &\le \exp(t\,\mathrm{Osc}(\boldsymbol{V}))\langle \mathfrak{P}_t^T \boldsymbol{f}, \boldsymbol{\nu}\rangle\\ &\le \exp(t\,\mathrm{Osc}(\boldsymbol{V}))\left(e^{\frac{\gamma_0 R^2}{2}} + R^{-2}C_{\mathcal{B}_0,h,t,T} l\right).\end{aligned}$$

Together with choosing R, l sufficiently large so that

$$R^{-2}C_{\mathcal{B}_0,h,t,T}\exp(t\,\mathrm{Osc}(\boldsymbol{V})) \le \frac{1}{2}, \qquad \exp(t\,\mathrm{Osc}(\boldsymbol{V}))e^{\frac{\gamma_0 R^2}{2}} \le \frac{l}{2},$$

this yields

$$\langle \boldsymbol{f}, G_{t,\boldsymbol{V}}\boldsymbol{\nu}\rangle \le l,$$

and thus $G_{t,\boldsymbol{V}}\boldsymbol{\nu} \in D_l$.

In view of the Schauder fixed point theorem, there is a measure $\boldsymbol{\nu} \in D_l$ satisfying (4.57) with

$$c_{t,\boldsymbol{V}} := \mathfrak{P}_t^{T,\boldsymbol{V}*}\boldsymbol{\nu}(X^T) > 0,$$

provided that $G_{t,\boldsymbol{V}}(D_l)$ is pre-compact. To see this, notice that from (4.56), it follows

$$\begin{aligned}&\int_{X^T} \log\left(1+\log\left(1+\|\boldsymbol{u}\|_{X_\infty^T}\right)\right) G_{t,\boldsymbol{V}}\boldsymbol{\nu}(\mathrm{d}\boldsymbol{u})\\ &\qquad \lesssim_{t,\boldsymbol{V}} \int_{X^T} \mathbb{E}_{\boldsymbol{u}} \log\left(1+\log\left(1+\|\boldsymbol{u}_t\|_{X_\infty^T}\right)\right)\boldsymbol{\nu}(\mathrm{d}\boldsymbol{u})\\ &\qquad \lesssim_{t,\boldsymbol{V},\mathcal{B}_1,h,T} \int_{X^T} \mathfrak{w}_1(\boldsymbol{u})\boldsymbol{\nu}(\mathrm{d}\boldsymbol{u}) \lesssim_{\gamma_0,t,\boldsymbol{V},\mathcal{B}_1,h,T} l\end{aligned}$$

for any $\boldsymbol{\nu} \in D_l$. Combining this with the Prokhorov theorem and the Chebyshev inequality, one gets the pre-compactness of $G_{t,\boldsymbol{V}}(D_l)$. Therefore, there exists an eigenvector $\mu_{t,\boldsymbol{V}}$ satisfying (4.58) for $t \ge T+1$.

Step 2: Existence of an eigenvector for $t < T+1$. Let $k \ge 1$ be such that $kt \ge T+1$. Denote by $\mu_{kt,\boldsymbol{V}} \in \mathcal{P}(X^T)$ the measure satisfying (4.57) and (4.58) with $c_{kt,\boldsymbol{V}} > 0$. Consider the following Bogolyubov averaging

$$\mu_{t,\boldsymbol{V}} = \frac{\hat{\mu}_{t,\boldsymbol{V}}}{\hat{\mu}_{t,\boldsymbol{V}}(X^T)}, \qquad \hat{\mu}_{t,\boldsymbol{V}} := \sum_{l=0}^{k-1}\left(c_{kt,\boldsymbol{V}}^{-\frac{1}{k}}\mathfrak{P}_t^{T,\boldsymbol{V}*}\right)^l \mu_{kt,\boldsymbol{V}}.$$

Applying (4.57) combined with the above definition, one has

$$\mathfrak{P}_t^{T,\boldsymbol{V}*}\mu_{t,\boldsymbol{V}} = c_{kt,\boldsymbol{V}}^{\frac{1}{k}}\mu_{t,\boldsymbol{V}}.$$

To see $\mu_{t,\boldsymbol{V}}$ satisfies the exponential moment estimate (4.58), note that from (A.6), it follows

$$\mathbb{E}_{\boldsymbol{u}}\boldsymbol{f}(\boldsymbol{u}_t) \lesssim_{\gamma_0,\mathcal{B}_0,h,T} \boldsymbol{f}(\boldsymbol{u}), \qquad \forall t \ge 0,$$

which implies

$$\begin{aligned}\langle \boldsymbol{f}, \mathfrak{P}_{lt}^{T,\boldsymbol{V}*}\mu_{kt,\boldsymbol{V}}\rangle &= \langle \mathfrak{P}_{lt}^{T,\boldsymbol{V}}\boldsymbol{f}, \mu_{kt,\boldsymbol{V}}\rangle \\ &\le \exp\left(\|V\|_\infty lt\right)\int_{X^T}\mathbb{E}_{\boldsymbol{u}}\boldsymbol{f}(\boldsymbol{u}_{lt})\mu_{kt,\boldsymbol{V}}(\mathrm{d}\boldsymbol{u}) \\ &\lesssim_{\boldsymbol{V},\gamma_0,\mathcal{B}_0,h,t,T} \langle \boldsymbol{f}, \mu_{kt,\boldsymbol{V}}\rangle < \infty.\end{aligned}$$

Consequently, by the definition of $\mu_{t,\boldsymbol{V}}$, (4.58) holds. This completes the proof. □

4.8 Multiplicative ergodic theorem

Combining the results established previously, we are in a position to apply Proposition E.2. As a result, we obtain a detailed description of the large-time behaviour of the Feynman–Kac semigroup. This result will be used to verify the conditions in the Kifer-type criterion Proposition D.1.

By applying Proposition 4.2, $\{\mathfrak{P}_t^{T,\boldsymbol{V}}\}_{t\ge 0}$ forms a semigroup of bounded linear operators in $C_{\mathfrak{w},m}(X^T)$ for sufficiently large m. As introduced in Section 2, a function $h_{\boldsymbol{V}} \in C_{\mathfrak{w},m}(X^T)$ is called an eigenfunction of $\{\mathfrak{P}_t^{T,\boldsymbol{V}}\}_{t\ge 0}$, if there is an eigenvalue $c_{\boldsymbol{V}} \ge 0$ such that

$$\mathfrak{P}_t^{T,V}h_V = c_V^t h_V, \qquad \forall t > 0.$$

A measure $\mu_{\boldsymbol{V}} \in \mathcal{M}^+(X^T)$ is called an eigenvector of the dual semigroup $\{\mathfrak{P}_t^{T,\boldsymbol{V}*}\}_{t\ge 0}$, if there is an eigenvalue $\tilde{c}_{\boldsymbol{V}} \ge 0$ such that

$$\mathfrak{P}_t^{T,\boldsymbol{V}*}\mu_{\boldsymbol{V}} = \tilde{c}_{\boldsymbol{V}}^t\mu_{\boldsymbol{V}}, \qquad \forall t > 0.$$

Recall that $\mathcal{V}^T$ is the space of functions in the form (4.45), and define

$$\mathcal{P}_{\mathfrak{w},m}(X^T) := \{\boldsymbol{\nu} \in \mathcal{P}(X^T) | \langle \mathfrak{w}_m, \boldsymbol{\nu}\rangle < \infty\},$$

and

$$C_+(X^T) := \{\boldsymbol{f} \in C(X^T) | \boldsymbol{f}(\boldsymbol{u}) \ge 0,\ \forall \boldsymbol{u} \in X^T\}.$$

Theorem 4.13. *For any $\boldsymbol{V} \in \mathcal{V}^T$, there is an integer $m = m(\boldsymbol{V}) \ge 1$ such that the following assertions hold.*

1. *There is a number $c_{\boldsymbol{V}} > 0$, a measure $\mu_{\boldsymbol{V}} \in \mathcal{P}_{\mathfrak{w},m}(X^T)$, and a function $h_{\boldsymbol{V}} \in C_{\mathfrak{w},m}(X^T) \cap C_+(X^T)$ such that*

$$\mathfrak{P}_t^{T,\boldsymbol{V}}h_{\boldsymbol{V}} = c_V^t h_{\boldsymbol{V}}, \qquad \mathfrak{P}_t^{T,\boldsymbol{V}*}\mu_{\boldsymbol{V}} = c_{\boldsymbol{V}}^t\mu_{\boldsymbol{V}}, \qquad \forall t > 0.$$

2. *The measure $\mu_{\boldsymbol{V}}$ is the unique eigenvector of $\{\mathfrak{P}_t^{T,\boldsymbol{V}*}\}_{t\ge 0}$ in $\mathcal{P}_{\mathfrak{w},m}(X^T)$, while $h_{\boldsymbol{V}}$ is the unique eigenfunction of $\{\mathfrak{P}_t^{T,\boldsymbol{V}}\}_{t\ge 0}$ in $C_{\mathfrak{w},m}(X^T)\cap C_+(X^T)$ satisfying the normalization condition*

$$\langle h_{\boldsymbol{V}}, \mu_{\boldsymbol{V}}\rangle = 1.$$

   Moreover, $\mu_{\boldsymbol{V}}$ satisfies the exponential moment estimate (4.58), and $h_{\boldsymbol{V}}$ is strictly positive.
3. *For any $\boldsymbol{f} \in C_{\mathfrak{w},m}(X^T)$, $\boldsymbol{\nu} \in \mathcal{P}_{\mathfrak{w},m}(X^T)$, and $R > 0$,*

$$c_{\boldsymbol{V}}^{-t}\mathfrak{P}_t^{T,\boldsymbol{V}}\boldsymbol{f} \to \langle \boldsymbol{f}, \mu_{\boldsymbol{V}}\rangle h_{\boldsymbol{V}}, \qquad \text{in } C_b(\overline{B_{X^T}(R)}) \bigcap L^1(X^T, \mu_{\boldsymbol{V}}), \tag{4.60}$$

$$c_{\boldsymbol{V}}^{-t}\mathfrak{P}_t^{T,\boldsymbol{V}*}\boldsymbol{\nu} \to \langle h_{\boldsymbol{V}}, \boldsymbol{\nu}\rangle\mu_{\boldsymbol{V}}, \qquad \text{in } \mathcal{M}^+(X^T), \tag{4.61}$$

   as $t \to +\infty$. Moreover, for any $\gamma, M > 0$, the convergence

$$c_{\boldsymbol{V}}^{-t}\mathbb{E}_{\boldsymbol{\nu}}\left(\boldsymbol{f}(\boldsymbol{u}_t)\exp\left(\int_0^t \boldsymbol{V}(\boldsymbol{u}_s)\mathrm{d}s\right)\right) \to \langle \boldsymbol{f}, \mu_{\boldsymbol{V}}\rangle\langle h_{\boldsymbol{V}}, \boldsymbol{\nu}\rangle, \tag{4.62}$$

as $t \to +\infty$, holds uniformly in

$$\boldsymbol{\nu} \in \boldsymbol{\Lambda}(\gamma, M) := \left\{ \boldsymbol{\nu} \in \mathcal{P}(X^T) \Big| \int_{X^T} \exp\left(\gamma \|\boldsymbol{u}\|_{X^T}^2\right) \boldsymbol{\nu}(\mathrm{d}\boldsymbol{u}) \le M \right\}. \tag{4.63}$$

Proof. Step 1. Checking the conditions of Proposition E.2. By Propositions 4.2–4.4, 4.8, 4.9, and 4.11, the conditions of Proposition E.2 are satisfied, provided that $\mathcal{V}^T$ is a determining family[3].

To see this, we apply the Dini theorem, cf. Theorem 1.7.10 in [Bog07], to have $\mathsf{P}_N \boldsymbol{u} \to \boldsymbol{u}$ in X^T. Consequently, for any $\boldsymbol{F} \in L_b(X^T)$, $\boldsymbol{F}(\mathsf{P}_N \boldsymbol{u}) \to \boldsymbol{F}(\boldsymbol{u})$ in the sense of buc-convergence[4] and pointwise convergence. It is known that any function $\boldsymbol{V} \in C_b(X^T)$ can be approximated by functions $\boldsymbol{F}_n \in L_b(X^T)$ in the sense of these types of convergence. Combining this with the dominated convergence theorem, we see that $\mathcal{V}^T$ is a determining family.

Let $\mathfrak{p} := C_{\boldsymbol{V}}^{-1}(C(R+1))^{-C_V} p$, where C_V and p are given in Proposition 4.8 and C is the integer in Proposition 4.4. For the eigenvector $\mu_{t,\boldsymbol{V}}$ constructed in Proposition 4.12, by applying the exponential moment estimate (4.58), one gets

$$\begin{aligned}
\mathfrak{p}^{-1}(R^{2m}+1) \int_{\overline{B_{X^T}(R)}^c} \mathfrak{w}_m(\boldsymbol{u}) \mu_{t,\boldsymbol{V}}(\mathrm{d}\boldsymbol{u}) &\lesssim_{\boldsymbol{V},\rho,r} R^{-1} \int_{X^T} \|\boldsymbol{u}\|_{X^T}^{C_V+1} \mathfrak{w}_{2m}(\boldsymbol{u}) \mu_{t,\boldsymbol{V}}(\mathrm{d}\boldsymbol{u}) \\
&\lesssim_{\boldsymbol{V},\rho,r,\gamma_0} R^{-1} \int_{X^T} \exp\left(\frac{\gamma_0}{2}\|\boldsymbol{u}\|_{X^T}^2\right) \mu_{t,\boldsymbol{V}}(\mathrm{d}\boldsymbol{u}) \\
&\to 0,
\end{aligned}$$

as $R \to \infty$. This verifies the tail probability estimate (E.12).

Consequently, by applying Proposition E.2 and enlarging m if necessary, it follows that $\mu_{t,\boldsymbol{V}}$ is independent of t; we thus set $\mu_{\boldsymbol{V}} := \mu_{t,\boldsymbol{V}}$. The corresponding eigenvalue $c_{\boldsymbol{V}}$ is positive, and there is a positive function $h_{\boldsymbol{V}} \in C_{\mathfrak{w},m}(X^T)$ satisfying

$$\langle h_{\boldsymbol{V}}, \mu_{\boldsymbol{V}} \rangle = 1, \qquad \mathfrak{P}_t^{T,\boldsymbol{V}} h_{\boldsymbol{V}} = c_V^t h_{\boldsymbol{V}}.$$

Moreover, the convergence (4.60) holds for any $\boldsymbol{f} \in \mathcal{V}^T$ and $R > 0$.

As the uniqueness of $h_{\boldsymbol{V}}, \mu_{\boldsymbol{V}}$ follows directly from (4.60) and (4.61), we turn to the proof of (4.60)–(4.62), which is divided into the following four steps.

Step 2: Proof of (4.60) *for* $\boldsymbol{f} \in C_b(X^T)$. We claim that the convergence (4.60) holds for all functions $\boldsymbol{f}$ that can be approximated, in the sense of buc-convergence and pointwise convergence, by a sequence $\{\boldsymbol{f}_n\}_{n\ge1}$ satisfying (4.60). Indeed, for fixed $R > 0$, let

$$\Delta_t(\boldsymbol{f}) := \left\| c_{\boldsymbol{V}}^{-t} \mathfrak{P}_t^{T,\boldsymbol{V}} \boldsymbol{f} - \langle \boldsymbol{f}, \mu_{\boldsymbol{V}} \rangle h_{\boldsymbol{V}} \right\|_{0,R}, \qquad \|\boldsymbol{f}\|_{0,R} := \sup_{\boldsymbol{u} \in \overline{B_{X^T}(R)}} |\boldsymbol{f}(\boldsymbol{u})|.$$

Then, one has

$$\Delta_t(\boldsymbol{f}) \le \Delta_t(\boldsymbol{f}_n) + |\langle \boldsymbol{f} - \boldsymbol{f}_n, \mu_{\boldsymbol{V}} \rangle| \|h_{\boldsymbol{V}}\|_{0,R} + c_{\boldsymbol{V}}^{-t} \|\mathfrak{P}_t^{T,\boldsymbol{V}}(\boldsymbol{f} - \boldsymbol{f}_n)\|_{0,R}.$$

Since for any fixed integer $n \ge 1$, $\lim_{t\to+\infty} \Delta_t(\boldsymbol{f}_n) = 0$, and

$$\lim_{n\to+\infty} |\langle \boldsymbol{f} - \boldsymbol{f}_n, \mu_{\boldsymbol{V}} \rangle| = 0,$$

it remains to establish

$$\lim_{n\to+\infty} \sup_{t \ge T+1} c_{\boldsymbol{V}}^{-t} \|\mathfrak{P}_t^{T,\boldsymbol{V}}(\boldsymbol{f} - \boldsymbol{f}_n)\|_{0,R} = 0. \tag{4.64}$$

[3] Recall that a family $\mathcal{C} \subset C_b(X^T)$ is determining, if any two measures $\mu, \nu \in \mathcal{M}^+(X^T)$ satisfying $\langle f, \mu \rangle = \langle f, \nu \rangle$ for any $f \in \mathcal{C}$ coincide.

[4] Here, we recall that the definition of buc-convergence is given in (2.3).

To this end, notice that for any integer $\rho \geq 1$,

$$\begin{aligned} c_{\boldsymbol{V}}^{-t}\|\mathfrak{P}_t^{T,\boldsymbol{V}}(\boldsymbol{f}-\boldsymbol{f}_n)\|_{0,R} \\ &\leq c_{\boldsymbol{V}}^{-t}\|\mathfrak{P}_t^{T,\boldsymbol{V}}(\mathbb{I}_{X_\rho^T}(\boldsymbol{f}-\boldsymbol{f}_n))\|_{0,R} + c_{\boldsymbol{V}}^{-t}\|\mathfrak{P}_t^{T,\boldsymbol{V}}(\mathbb{I}_{(X_\rho^T)^c}(\boldsymbol{f}-\boldsymbol{f}_n))\|_{0,R} \\ &=: J_1(n,t,\rho)+J_2(n,t,\rho), \end{aligned} \tag{4.65}$$

where, for J_1, by applying (4.23), one has

$$\begin{aligned} J_1(n,t,\rho) &\leq c_{\boldsymbol{V}}^{-t}\|\boldsymbol{f}-\boldsymbol{f}_n\|_\rho\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}\|_{0,R} \lesssim_R c_{\boldsymbol{V}}^{-t}\|\boldsymbol{f}-\boldsymbol{f}_n\|_\rho\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}\|_{L^\infty_{\mathfrak{w},m}} \\ &\lesssim_R c_{\boldsymbol{V}}^{-t}\|\boldsymbol{f}-\boldsymbol{f}_n\|_\rho\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}\|_{R_0}. \end{aligned} \tag{4.66}$$

Convergence (4.60) with $\boldsymbol{f}=\mathbf{1}$ gives

$$\sup_{t\geq 0} c_{\boldsymbol{V}}^{-t}\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}\|_{R_0} < \infty, \tag{4.67}$$

which, together with (4.66), implies

$$\lim_{n\to+\infty}\sup_{t\geq 0} J_1(n,t,\rho) = 0, \qquad \forall \rho \geq 1. \tag{4.68}$$

We turn to the estimate of J_2. Let

$$\mathbf{g}(\boldsymbol{u}) := \log(1+\log(1+\|\boldsymbol{u}\|_{X_\infty^T})), \qquad g(x) := \log(1+\log(1+x)).$$

Notice that

$$J_2(n,t,\rho) \lesssim_{\boldsymbol{f}} \frac{1}{g(\rho)} c_{\boldsymbol{V}}^{-t}\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{g}\|_{0,R}, \tag{4.69}$$

where, by applying the Markov property combined with the estimates (4.18) and (4.56), for $\boldsymbol{u}\in\overline{B_{X^T}(R)}$ and $t\geq T+1$, it follows

$$\begin{aligned} \mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{g}(\boldsymbol{u}) &\lesssim_{\boldsymbol{V},T} \mathbb{E}_{\boldsymbol{u}}\left(\exp\left(\int_0^{t-T-1}\boldsymbol{V}(\boldsymbol{u}_s)\mathrm{d}s\right)\mathbb{E}_{\boldsymbol{u}_{t-T-1}}\mathbf{g}(\boldsymbol{u}_{T+1})\right) \\ &\lesssim_{\boldsymbol{V},T,\mathcal{B}_1,h} \mathbb{E}_{\boldsymbol{u}}\left(\exp\left(\int_0^{t-T-1}\boldsymbol{V}(\boldsymbol{u}_s)\mathrm{d}s\right)\mathfrak{w}_m(\boldsymbol{u}_{t-T-1})\right) \\ &\lesssim_{\boldsymbol{V},T,\mathcal{B}_1,h,R} \|\mathfrak{P}_{t-T-1}^{T,\boldsymbol{V}}\mathfrak{w}_m\|_{L^\infty_{\mathfrak{w},m}} \lesssim_{\boldsymbol{V},T,\mathcal{B}_1,h,R} \|\mathfrak{P}_{t-T-1}^{T,\boldsymbol{V}}\mathbf{1}\|_{R_0}. \end{aligned}$$

This, together with (4.67) and (4.69), gives

$$\sup_{n\geq 1}\sup_{t\geq T+1} J_2(n,t,\rho) \lesssim_{\boldsymbol{f},\boldsymbol{V},T,\mathcal{B}_1,h,R} \frac{1}{g(\rho)}\sup_{t\geq 0} c_{\boldsymbol{V}}^{-t}\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}\|_{R_0} \to 0, \tag{4.70}$$

as $\rho\to+\infty$. Combining (4.65), (4.68), and (4.70), we obtain (4.64) and thus the claimed result. As established in Step 1, any function $\boldsymbol{f}\in C_b(X^T)$ can be buc-approximated and pointwise approximated by functions in $L_b(X^T)$, while the latter can be approximated in the same way by functions in $\mathcal{V}^T$. Therefore, (4.60) holds for any $\boldsymbol{f}\in C_b(X^T)$.

Step 3: Proof of (4.60) *for* $\boldsymbol{f}\in C_{\mathfrak{w},m}(X^T)$. Define

$$\boldsymbol{f}_n := \boldsymbol{f}^+\wedge n - \boldsymbol{f}^-\wedge n.$$

Notice that

$$\boldsymbol{f}_n \in C_b(X^T), \qquad |\boldsymbol{f}_n|\leq|\boldsymbol{f}|, \qquad \forall n\geq 1,$$

and $\boldsymbol{f}_n \to \boldsymbol{f}$ in $C_{\mathfrak{w},m+1}(X^T)$, as $n \to \infty$. This implies

$$\lim_{n\to+\infty} |\langle \boldsymbol{f} - \boldsymbol{f}_n, \mu_{\boldsymbol{V}}\rangle| = 0.$$

Consequently, as in the previous step, it suffices to establish (4.64). Indeed, by applying (4.18) with $l = m+1$, one has

$$\begin{aligned} c_{\boldsymbol{V}}^{-t}\|\mathfrak{P}_t^{T,\boldsymbol{V}}(\boldsymbol{f}-\boldsymbol{f}_n)\|_{0,R} &\lesssim_R \|\boldsymbol{f}-\boldsymbol{f}_n\|_{L^\infty_{\mathfrak{w},m+1}} c_{\boldsymbol{V}}^{-t}\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathfrak{w}_{m+1}\|_{L^\infty_{\mathfrak{w},m+1}} \\ &\lesssim_R \|\boldsymbol{f}-\boldsymbol{f}_n\|_{L^\infty_{\mathfrak{w},m+1}} c_{\boldsymbol{V}}^{-t}\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}\|_{R_0}. \end{aligned}$$

Together with (4.67), we obtain (4.64). Thus, (4.60) holds for $\boldsymbol{f} \in C_{\mathfrak{w},m}(X^T)$.

Step 4: Proof of (4.61) *and* (4.62). Let $\boldsymbol{f} \in C_{\mathfrak{w},m}(X^T)$. Applying (4.18), one has

$$\begin{aligned} c_{\boldsymbol{V}}^{-t}\left|\mathfrak{P}_t^{T,\boldsymbol{V}}\boldsymbol{f}(\boldsymbol{u})\right| &\le c_{\boldsymbol{V}}^{-t}\|\boldsymbol{f}\|_{L^\infty_{\mathfrak{w},m}}\mathfrak{P}_t^{T,\boldsymbol{V}}\mathfrak{w}_m(\boldsymbol{u}) \\ &\le \|\boldsymbol{f}\|_{L^\infty_{\mathfrak{w},m}}\mathfrak{w}_m(\boldsymbol{u})c_{\boldsymbol{V}}^{-t}\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathfrak{w}_m\|_{L^\infty_{\mathfrak{w},m}} \\ &\lesssim \|\boldsymbol{f}\|_{L^\infty_{\mathfrak{w},m}}\mathfrak{w}_m(\boldsymbol{u})\sup_{t\ge0}\left(c_{\boldsymbol{V}}^{-t}\|\mathfrak{P}_t^{T,\boldsymbol{V}}\mathbf{1}\|_{R_0}\right), \qquad \forall \boldsymbol{u}\in X^T, \end{aligned}$$

which, together with the fact $h_{\boldsymbol{V}} \in C_{\mathfrak{w},m}(X^T)$, implies

$$\begin{aligned} &|\langle c_{\boldsymbol{V}}^{-t}\mathfrak{P}_t^{T,\boldsymbol{V}}\boldsymbol{f},\boldsymbol{\nu}\rangle - \langle h_{\boldsymbol{V}},\boldsymbol{\nu}\rangle\langle \boldsymbol{f},\mu_{\boldsymbol{V}}\rangle| \\ &\lesssim_{\boldsymbol{V},\boldsymbol{f}} \left|\langle \mathbb{I}_{B_{X^T}(R)}\left(c_{\boldsymbol{V}}^{-t}\mathfrak{P}_t^{T,\boldsymbol{V}}\boldsymbol{f} - \langle \boldsymbol{f},\mu_{\boldsymbol{V}}\rangle h_{\boldsymbol{V}}\right),\boldsymbol{\nu}\rangle\right| \\ &\quad + |\langle \mathbb{I}_{B_{X^T}(R)^c}\mathfrak{w}_m,\boldsymbol{\nu}\rangle| + |\langle \boldsymbol{f},\mu_{\boldsymbol{V}}\rangle||\langle \mathbb{I}_{B_{X^T}(R)^c}h_{\boldsymbol{V}},\boldsymbol{\nu}\rangle| \\ &\lesssim_{\boldsymbol{V},\boldsymbol{f}} \sup_{\boldsymbol{u}\in B_{X^T}(R)}\left|c_{\boldsymbol{V}}^{-t}\mathfrak{P}_t^{T,\boldsymbol{V}}\boldsymbol{f}(\boldsymbol{u}) - \langle \boldsymbol{f},\mu_{\boldsymbol{V}}\rangle h_{\boldsymbol{V}}(\boldsymbol{u})\right| + \langle \mathbb{I}_{B_{X^T}(R)^c}\mathfrak{w}_m,\boldsymbol{\nu}\rangle. \end{aligned} \tag{4.71}$$

The first term on the right-hand side of (4.71) vanishes, as $t \to +\infty$, because of (4.60). For $\boldsymbol{\nu} \in \mathcal{P}_{\mathfrak{w},m}(X^T)$, the second term can be made arbitrarily small by choosing $R > 0$ sufficiently large. Therefore, the convergence (4.61) holds.

Finally, to get (4.62), notice that for $\boldsymbol{\nu} \in \boldsymbol{\Lambda}(\gamma, M)$,

$$\begin{aligned} \langle \mathbb{I}_{B_{X^T}(R)^c}\mathfrak{w}_m,\boldsymbol{\nu}\rangle &\le \frac{1}{1+R^2}\langle \mathfrak{w}_1\mathfrak{w}_m,\boldsymbol{\nu}\rangle \\ &\lesssim_\gamma \frac{1}{1+R^2}\int_{X^T}\exp\left(\gamma\|\boldsymbol{u}\|_{X^T}^2\right)\boldsymbol{\nu}(\mathrm{d}\boldsymbol{u}) \lesssim_\gamma \frac{M}{1+R^2}. \end{aligned}$$

Together with taking supremum in $\boldsymbol{\nu}$ in both sides of (4.71), this ensures (4.62). □

5 Proof of Theorem 1.2

First, we reduce the proof to a uniform LDP for appropriately shifted empirical measures. Let $(u_t, \mathbb{P}_u)$ be the Markov family associated with (1.1), and let $\boldsymbol{\zeta}_t^T$ be defined in (1.14). We introduce a shifted version of $\boldsymbol{\zeta}_t^T$, that is,

$$\bar{\boldsymbol{\zeta}}_t^T := \frac{1}{t}\int_0^t \delta_{u_{[s-T,s]}}\mathrm{d}s, \tag{5.1}$$

where

$$u_t \equiv u_0, \qquad t \in [-T, 0].$$

By arguments similar to those in Section A.2 of [JNPS15], for any $\gamma, M > 0$, $\{\boldsymbol{\zeta}_t^T\}_{t>0}$ and $\{\bar{\boldsymbol{\zeta}}_t^T\}_{t>0}$ are uniformly exponentially equivalent with respect to the initial distributions in

$\Lambda(\gamma, M)$, where $\Lambda(\gamma, M)$ is given in (1.9). Consequently, it suffices to prove a uniform LDP for $\{\bar{\zeta}_t^T\}_{t>0}$.

On the other hand, notice that the distribution of $\{u_{[s-T,s]}\}_{s\geq 0}$ under $\mathbb{P}_u$ coincides with that of $\{\boldsymbol{u}_t\}_{t\geq 0}$ under $\mathbb{P}_{\boldsymbol{u}}$, where $\boldsymbol{u} \equiv u$ and $(\boldsymbol{u}_t, \mathbb{P}_{\boldsymbol{u}})$ is the T-Markov family defined by (4.1)–(4.3). This further reduces the problem to a uniform LDP for

$$\tilde{\boldsymbol{\zeta}}_t^T := \frac{1}{t}\int_0^t \delta_{\boldsymbol{u}_s} \mathrm{d}s,$$

with respect to the initial distribution $\boldsymbol{\nu} \in \mathcal{P}_c(X^T) \bigcap \boldsymbol{\Lambda}(\gamma, M)$. Here, $\boldsymbol{\Lambda}(\gamma, M)$ is given in (4.63) and $\mathcal{P}_c(X^T)$ denotes the set of all Borel probability measures $\boldsymbol{\nu}$ satisfying

$$\boldsymbol{\nu}\{\boldsymbol{u} \in X^T | \boldsymbol{u} \equiv \boldsymbol{u}(T)\} = 1.$$

We apply the Kifer-type criterion Proposition D.1 to derive a uniform LDP for $\{\tilde{\boldsymbol{\zeta}}_t^T\}_{t>0}$. To this end, we introduce

$$\Theta := \{\theta = (t, \boldsymbol{\nu}) | t \geq 0, \boldsymbol{\nu} \in \mathcal{P}_c(X^T) \bigcap \boldsymbol{\Lambda}(\gamma, M)\}, \qquad r(\theta) = t$$

with an order relation $\prec$ such that $(t_1, \boldsymbol{\nu}_1) \prec (t_2, \boldsymbol{\nu}_2)$ if and only if $t_1 \leq t_2$. By definition, the convergence in $\theta \in \Theta$ is equivalent to the uniform convergence in $\boldsymbol{\nu} \in \mathcal{P}_c(X^T) \bigcap \boldsymbol{\Lambda}(\gamma, M)$, as $t \to +\infty$. We divide the verification of the conditions in Proposition D.1 into three steps.

Step 1: Exponential tightness along any subsequence. The first condition in Proposition D.1 is verified by Proposition 3.7 combined with the uniform exponential equivalence of $\{\boldsymbol{\zeta}_t^T\}_{t>0}$ and $\{\tilde{\boldsymbol{\zeta}}_t^T\}_{t>0}$.

Step 2: Existence of pressure. We turn to the second condition. By taking $\boldsymbol{f} = \mathbf{1}$ in (4.62), one has

$$Q_T(\boldsymbol{V}) = \log c_{\boldsymbol{V}}, \qquad \boldsymbol{V} \in \mathcal{V}^T. \tag{5.2}$$

Since the set of functions $\boldsymbol{V}$ for which the pressure function Q_T exists is closed under buc-convergence, and any function in $C_b(X^T)$ can be buc-approximated by functions in $\mathcal{V}^T$, the pressure function Q_T exists for general $\boldsymbol{V} \in C_b(X^T)$. This verifies the second condition.

Step 3: Uniqueness of equilibrium. In this step, we aim to show that for any $\boldsymbol{V} \in \mathcal{V}^T$, there is only one equilibrium state $\nu_{\boldsymbol{V}}$. To this end, we introduce the following generalized Markov semigroup

$$\mathscr{S}_t^{T,\boldsymbol{V},\boldsymbol{F}}\boldsymbol{f} := c_{\boldsymbol{V}}^{-t} h_{\boldsymbol{V}}^{-1} \mathfrak{P}_t^{T,\boldsymbol{V}+\boldsymbol{F}}(h_{\boldsymbol{V}}\boldsymbol{f}), \qquad \boldsymbol{f}, \boldsymbol{F} \in C_b(X^T),$$

and $\mathscr{S}_t^{T,\boldsymbol{V}} := \mathscr{S}_t^{T,\boldsymbol{V},\mathbf{0}}$. Combining (4.60) with the argument used in Step 2, we see that the pressure function is well-defined:

$$Q_T^{\boldsymbol{V}}(\boldsymbol{F}) := \lim_{t\to+\infty} \frac{1}{t} \log \mathscr{S}_t^{T,\boldsymbol{V},\boldsymbol{F}}\mathbf{1}, \qquad \boldsymbol{F} \in C_b(X^T). \tag{5.3}$$

Define $I_T, I_T^{\boldsymbol{V}}$ as Legendre transforms[5] of $Q_T, Q_T^{\boldsymbol{V}}$, respectively. Applying (4.60), (5.2), and (5.3), one has

$$Q_T^{\boldsymbol{V}}(\boldsymbol{F}) = Q_T(\boldsymbol{V}+\boldsymbol{F}) - Q_T(\boldsymbol{V}), \qquad \forall \boldsymbol{F} \in \mathcal{V}^T,$$

which implies[6]

$$I_T^{\boldsymbol{V}}(\boldsymbol{\nu}) = I_T(\boldsymbol{\nu}) + Q_T(\boldsymbol{V}) - \langle \boldsymbol{V}, \boldsymbol{\nu}\rangle.$$

[5]For the definition of Legendre transform, we refer to the formula (D.2).

[6]The supremum in the definition of Legendre transform can be taken over $\mathcal{V}^T$, which can be proved as in Lemma 2.1.

Therefore, any equilibrium state $\nu_{\boldsymbol{V}}$ is a zero of $I_T^{\boldsymbol{V}}$.

We claim that $I_T^{\boldsymbol{V}}$ has a unique zero

$$\nu_{\boldsymbol{V}} = h_{\boldsymbol{V}}\mu_{\boldsymbol{V}}. \tag{5.4}$$

To see this, we introduce the generator $\mathbb{L}_{\boldsymbol{V}}$ of the semigroup $\{\mathscr{S}_t^{T,\boldsymbol{V}}\}_{t\geq 0}$ defined on the domain

$$\mathcal{D}(\mathbb{L}_{\boldsymbol{V}}) := \left\{ \boldsymbol{f} \in C_b(X^T) \middle| \exists \mathbf{g} \in C_b(X^T),\, \mathscr{S}_t^{T,\boldsymbol{V}}\boldsymbol{f} - \boldsymbol{f} = \int_0^t \mathscr{S}_s^{T,\boldsymbol{V}}\mathbf{g}\mathrm{d}s,\, \forall t \geq 0 \right\}$$

with $\mathbb{L}_{\boldsymbol{V}}\boldsymbol{f} := \mathbf{g}$ for $\boldsymbol{f} \in \mathcal{D}(\mathbb{L}_{\boldsymbol{V}})$. We need the following auxiliary lemmas.

Lemma 5.1. *For $\boldsymbol{F}, \boldsymbol{f} \in C_b(X^T)$,*

$$\mathscr{S}_t^{T,\boldsymbol{V},\boldsymbol{F}}\boldsymbol{f} - \mathscr{S}_t^{T,\boldsymbol{V}}\boldsymbol{f} = \int_0^t \mathscr{S}_{t-s}^{T,\boldsymbol{V}}(\boldsymbol{F}\mathscr{S}_s^{T,\boldsymbol{V},\boldsymbol{F}}\boldsymbol{f})\mathrm{d}s, \qquad \forall t \geq 0.$$

Lemma 5.2. $\mathcal{D}_+(\mathbb{L}_{\boldsymbol{V}}) := \{\boldsymbol{f} \in \mathcal{D}(\mathbb{L}_{\boldsymbol{V}}) | \inf_{X^T} \boldsymbol{f} > 0\}$ *is a determining family.*

Lemma 5.3. $\{\mathscr{S}_t^{T,\boldsymbol{V}}\}_{t\geq 0}$ *has a unique invariant measure given by* (5.4).

Lemma 5.1 follows in the same way as in Lemma 2.2, while Lemma 5.2 can be established as in Lemma A.7 in [MN18]. Moreover, Lemma 5.3 is a direct consequence of (4.60), see Lemma 5.9 in [JNPS18]. Thus, we omit the proofs.

Next, we show that for any $\boldsymbol{f} \in \mathcal{D}_+(\mathbb{L}_{\boldsymbol{V}})$,

$$Q_T^{\boldsymbol{V}}(\boldsymbol{F}_{\boldsymbol{f}}) = 0, \tag{5.5}$$

where $\boldsymbol{F}_{\boldsymbol{f}} := -\mathbb{L}_{\boldsymbol{V}}\boldsymbol{f}/\boldsymbol{f}$. Indeed, by Lemma 5.1 and the Gronwall inequality, one has

$$\mathscr{S}_t^{T,\boldsymbol{V},\boldsymbol{F}_{\boldsymbol{f}}}\boldsymbol{f} \equiv \boldsymbol{f}, \qquad \forall t > 0,$$

which implies

$$\lim_{t\to+\infty} \frac{1}{t} \log \mathscr{S}_t^{T,\boldsymbol{V},\boldsymbol{F}_{\boldsymbol{f}}}\boldsymbol{f} = 0.$$

As $c \leq \boldsymbol{f} \leq C$ for some constants $c, C > 0$, we obtain (5.5).

Finally, let $\nu_{\boldsymbol{V}}$ be a zero of $I_T^{\boldsymbol{V}}$. Then, for any $\boldsymbol{f} \in \mathcal{D}_+(\mathbb{L}_{\boldsymbol{V}})$, it follows

$$0 = I_T^{\boldsymbol{V}}(\nu_{\boldsymbol{V}}) \geq \langle \boldsymbol{F}_{\boldsymbol{f}}, \nu_{\boldsymbol{V}} \rangle = -\int_{X^T} \frac{\mathbb{L}_{\boldsymbol{V}}\boldsymbol{f}}{\boldsymbol{f}} \nu_{\boldsymbol{V}}(\mathrm{d}\boldsymbol{u}).$$

This, together with the facts $\mathbb{L}_{\boldsymbol{V}}\mathbf{1} \equiv 0$ and $\mathbf{1} + c\boldsymbol{f} \in \mathcal{D}_+(\mathbb{L}_{\boldsymbol{V}})$ for sufficiently small c, implies that

$$\psi(c) := \int_{X^T} \frac{\mathbb{L}_{\boldsymbol{V}}(\mathbf{1} + c\boldsymbol{f})}{(\mathbf{1} + c\boldsymbol{f})} \nu_{\boldsymbol{V}}(\mathrm{d}\boldsymbol{u})$$

has a local minimum at $c = 0$. Therefore,

$$0 = \psi'(0) = \int_{X^T} \mathbb{L}_{\boldsymbol{V}}\boldsymbol{f}\nu_{\boldsymbol{V}}(\mathrm{d}\boldsymbol{u})$$

for any $\boldsymbol{f} \in \mathcal{D}_+(\mathbb{L}_{\boldsymbol{V}})$. Consequently, by integrating in the time variable t, it follows

$$\int_{X^T} \mathscr{S}_t^{T,\boldsymbol{V}}\boldsymbol{f}\nu_{\boldsymbol{V}}(\mathrm{d}\boldsymbol{u}) - \int_{X^T} \boldsymbol{f}\nu_{\boldsymbol{V}}(\mathrm{d}\boldsymbol{u}) = 0, \qquad \forall t \geq 0,$$

which, together with Lemma 5.2, implies that $\nu_{\boldsymbol{V}}$ is an invariant measure of $\{\mathscr{S}_t^{T,\boldsymbol{V}}\}_{t\geq 0}$. Thus, an application of Lemma 5.3 leads to $\nu_{\boldsymbol{V}} = h_{\boldsymbol{V}}\mu_{\boldsymbol{V}}$. This proves the uniqueness of equilibrium state.

6 Proof of Theorem 1.3

The LD upper and lower bounds with a good rate function $\boldsymbol{I}$ can be established by the Dawson–Gärtner type argument as in Section A.3 in [JNPS15]. Thus, we omit the proof, and turn to the Donsker–Varadhan entropy formula (1.17). The proof is divided into three steps.

Step 1. Let I denote the rate function given by the Donsker–Varadhan variational formula (1.12), and let $\mathfrak{P}_t$ denote the transition operator defined in (1.5). We claim that

$$I(\lambda) \le \liminf_{t\to 0^+} \frac{1}{t} \sup_{\substack{f\in C_b(H)\\ \inf_H f>0}} \int_H \log \frac{f}{\mathfrak{P}_t f} \mathrm{d}\lambda. \tag{6.1}$$

To see this, let $J(\lambda)$ denote the right-hand side of the above inequality. For any $f \in \mathcal{D}(\mathfrak{L})$ satisfying $\inf_H f > 0$, by applying the continuity of the map $s \mapsto \frac{-\mathfrak{P}_s \mathfrak{L} f}{\mathfrak{P}_s f}$ and the dominated convergence theorem, it follows

$$\begin{aligned}
\int_H \frac{-\mathfrak{L} f}{f} \mathrm{d}\lambda &= \int_H \lim_{t\to 0^+} \frac{1}{t} \int_0^t \frac{-\mathfrak{P}_s \mathfrak{L} f}{\mathfrak{P}_s f} \mathrm{d}s \mathrm{d}\lambda \\
&= \lim_{t\to 0^+} \frac{1}{t} \int_H \int_0^t \frac{-\mathfrak{P}_s \mathfrak{L} f}{\mathfrak{P}_s f} \mathrm{d}s \mathrm{d}\lambda \\
&= \lim_{t\to 0^+} \frac{1}{t} \int_H (-\log \mathfrak{P}_t f + \log f) \mathrm{d}\lambda \le J(\lambda).
\end{aligned}$$

Taking supremum in f, we obtain (6.1).

Step 2. Let $\mathfrak{P}_t^T$ denote the transition operator defined in (4.5). Repeating the derivation of (6.1) and the proof of Theorem 1.1, we see that the rate function I_T in Theorem 1.2 satisfies

$$I_T(\lambda_{[0,T]}) \le \liminf_{t\to 0^+} \frac{1}{t} \sup_{\substack{\boldsymbol{f}\in C_b(X^T)\\ \inf_{X^T} \boldsymbol{f}>0}} \int_{X^T} \log \frac{\boldsymbol{f}}{\mathfrak{P}_t^T \boldsymbol{f}} \mathrm{d}\lambda_{[0,T]}, \qquad \forall \lambda_{[0,T]} \in \mathcal{P}(X^T).$$

Hence, by the Dawson–Gärtner theorem, cf. Theorem A.3 in [JNPS15], one has

$$\boldsymbol{I}(\boldsymbol{\lambda}) \le \sup_{T>0} \liminf_{t\to 0^+} \frac{1}{t} \sup_{\substack{\boldsymbol{f}\in C_b(X^T)\\ \inf_{X^T} \boldsymbol{f}>0}} \int_{X^T} \log \frac{\boldsymbol{f}}{\mathfrak{P}_t^T \boldsymbol{f}} \mathrm{d}\pi^*_{[0,T]}\boldsymbol{\lambda}, \qquad \forall \boldsymbol{\lambda} \in \mathcal{P}(X).$$

Let $\boldsymbol{\lambda} \in \mathcal{P}_s(X)$. For any $\boldsymbol{f} \in C_b(X^T)$ satisfying $\inf_{X^T} \boldsymbol{f} > 0$, define

$$\mathbf{g} := \log \boldsymbol{f} - \log \inf_{X^T} \boldsymbol{f} + 1.$$

Then, $\inf_{X^T} \mathbf{g} = 1 > 0$, and

$$\log \frac{\boldsymbol{f}}{\mathfrak{P}_t^T \boldsymbol{f}} = \mathbf{g} - \log \mathfrak{P}_t^T e^{\mathbf{g}}.$$

Applying this change of test functions and the translation-invariance of $\boldsymbol{\lambda}$, and enlarging the class of test functions from bounded continuous functions to bounded measurable functions, one has

$$\boldsymbol{I}(\boldsymbol{\lambda}) \le \sup_{T>0} \liminf_{t\to 0^+} \frac{1}{t} \sup_{\boldsymbol{f}\in \mathcal{B}^+(X^T)} \int_{X^{\mathbb{R}}} \left(\boldsymbol{f} - \log \mathfrak{P}_t^T e^{\boldsymbol{f}} \right) (u_{[t-T,t]}) \mathrm{d}\boldsymbol{\lambda}.$$

Here, $\mathcal{B}^+(X^T)$ stands for the set of all bounded measurable functions $\boldsymbol{f}$ on X^T satisfying $\inf_{X^T} \boldsymbol{f} > 0$, and for simplicity, we use the notation $\boldsymbol{f}$ instead of $\mathbf{g}$ in the supremum. Now notice that by applying the conditional expectation, it follows

$$\int_{X^{\mathbb{R}}} \boldsymbol{f}(u_{[t-T,t]}) \mathrm{d}\boldsymbol{\lambda} = \int_{X^{\mathbb{R}}} \int_{X^t} \boldsymbol{f}(u_{[t-T,0]}; v_{[0,t]}) \pi^*_{[0,t]} \boldsymbol{\lambda}(u_{(-\infty,0]}, \mathrm{d}v_{[0,t]}) \mathrm{d}\boldsymbol{\lambda}.$$

On the other hand, by utilizing the definition of $\mathfrak{P}_t^T$ and the translation-invariance of $\boldsymbol{\lambda}$ again, one has

$$\begin{aligned}
&\int_{X^{\mathbb{R}}} \log \mathfrak{P}_t^T e^{\boldsymbol{f}}(u_{[t-T,t]}) \mathrm{d}\boldsymbol{\lambda} \\
&\qquad = \int_{X^{\mathbb{R}}} \log \left(\int_{X^t} e^{\boldsymbol{f}(u_{[2t-T,t]}; v_{[0,t]})} \pi_{[0,t]}^* \mathbb{P}_{u(t)}(\mathrm{d}v_{[0,t]}) \right) \mathrm{d}\boldsymbol{\lambda} \\
&\qquad = \int_{X^{\mathbb{R}}} \log \left(\int_{X^t} e^{\boldsymbol{f}(u_{[t-T,0]}; v_{[0,t]})} \pi_{[0,t]}^* \mathbb{P}_{u(0)}(\mathrm{d}v_{[0,t]}) \right) \mathrm{d}\boldsymbol{\lambda}.
\end{aligned}$$

Define

$$\begin{aligned}
F_{\boldsymbol{f},T,t}(u_{(-\infty,0]}) := &\int_{X^t} \boldsymbol{f}(u_{[t-T,0]}; v_{[0,t]}) \pi_{[0,t]}^* \boldsymbol{\lambda}(u_{(-\infty,0]}, \mathrm{d}v_{[0,t]}) \\
&- \log \left(\int_{X^t} e^{\boldsymbol{f}(u_{[t-T,0]}; v_{[0,t]})} \pi_{[0,t]}^* \mathbb{P}_{u(0)}(\mathrm{d}v_{[0,t]}) \right).
\end{aligned}$$

Combining the above results, it follows

$$\boldsymbol{I}(\boldsymbol{\lambda}) \le \sup_{T>0} \liminf_{t\to 0^+} \frac{1}{t} \sup_{\boldsymbol{f}\in\mathcal{B}^+(X^T)} \int_{X^{\mathbb{R}}} F_{\boldsymbol{f},T,t}(u_{(-\infty,0]}) \mathrm{d}\boldsymbol{\lambda}.$$

By the definition (1.23) of the relative entropy,

$$F_{\boldsymbol{f},T,t}(u_{(-\infty,0]}) \le \mathrm{Ent}(\pi_{[0,t]}^* \boldsymbol{\lambda}(u_{(-\infty,0]}, \cdot) | \pi_{[0,t]}^* \mathbb{P}_{u(0)}).$$

This, together with Proposition C.1, implies

$$\boldsymbol{I}(\boldsymbol{\lambda}) \le \liminf_{t\to 0^+} \frac{1}{t} \int_{X^{\mathbb{R}}} \mathrm{Ent}(\pi_{[0,t]}^* \boldsymbol{\lambda}(u_{(-\infty,0]}, \cdot) | \pi_{[0,t]}^* \mathbb{P}_{u(0)}) \mathrm{d}\boldsymbol{\lambda} = \mathcal{I}(\boldsymbol{\lambda}),$$

where $\mathcal{I}(\boldsymbol{\lambda})$ denotes the right-hand side of (1.17). Since $\mathcal{I}(\boldsymbol{\lambda}) = +\infty$ for $\boldsymbol{\lambda} \notin \mathcal{P}_s(X)$, we conclude that

$$\boldsymbol{I}(\boldsymbol{\lambda}) \le \mathcal{I}(\boldsymbol{\lambda}). \tag{6.2}$$

Step 3. It remains to prove (6.2) in the other direction. By contradiction, we assume that there is a probability measure $\boldsymbol{\lambda} \in \mathcal{P}(X)$ such that $\boldsymbol{I}(\boldsymbol{\lambda}) < \mathcal{I}(\boldsymbol{\lambda})$. For $\delta > 0$, let A be an open set such that $\boldsymbol{\lambda} \in A$ and

$$\inf_{\boldsymbol{\sigma}\in\bar{A}} \mathcal{I}(\boldsymbol{\sigma}) \ge (\mathcal{I}(\boldsymbol{\lambda}) - \delta) \wedge \frac{1}{\delta}.$$

The existence of such an open set A is due to the lower semi-continuity of $\mathcal{I}$. Utilizing this, the LD lower bounds with the rate function $\boldsymbol{I}$, and Proposition 3.6, we have

$$\boldsymbol{I}(\boldsymbol{\lambda}) \ge \inf_{\boldsymbol{\sigma}\in A} \boldsymbol{I}(\boldsymbol{\sigma}) \ge \inf_{\boldsymbol{\sigma}\in\bar{A}} \mathcal{I}(\boldsymbol{\sigma}) \ge (\mathcal{I}(\boldsymbol{\lambda}) - \delta) \wedge \frac{1}{\delta}.$$

Sending $\delta \to 0^+$, we obtain $\boldsymbol{I}(\boldsymbol{\lambda}) \ge \mathcal{I}(\boldsymbol{\lambda})$, which is a contradiction.

A A priori estimates

The following lemma gathers standard a priori estimates for the stochastic NS system.

Lemma A.1. *Let $u(t)$ be the solution of (1.1) issued from $u_0 \in H$. Suppose that $h \in U$ and $\mathcal{B}_1 < \infty$. Then, there are positive constants γ_0, K depending on h and $\mathcal{B}_0 := \sum_{j=1}^{\infty} b_j^2$ such that the following estimates hold.*

1. ***Energy equality.*** *For any $t \geq 0$,*

$$\mathbb{E}\|u(t)\|^2 + 2\mathbb{E}\int_0^t \|\nabla u(s)\|^2 \mathrm{d}s = 2\mathbb{E}\int_0^t \langle u(s), h\rangle \mathrm{d}s + \|u_0\|^2 + \mathcal{B}_0 t. \tag{A.1}$$

2. ***Moment estimates for the energy.*** *For any $\kappa \in (0, \gamma_0)$,*

$$\mathbb{P}\left\{\sup_{t\geq 0}(\mathcal{E}(t) - Kt) \geq \|u_0\|^2 + \rho\right\} \leq e^{-\gamma_0 \rho}, \qquad \forall \rho \geq 0, \tag{A.2}$$

$$\mathbb{E}e^{\kappa \mathcal{E}(t)} \lesssim_{\kappa, \mathcal{B}_0, h} e^{\kappa(Kt + \|u_0\|^2)}, \tag{A.3}$$

where

$$\mathcal{E}(t) := \|u(t)\|^2 + \int_0^t \|u(s)\|_1^2 \mathrm{d}s.$$

3. ***Moment estimates for the L^2-norm.*** *For any $\kappa \in (0, \gamma_0)$ and $m \geq 1$,*

$$\mathbb{E}e^{\kappa\|u(t)\|^2} \leq e^{-\kappa t} e^{\kappa\|u_0\|^2} + C_{\kappa, \mathcal{B}_0, h}, \tag{A.4}$$

$$\mathbb{E}\|u(t)\|^{2m} \leq e^{-m\alpha_1 t}\|u_0\|^{2m} + C_{m, \mathcal{B}_0, h}. \tag{A.5}$$

4. ***Moment estimates for the $L_t^\infty L_x^2$-norm.*** *For any $\kappa \in (0, \gamma_0)$ and $m \geq 1$,*

$$\mathbb{E}\exp\left(\kappa\|u(\cdot)\|^2_{C([t,t+T];H)}\right) \leq 4e^{-\kappa t} e^{\kappa\|u_0\|^2} + C_{\kappa, \mathcal{B}_0, h, T}, \tag{A.6}$$

$$\mathbb{E}\|u(\cdot)\|^{2m}_{C([t,t+T];H)} \leq 4e^{-m\alpha_1 t}\|u_0\|^{2m} + C_{m, \mathcal{B}_0, h, T}. \tag{A.7}$$

Moreover, for any $t > 0$,

$$\mathbb{E}\sup_{s\in[t,t+T]}(\|u(s)\|^2 \exp(\kappa\|u(s)\|^2)) \lesssim_{\kappa, \mathcal{B}_0, h, t, T} e^{\kappa\|u_0\|^2}. \tag{A.8}$$

5. ***Doubly-logarithmic moment estimate for the $L_t^\infty H_x^1$-norm.*** *For any $t > 0$,*

$$\mathbb{E}\log\left(1 + \log\left(1 + t\|u(\cdot)\|^2_{C([t,t+T];U)}\right)\right) \lesssim_{\mathcal{B}_1, h, T} \|u_0\|^2 + 1 + t. \tag{A.9}$$

Proof. The estimates (A.1)–(A.5) are proved in [KS12] and [Ner19] in the case of periodic boundary conditions, and the proof is similar in the case of Dirichlet boundary conditions.

The proof of (A.6)–(A.8) is based on (A.4), (A.5), and (6.22) in [Ner19] combined with the Markov property. We present only the proof of (A.7), since the other two follow similarly. First, we establish (A.7) for $t = 0$. Indeed, by using the Itô formula, it follows

$$\begin{aligned}\mathrm{d}\|u(t)\|^{2m} = {} & m\|u(t)\|^{2(m-1)}\left(-2\|\nabla u\|^2 + 2\langle u, h\rangle + \mathcal{B}_0\right)\mathrm{d}t \\ & + 2m\|u(t)\|^{2(m-1)}\sum_{j\geq 1} b_j\langle u, e_j\rangle \mathrm{d}\beta_j \\ & + 2m(m-1)\|u\|^{2(m-2)}\sum_{j\geq 1} b_j^2\langle u, e_j\rangle^2 \mathrm{d}t,\end{aligned}$$

from which, by using the Young inequality, one has

$$\begin{aligned}\|u(t)\|^{2m} - \|u_0\|^{2m} + m\alpha_1\int_0^t \|u\|^{2m}\mathrm{d}s & \\ \leq C_{m, \mathcal{B}_0, h} t + 2m\int_0^t \|u\|^{2(m-1)}\sum_{j\geq 1} b_j\langle u, e_j\rangle \mathrm{d}\beta_j.&\end{aligned}$$

Therefore,

$$\sup_{t\le T}\|u(t)\|^{2m}+m\alpha_1\int_0^T\|u\|^{2m}\mathrm{d}s \\ \le 2\|u_0\|^{2m}+C_{m,\mathcal{B}_0,h}T+4m\sup_{t\le T}\int_0^t\|u\|^{2(m-1)}\sum_{j\ge1}b_j\langle u,e_j\rangle\mathrm{d}\beta_j,$$

which, by using the Burkholder–Davis–Gundy inequality, leads to

$$\mathbb{E}\sup_{t\le T}\|u(t)\|^{2m}+m\alpha_1\mathbb{E}\int_0^T\|u\|^{2m}\mathrm{d}s \\ \le 2\|u_0\|^{2m}+C_{m,\mathcal{B}_0,h}T+4m\mathbb{E}\left(\int_0^T\|u\|^{4(m-1)}\sum_{j\ge1}b_j^2\langle u,e_j\rangle^2\mathrm{d}s\right)^{\frac12}. \tag{A.10}$$

Notice that by the Young inequality,

$$\begin{aligned}\left(\int_0^T\|u\|^{4(m-1)}\sum_{j\ge1}b_j^2\langle u,e_j\rangle^2\mathrm{d}s\right)^{\frac12}&\le\mathcal{B}_0^{\frac12}\left(\int_0^T\|u\|^{4m-2}\mathrm{d}s\right)^{\frac12}\\&\le\mathcal{B}_0^{\frac12}\left(\int_0^T\|u\|^{2m-2}\mathrm{d}s\right)^{\frac12}\left(\sup_{t\le T}\|u(t)\|^{2m}\right)^{\frac12}\\&\le\frac{1}{8m}\sup_{t\le T}\|u(t)\|^{2m}+C_{m,\mathcal{B}_0}\int_0^T\|u\|^{2m-2}\mathrm{d}s\\&\le\frac{1}{8m}\sup_{t\le T}\|u(t)\|^{2m}+\frac{\alpha_1}{8}\int_0^T\|u\|^{2m}\mathrm{d}s+C_{m,\mathcal{B}_0}T.\end{aligned}$$

This, combined with (A.10), implies

$$\mathbb{E}\sup_{t\le T}\|u(t)\|^{2m}+m\alpha_1\mathbb{E}\int_0^T\|u\|^{2m}\mathrm{d}s\le4\|u_0\|^{2m}+C_{m,\mathcal{B}_0,h}T, \tag{A.11}$$

and thus (A.7) with $t=0$. For $t>0$, notice that by applying (A.11), (A.5), and the Markov property, it follows

$$\begin{aligned}\mathbb{E}\|u(\cdot)\|^{2m}_{C([t,t+T];H)}&=\mathbb{E}\left(\mathbb{E}\left(\|u(\cdot)\|^{2m}_{C([t,t+T];H)}\middle|\mathscr{F}_t\right)\right)\\&=\mathbb{E}\left(\mathbb{E}_{u_t}\left(\|u(\cdot)\|^{2m}_{C([0,T];H)}\right)\right)\\&\le\mathbb{E}\left(4\|u(t)\|^{2m}+C_{m,\mathcal{B}_0,h,T}\right)\\&\le4e^{-m\alpha_1t}\|u_0\|^{2m}+C_{m,\mathcal{B}_0,h,T}\end{aligned}$$

as desired.

The estimate (A.9) in the case of multiplicative noise is obtained in [KV14]. In the additive case, the proof is simpler and we give it below for completeness. We decompose

$$u(t)=v(t)+z(t), \tag{A.12}$$

where $v(t)$ is the solution of the following NS-type equation

$$\begin{cases} \partial_t v + Lv + B(v+z) = h, \\ v(0) = u_0, \end{cases} \tag{A.13}$$

and $z(t)$ solves the following stochastic Stokes system

$$\begin{cases} \partial_t z + Lz = \eta, \\ z(0) = 0. \end{cases} \tag{A.14}$$

Applying the inner product in U to both sides of (A.13), one obtains

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|L^{\frac{1}{2}}v(t)\|^2 + \|Lv\|^2 = \langle h, Lv\rangle - \langle B(v+z), Lv\rangle.$$

Utilizing the boundedness of the Leray projection Π combined with the Ladyzhenskaya inequality, it follows

$$\begin{aligned} -\langle B(v+z), Lv\rangle &\lesssim \|Lv\|\|v+z\|_{L^4}\|\nabla(v+z)\|_{L^4} \\ &\lesssim \|v\|_2\|v+z\|^{\frac{1}{2}}\|v+z\|_1\|v+z\|_2^{\frac{1}{2}} \\ &\lesssim \|v\|_2^{\frac{3}{2}}\|v+z\|^{\frac{1}{2}}\|v+z\|_1 + \|v\|_2\|v+z\|^{\frac{1}{2}}\|v+z\|_1\|z\|_2^{\frac{1}{2}} \\ &\leq \frac{1}{8}\|Lv\|^2 + C\|v+z\|_1^4\|v+z\|^2 + C\|z\|_2^2. \end{aligned}$$

Consequently, one has

$$\frac{\mathrm{d}}{\mathrm{d}t}\|v(t)\|_1^2 + \|v\|_2^2 \lesssim \|h\|^2 + \|v+z\|_1^4\|v+z\|^2 + \|z\|_2^2. \tag{A.15}$$

Next, we establish estimates for $\|v\|$. Applying the inner product in H to both sides of (A.13), the cancellation property of the nonlinearity B, and the Ladyzhenskaya inequality, one has

$$\frac{\mathrm{d}}{\mathrm{d}s}\|v(s)\|^2 + \|\nabla v\|^2 \lesssim \|h\|^2 + \|z\|_1^2\|v\|^2 + \|z\|^2\|z\|_1^2,$$

which gives

$$\|v(t)\|^2 + \int_0^t \|v\|_1^2 \mathrm{d}s \lesssim_h (\|u_0\|^2 + t + 1)\exp\left(C\|z_{[0,t]}\|_{\mathcal{X}^0}^2\right), \tag{A.16}$$

where for $j \in \mathbb{N}$,

$$\|z_{[0,t]}\|_{\mathcal{X}^j}^2 := \sup_{s\leq t}\|z(s)\|_j^2 + \int_0^t \|z\|_{j+1}^2 \mathrm{d}s. \tag{A.17}$$

To close the estimate for v, from (A.15), one obtains

$$\frac{\mathrm{d}}{\mathrm{d}t}(t\|v(t)\|_1^2) + t\|v\|_2^2 \lesssim t\|h\|^2 + t(\|v+z\|_1^4\|v+z\|^2 + \|z\|_2^2) + \|v(t)\|_1^2,$$

which, together with the Gronwall inequality and (A.16), leads to

$$t\|v(t)\|_1^2 + \int_0^t s\|v\|_2^2 \mathrm{d}s \lesssim_h \exp\left(C_h\mathfrak{F}_1(t, u_0, z_{[0,t]})\right), \tag{A.18}$$

where

$$\mathfrak{F}_1(t, u_0, z_{[0,t]}) := (\|u_0\|^4 + t^2 + 1)\exp(C\|z_{[0,t]}\|_{\mathcal{X}^1}^2). \tag{A.19}$$

By first taking double logarithm and then expectation, one gets

$$\mathbb{E}\log\left(1+\log\left(1+t\|v(\cdot)\|^2_{C([t,t+T];U)}\right)\right) \lesssim_{h,T} 1+t+\mathbb{E}\|u_0\|^2+\mathbb{E}\|z_{[0,t+T]}\|^2_{\mathcal{X}^1}. \tag{A.20}$$

It remains to estimate the solutions of the stochastic Stokes system (A.14). By the Itô formula,

$$\mathrm{d}\|L^{\frac{1}{2}}z(t)\|^2+2\|Lz\|^2\mathrm{d}t=2\sum_{j\geq 1}b_j\alpha_j\langle z,e_j\rangle\mathrm{d}\beta_j+\mathcal{B}_1\mathrm{d}t.$$

An application of the Burkholder–Davis–Gundy inequality yields

$$\mathbb{E}\sup_{s\leq t}\|z(s)\|_1^2+\mathbb{E}\int_0^t\|z\|_2^2\mathrm{d}s\lesssim_{\mathcal{B}_1} t+1. \tag{A.21}$$

Combining this with (A.20), one has

$$\mathbb{E}\log\left(1+\log\left(1+t\|v(\cdot)\|^2_{C([t,t+T];U)}\right)\right)\lesssim_{\mathcal{B}_1,h,T}\mathbb{E}\|u_0\|^2+1+t. \tag{A.22}$$

Notice that for $g(x):=\log(1+\log(1+x))$,

$$g(xy)+g(x+y)\leq 2(g(x)+g(y)) \tag{A.23}$$

and $g(x)\leq x$ for any $x,y\geq 0$. This, together with (A.12), (A.21), and (A.22), implies (A.9) as desired. □

Corollary A.2. *Under the conditions of Lemma A.1, for any $t>0$,*

$$\mathbb{E}\log\left(1+\log\left(1+t\|u(\cdot)\|^2_{C^{\frac{1}{4}}([t,t+T];U^*)}\right)\right)\lesssim_{\mathcal{B}_1,h,T}1+t+\frac{1}{t}+\|u_0\|^2. \tag{A.24}$$

Proof. Note that

$$u(t)=u_0+\int_0^t(h-B(u)-Lu)\mathrm{d}s+W(t),$$

where

$$W(t):=\sum_{j\geq 1}b_je_j\beta_j(t). \tag{A.25}$$

Then,

$$\|u(\cdot)\|_{C^{\frac{1}{4}}([t,t+T];U^*)} \leq\|u_0\|+\left\|\int_0^{\cdot}(h-B(u)-Lu)\mathrm{d}s\right\|_{C^1([t,t+T];U^*)}+\|W(\cdot)\|_{C^{\frac{1}{4}}([t,t+T];H)},$$

where

$$\begin{aligned}&\left\|\int_0^{\cdot}(h-B(u)-Lu)\mathrm{d}s\right\|_{C^1([t,t+T];U^*)}\\ &\quad\leq\|u(\cdot)-u_0-W(\cdot)\|_{C([t,t+T];H)}+\|h-B(u)-Lu\|_{C([t,t+T];U^*)}\\ &\quad\lesssim\|u_0\|+\|h\|+1+\|u(\cdot)\|^2_{C([t,t+T];U)}+\|W(\cdot)\|_{C^{\frac{1}{4}}([t,t+T];H)}.\end{aligned}$$

Consequently,

$$\|u(\cdot)\|_{C^{\frac{1}{4}}([t,t+T];U^*)}\lesssim\|u_0\|+\|h\|+1+\|u(\cdot)\|^2_{C([t,t+T];U)}+\|W(\cdot)\|_{C^{\frac{1}{4}}([t,t+T];H)}.$$

Combining this with (A.23), one has

$$g\left(t\|u(\cdot)\|^2_{C^{\frac{1}{4}}([t,t+T];U^*)}\right) \lesssim_h g\left(t\|u(\cdot)\|^2_{C([t,t+T];U)}\right) + t + \frac{1}{t} + \|u_0\|^2 + \|W(\cdot)\|^2_{C^{\frac{1}{4}}([t,t+T];H)}. \tag{A.26}$$

It remains to establish moment estimates on the Hölder norm of the Wiener process W. By the embedding

$$W^{\frac{3}{8},8}(0,T;H) \subset C^{\frac{1}{4}}([0,T];H),$$

and Lemma 2.1 in [FG95], it follows

$$\mathbb{E}\|W(\cdot)\|^2_{C^{\frac{1}{4}}([t,t+T];H)} \lesssim 1 + \left(\mathbb{E}\|W(\cdot)\|^8_{W^{\frac{3}{8},8}(t,t+T;H)}\right)^{\frac{1}{4}} \lesssim_{\mathcal{B}_0,T} t + 1.$$

This, together with (A.9) and (A.26), implies (A.24). □

For $t, T > 0$, we define the solution operator $\mathfrak{R}_t^{t+T}$ of (A.13) by

$$\begin{aligned} \mathfrak{R}_t^{t+T} : H \times \mathcal{X}^1(0,t+T) &\to C([0,T];U) \bigcap C^1([0,T];U^*), \\ (u_0, z_{[0,t+T]}) &\mapsto v_{[t,t+T]}, \end{aligned} \tag{A.27}$$

where for $j \in \mathbb{N}$ and $t > 0$,

$$\mathcal{X}^j(0,t) := C_0([0,t];U^j) \bigcap L^2(0,t;U^{j+1}) \tag{A.28}$$

with the norm $\|\cdot\|_{\mathcal{X}^j}$ defined in (A.17) and $C_0([0,t];U^j)$ denoting the space of all continuous functions valued in U^j and vanishing at $t = 0$.

Proposition A.3. *Suppose that $h \in U$ and $\mathcal{B}_1 < \infty$. Then, for any $t, T > 0$, $\mathfrak{R}_t^{t+T}$ is locally Lipschitz.*

Proof. Let $u_0^i \in H$, $z^i \in \mathcal{X}^1(0,t+T)$, and let v^i be the solution of (A.13) with the initial data u_0^i and the trajectory z^i. Write

$$v := v^1 - v^2, \qquad z := z^1 - z^2.$$

Then, v satisfies the equation

$$\partial_t v + B(v+z, v^1+z^1) + B(v^2+z^2, v+z) + Lv = 0. \tag{A.29}$$

Applying the inner product in H and the cancellation property of the convection term B, it follows

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}s}\|v(s)\|^2 + \|\nabla v\|^2 = -\langle B(v+z, v^1+z^1), v\rangle - \langle B(v^2+z^2, z), v\rangle.$$

Using the embedding $H^{\frac{1}{2}} \subset L^4$, the Ladyzhenskaya inequality, and the Cauchy–Schwarz inequality, one has

$$\begin{aligned} -\langle B(v+z, v^1+z^1), v\rangle &\lesssim \|v+z\|_{L^4}\|v\|_{L^4}\|v^1+z^1\|_1 \\ &\lesssim \|v\|\|v\|_1\|v^1+z^1\|_1 + \|z\|_{\frac{1}{2}}\|v\|_{\frac{1}{2}}\|v^1+z^1\|_1 \\ &\leq \frac{1}{8}\|\nabla v\|^2 + C\|v\|^2\|v^1+z^1\|_1^2 + C\|z\|_1^2\|v^1+z^1\|_1^2, \end{aligned}$$

and

$$-\langle B(v^2+z^2,z),v\rangle \le \|v\|_1\|v^2+z^2\|_{\frac12}\|z\|_{\frac12} \le \frac18\|\nabla v\|^2 + C\|z\|_1^2\|v^2+z^2\|_1^2.$$

This implies

$$\frac{\mathrm{d}}{\mathrm{d}s}\|v(s)\|^2 + \|v\|_1^2 \lesssim \|v\|^2\|v^1+z^1\|_1^2 + \|z\|_1^2\|v^1+z^1\|_1^2 + \|z\|_1^2\|v^2+z^2\|_1^2.$$

An application of the Gronwall inequality gives

$$\|v(s)\|^2 + \int_0^s \|v\|_1^2\mathrm{d}r \lesssim \left(\|v_0\|^2 + \|z_{[0,s]}\|_{\mathcal{X}^1}^2\right)\exp\left(C\sum_{i=1}^2\left(\|v^i_{[0,s]}\|_{\mathcal{X}^0}^2 + \|z^i_{[0,s]}\|_{\mathcal{X}^0}^2\right)\right). \tag{A.30}$$

Next, we move to the H^1-estimate. Taking the inner product in U, it follows

$$\frac12\frac{\mathrm{d}}{\mathrm{d}t}\|L^{\frac12}v(t)\|^2 + \|Lv\|^2 = -\langle B(v+z,v^1+z^1),Lv\rangle - \langle B(v^2+z^2,v+z),Lv\rangle,$$

where, by using the interpolation inequalities

$$\|f\|_1 \lesssim \|f\|^{\frac12}\|f\|_2^{\frac12}, \qquad \|f\|_{L^\infty} \lesssim \|f\|^{\frac12}\|f\|_2^{\frac12}, \tag{A.31}$$

and the Cauchy–Schwarz inequality, one has

$$\begin{aligned}\langle B(v+z,v^1+z^1),Lv\rangle &\lesssim \|v+z\|^{\frac12}\|v+z\|_2^{\frac12}\|v^1+z^1\|^{\frac12}\|v^1+z^1\|_2^{\frac12}\|Lv\| \\ &\lesssim \frac18\|Lv\|^2 + C\|v+z\|^2\|v^1+z^1\|^2\|v^1+z^1\|_2^2 + C\|z\|_2^2,\end{aligned}$$

and

$$\begin{aligned}\langle B(v^2+z^2,v+z),Lv\rangle &\lesssim \|v+z\|^{\frac12}\|v+z\|_2^{\frac12}\|v^2+z^2\|^{\frac12}\|v^2+z^2\|_2^{\frac12}\|Lv\| \\ &\lesssim \frac18\|Lv\|^2 + C\|v+z\|^2\|v^2+z^2\|^2\|v^2+z^2\|_2^2 + C\|z\|_2^2.\end{aligned}$$

Therefore,

$$\frac{\mathrm{d}}{\mathrm{d}s}(s\|v(s)\|_1^2) + s\|v\|_2^2 \lesssim \|v\|_1^2 + \sum_{i=1}^2 s\left(\|v^i+z^i\|^2\|v^i+z^i\|_2^2\right)\|v+z\|^2 + s\|z\|_2^2.$$

Combining this and (A.30), one obtains

$$\begin{aligned}s\|v(s)\|_1^2 + \int_0^s r\|v\|_2^2\mathrm{d}r \lesssim (s+1)\left(\|v_0\|^2 + \|z_{[0,s]}\|_{\mathcal{X}^1}^2\right)\left(1+\sum_{i=1}^2\int_0^s r\|v^i+z^i\|_2^2\mathrm{d}r\right) \\ \times\exp\left(C\sum_{i=1}^2\left(\|v^i_{[0,s]}\|_{\mathcal{X}^0}^2 + \|z^i_{[0,s]}\|_{\mathcal{X}^0}^2\right)\right).\end{aligned} \tag{A.32}$$

Notice that from (A.16) and (A.18), it follows

$$\left(\|v^i_{[0,s]}\|_{\mathcal{X}^0}^2 + \int_0^s r\|v^i\|_2^2\mathrm{d}r\right) \lesssim_h \exp(C_h\mathfrak{F}_1(s,u_0^i,z^i_{[0,s]})), \qquad i=1,2,$$

where $\mathfrak{F}_1$ is given in (A.19). Plugging this estimate into (A.32) gives

$$\|v(\cdot)\|^2_{C([t,t+T];U)} \lesssim_{h,t,T,M} \left(\|u_0^1 - u_0^2\|^2 + \|z^1_{[0,t+T]} - z^2_{[0,t+T]}\|^2_{\mathcal{X}^1}\right)$$

for

$$\|u_0^i\| \le M, \qquad \|z^i_{[0,t+T]}\|_{\mathcal{X}^1} \le M, \qquad i = 1, 2.$$

This, together with the equation (A.29) and (A.18), implies

$$\|\partial_t v(\cdot)\|^2_{C([t,t+T];U^*)} \lesssim_{h,t,T,M} \left(\|u_0^1 - u_0^2\|^2 + \|z^1_{[0,t+T]} - z^2_{[0,t+T]}\|^2_{\mathcal{X}^1}\right),$$

and thus completes the proof. □

B Markov semigroups on weighted phase spaces

Let $\{\mathfrak{P}_t\}_{t\ge0}$ denote the Markov semigroup defined in (1.5). For $V \in C_b(H)$, we denote by $\{\mathfrak{P}_t^V\}_{t\ge0}$ the Feynman–Kac semigroup defined in (2.14). For $\kappa > 0$, we introduce the weighted space

$$C_{\boldsymbol{w},\kappa}(H) := \left\{ f \in C(H) \middle| \|f\|_{C_{\boldsymbol{w},\kappa}} := \sup_{u\in H} \frac{|f(u)|}{e^{\kappa\|u\|^2}} < \infty \right\}. \tag{B.1}$$

The goal of this section is twofold. First, we show that $\{\mathfrak{P}_t^V\}_{t\ge0}$ forms a semigroup of bounded linear operators in the weighted space $C_{\boldsymbol{w},\kappa}(H)$ for appropriately chosen $\kappa > 0$ and any $V \in C_b(H)$. Second, we introduce the resolvent operator $\mathcal{R}_\alpha$ of the Markov semigroup $\{\mathfrak{P}_t\}_{t\ge0}$ and establish some approximation properties for $\mathcal{R}_\alpha$. These properties are essential for the proof of Theorem 1.1.

Proposition B.1. *Let $V \in C_b(H)$ and $\kappa := \frac{\gamma_0}{4}$, where γ_0 is the constant in Lemma A.1. Then, $\{\mathfrak{P}_t^V\}_{t\ge0}$ forms a semigroup of bounded linear operators in $C_{\boldsymbol{w},\kappa}(H)$ satisfying the following assertions.*

1. *For any $T > 0$, $\sup_{0\le t\le T} \|\mathfrak{P}_t^V\|_{\mathcal{L}(C_{\boldsymbol{w},\kappa}(H))} \le \exp(T\|V\|_\infty) C_{\kappa,\mathcal{B}_0,h} < \infty$.*
2. *For any $u \in H$ and $f \in C_{\boldsymbol{w},\kappa}(H)$, $t \mapsto \mathfrak{P}_t^V f(u)$ is continuous.*

Proof. Step 1. Let $u(t;u_0)$ be the solution of (1.1) issued from $u_0 \in H$. For any $V \in C_b(H)$, $f \in C_{\boldsymbol{w},\kappa}(H)$, and $R, T > 0$, we claim that the family of random variables

$$\mathcal{F}_{V,f,R,T} := \left\{ \exp\left(\int_0^t V(u(s;u_0))\mathrm{d}s\right) f(u(t;u_0)) \middle| 0 \le t \le T, u_0 \in B_H(R) \right\}$$

is uniformly integrable. Indeed, by the exponential moment estimate (A.4), one has

$$\begin{aligned}
&\sup_{0\le t\le T, u_0\in B_H(R)} \mathbb{E}\left(\exp\left(2\int_0^t V(u(s;u_0))\mathrm{d}s\right)|f(u(t;u_0))|^2\right) \\
&\qquad \le \exp(2T\|V\|_\infty)\|f\|^2_{C_{\boldsymbol{w},\kappa}} \sup_{t\ge0, u_0\in B_H(R)} \mathbb{E}e^{2\kappa\|u(t;u_0)\|^2} \\
&\qquad \le \exp(2T\|V\|_\infty)\|f\|^2_{C_{\boldsymbol{w},\kappa}}(e^{2\kappa R^2} + C_{\kappa,\mathcal{B}_0,h}) < \infty,
\end{aligned}$$

which ensures the uniform integrability of $\mathcal{F}_{V,f,R,T}$ as claimed.
Step 2. We show that $\mathfrak{P}_t^V$ is a bounded linear operator in $C_{\boldsymbol{w},\kappa}(H)$ for any $t \ge 0$. Indeed, by the continuity of $u(\cdot,u_0)$ with respect to the initial data u_0, see Proposition 2.4.7 in [KS12], for any $f \in C_{\boldsymbol{w},\kappa}(H)$ and $V \in C_b(H)$, one has

$$\lim_{u\to u_0} \exp\left(\int_0^t V(u(s;u))\mathrm{d}s\right) f(u(t;u)) = \exp\left(\int_0^t V(u(s;u_0))\mathrm{d}s\right) f(u(t;u_0))$$

$\mathbb{P}$-almost surely. Therefore, by applying the uniform integrability of the family $\mathcal{F}_{V,f,R,T}$ combined with the Vitali convergence theorem, see Theorem 4.5.4. in [Bog07], the map $u \mapsto \mathfrak{P}_t^V f(u)$ is continuous. Moreover, using the exponential moment estimate (A.4) again, it follows

$$\begin{aligned}\|\mathfrak{P}_t^V f\|_{C_{\boldsymbol{w},\kappa}} &= \sup_{u\in H} \frac{|\mathfrak{P}_t^V f(u)|}{e^{\kappa\|u\|^2}} \leq \exp(t\|V\|_\infty) \sup_{u\in H}\left(\frac{\mathbb{E}|f(u(t;u))|}{e^{\kappa\|u\|^2}}\right)\\ &\leq \exp(t\|V\|_\infty)\|f\|_{C_{\boldsymbol{w},\kappa}} \sup_{u\in H} \frac{\mathbb{E}e^{\kappa\|u(t;u)\|^2}}{e^{\kappa\|u\|^2}}\\ &\leq \exp(t\|V\|_\infty) C_{\kappa,\mathcal{B}_0,h}\|f\|_{C_{\boldsymbol{w},\kappa}}.\end{aligned}$$

Hence, $\mathfrak{P}_t^V$ is a bounded linear operator in $C_{\boldsymbol{w},\kappa}(H)$ for any $t \geq 0$, and the first statement holds. The continuity of $t \mapsto \mathfrak{P}_t^V f(u)$ for fixed $u \in H$ follows similarly from the uniform integrability of $\mathcal{F}_{V,f,R,T}$ and the Vitali convergence theorem. The details are omitted. □

In what follows, we shall always take κ as in Proposition B.1. For $\alpha > 0$, we introduce the resolvent operator

$$\mathcal{R}_\alpha f(u) := \int_0^{+\infty} e^{-\alpha t}\mathfrak{P}_t f(u)\mathrm{d}t, \qquad f \in C_{\boldsymbol{w},\kappa}(H). \tag{B.2}$$

By Proposition B.1, this operator is well-defined.

Proposition B.2. *The following properties hold for the resolvent operator $\mathcal{R}_\alpha$.*

1. *For each $\alpha > 0$, $\mathcal{R}_\alpha$ is a bounded linear operator in $C_{\boldsymbol{w},\kappa}(H)$ satisfying*
$$\alpha\|\mathcal{R}_\alpha f\|_{C_{\boldsymbol{w},\kappa}} \lesssim_{\kappa,\mathcal{B}_0,h} \|f\|_{C_{\boldsymbol{w},\kappa}}.$$
2. *For any $f \in C_{\boldsymbol{w},\kappa}(H)$ and $u \in H$,*
$$\lim_{\alpha\to+\infty} \alpha\mathcal{R}_\alpha f(u) = f(u).$$
3. *For any $f \in C_{\boldsymbol{w},\kappa}(H)$, $u \in H$ and $\alpha, t > 0$,*
$$\mathfrak{P}_t\mathcal{R}_\alpha f(u) = \int_0^{+\infty} e^{-\alpha s}\mathfrak{P}_{t+s} f(u)\mathrm{d}s = \mathcal{R}_\alpha\mathfrak{P}_t f(u), \tag{B.3}$$
$$\mathfrak{P}_t\mathcal{R}_\alpha f(u) - \mathcal{R}_\alpha f(u) = \int_0^t \mathfrak{P}_s(\alpha\mathcal{R}_\alpha f - f)(u)\mathrm{d}s. \tag{B.4}$$
4. *If a sequence $\{f_n\}_{n=1}^\infty \subset C_{\boldsymbol{w},\kappa}(H)$ satisfies*
$$\sup_{n\geq 1}\|f_n\|_{C_{\boldsymbol{w},\kappa}} < \infty \tag{B.5}$$
and
$$\lim_{n\to+\infty} f_n(u) = f(u), \qquad \forall u \in H, \tag{B.6}$$
with $f \in C_{\boldsymbol{w},\kappa}(H)$, then for any $\alpha > 0$,
$$\lim_{n\to+\infty} \mathcal{R}_\alpha f_n(u) = \mathcal{R}_\alpha f(u), \qquad \forall u \in H.$$

Proof. We prove the first statement. Note that

$$e^{-\alpha t}|\mathfrak{P}_t f(u)| \leq e^{-\alpha t}e^{\kappa\|u\|^2}\|f\|_{C_{\boldsymbol{w},\kappa}} \sup_{t\geq 0}\|\mathfrak{P}_t\|_{\mathcal{L}(C_{\boldsymbol{w},\kappa}(H))}.$$

By Proposition B.1 and the dominated convergence theorem, one has

$$\lim_{u\to u_0}\mathcal{R}_\alpha f(u)=\mathcal{R}_\alpha f(u_0),$$

and thus the continuity of $\mathcal{R}_\alpha f$. Moreover, it follows

$$\begin{aligned}|\mathcal{R}_\alpha f(u)|&\le\int_0^{+\infty}e^{-\alpha t}|\mathfrak{P}_t f(u)|\mathrm{d}t\\&\le\left(\int_0^{+\infty}e^{-\alpha t}\mathrm{d}t\right)e^{\kappa\|u\|^2}\|f\|_{C_{w,\kappa}}\sup_{t\ge0}\|\mathfrak{P}_t\|_{\mathcal{L}(C_{w,\kappa}(H))}\\&\lesssim_{\kappa,\mathcal{B}_0,h}\alpha^{-1}e^{\kappa\|u\|^2}\|f\|_{C_{w,\kappa}}.\end{aligned}$$

Thus, the first statement holds.

The second statement follows from a change of variables, Proposition B.1, and the dominated convergence theorem:

$$\lim_{\alpha\to+\infty}\alpha\mathcal{R}_\alpha f(u)=\lim_{\alpha\to+\infty}\int_0^{+\infty}e^{-t}\mathfrak{P}_{t/\alpha}f(u)\mathrm{d}t=f(u).$$

We turn to the proof of the third statement. By the Fubini theorem, one has

$$\begin{aligned}\mathfrak{P}_t\mathcal{R}_\alpha f(u)&=\mathbb{E}_u\left(\int_0^{+\infty}e^{-\alpha s}\mathfrak{P}_s f(u_t)\mathrm{d}s\right)\\&=\int_0^{+\infty}e^{-\alpha s}\mathbb{E}_u\mathfrak{P}_s f(u_t)\mathrm{d}s=\int_0^{+\infty}e^{-\alpha s}\mathfrak{P}_{t+s}f(u)\mathrm{d}s=\mathcal{R}_\alpha\mathfrak{P}_t f(u).\end{aligned}$$

The identity (B.4) is established in the following way. Applying the Fubini theorem combined with a change of variables, it follows

$$\mathfrak{P}_t\mathcal{R}_\alpha f(u)=\int_0^{+\infty}e^{-\alpha s}\mathfrak{P}_{t+s}f(u)\mathrm{d}s=e^{\alpha t}\int_t^{+\infty}e^{-\alpha s}\mathfrak{P}_s f(u)\mathrm{d}s.$$

This, together with the Newton–Leibniz formula, gives

$$\begin{aligned}\mathfrak{P}_t\mathcal{R}_\alpha f(u)-\mathcal{R}_\alpha f(u)&=\int_0^t\frac{\mathrm{d}}{\mathrm{d}s}\left(e^{\alpha s}\int_s^{+\infty}e^{-\alpha r}\mathfrak{P}_r f(u)\mathrm{d}r\right)\mathrm{d}s\\&=\int_0^t\left(\alpha\mathfrak{P}_s\mathcal{R}_\alpha f(u)-\mathfrak{P}_s f(u)\right)\mathrm{d}s\\&=\int_0^t\mathfrak{P}_s\left(\alpha\mathcal{R}_\alpha f-f\right)(u)\mathrm{d}s\end{aligned}$$

as desired.

Finally, we prove the last statement. For each fixed $u\in H$ and $t\ge0$, by using (A.4), one has

$$\sup_{n\ge1}\mathbb{E}|f_n(u(t;u))|^2\le\left(\sup_{n\ge1}\|f_n\|^2_{C_{w,\kappa}}\right)\left(e^{2\kappa\|u\|^2}+C_{\kappa,\mathcal{B}_0,h}\right)<\infty,$$

which implies the uniform integrability of $\{f_n(u(t;u))\}_{n\ge1}$. Notice that

$$\lim_{n\to+\infty}f_n(u(t;u))=f(u(t;u))$$

$\mathbb{P}$-almost surely. Therefore, by the Vitali convergence theorem, one has

$$\lim_{n\to+\infty}\mathfrak{P}_t f_n(u)=\mathfrak{P}_t f(u)$$

for any $u \in H$ and $t \geq 0$. Moreover, since

$$e^{-\alpha t}|\mathfrak{P}_t f_n(u)| \leq e^{-\alpha t} e^{\kappa \|u\|^2} \sup_{n \geq 1} \|\mathfrak{P}_t f_n\|_{C_{\boldsymbol{w},\kappa}} \lesssim_{\kappa, \mathcal{B}_0, h} e^{-\alpha t} e^{\kappa \|u\|^2} \sup_{n \geq 1} \|f_n\|_{C_{\boldsymbol{w},\kappa}},$$

by the dominated convergence theorem, it follows

$$\lim_{n \to +\infty} \mathcal{R}_\alpha f_n(u) = \lim_{n \to +\infty} \int_0^{+\infty} e^{-\alpha t} \mathfrak{P}_t f_n(u) \mathrm{d}t = \int_0^{+\infty} e^{-\alpha t} \mathfrak{P}_t f(u) \mathrm{d}t = \mathcal{R}_\alpha f(u)$$

for any $\alpha > 0$ and $u \in H$. □

Let

$$\mathcal{D}_{\boldsymbol{w},\kappa}(\mathfrak{L}) := \left\{ f \in C_{\boldsymbol{w},\kappa}(H) \middle| \exists g \in C_{\boldsymbol{w},\kappa}(H),\ \mathfrak{P}_t f - f = \int_0^t \mathfrak{P}_s g \mathrm{d}s,\ \forall t \geq 0 \right\}, \tag{B.7}$$

and define the extended generator $\mathfrak{L} f := g$ for $f \in \mathcal{D}_{\boldsymbol{w},\kappa}(\mathfrak{L})$.

Proposition B.3. *The following properties hold.*

1. *For any $f \in C_{\boldsymbol{w},\kappa}(H)$ and $\alpha > 0$,*

$$\mathcal{R}_\alpha f \in \mathcal{D}_{\boldsymbol{w},\kappa}(\mathfrak{L}), \qquad \mathfrak{L}\mathcal{R}_\alpha f = \alpha \mathcal{R}_\alpha f - f. \tag{B.8}$$

2. *For any $f \in \mathcal{D}_{\boldsymbol{w},\kappa}(\mathfrak{L})$ and $\alpha > 0$,*

$$\mathfrak{L}\mathcal{R}_\alpha f = \mathcal{R}_\alpha \mathfrak{L} f. \tag{B.9}$$

3. *If a sequence $\{f_n\}_{n=1}^\infty \subset C_{\boldsymbol{w},\kappa}(H)$ converges to $f \in C_{\boldsymbol{w},\kappa}(H)$ in the sense of (B.5) and (B.6), then*

$$\lim_{n \to +\infty} \mathfrak{L}\mathcal{R}_\alpha f_n(u) = \mathfrak{L}\mathcal{R}_\alpha f(u)$$

for any $\alpha > 0$ and $u \in H$.

Proof. The first statement is valid because of (B.4), while the third follows from the assertion *4* in Proposition B.2 and (B.8). We turn to the second statement. By (B.3), the definitions (B.2) and (B.7), and the Fubini theorem, it follows

$$\begin{aligned}
\mathfrak{P}_t \mathcal{R}_\alpha f(u) - \mathcal{R}_\alpha f(u) &= \mathcal{R}_\alpha (\mathfrak{P}_t f - f)(u) \\
&= \int_0^{+\infty} e^{-\alpha r} \mathfrak{P}_r \left(\int_0^t \mathfrak{P}_s \mathfrak{L} f \mathrm{d}s \right)(u) \mathrm{d}r \\
&= \int_0^{+\infty} e^{-\alpha r} \left(\int_0^t \mathfrak{P}_{s+r} \mathfrak{L} f(u) \mathrm{d}s \right) \mathrm{d}r \\
&= \int_0^t \int_0^{+\infty} e^{-\alpha r} \mathfrak{P}_{s+r} \mathfrak{L} f(u) \mathrm{d}r \mathrm{d}s \\
&= \int_0^t \mathcal{R}_\alpha \mathfrak{P}_s \mathfrak{L} f(u) \mathrm{d}s = \int_0^t \mathfrak{P}_s \mathcal{R}_\alpha \mathfrak{L} f(u) \mathrm{d}s.
\end{aligned}$$

This implies (B.9). □

The following approximation result plays a crucial role in the proof of the Donsker–Varadhan variational formula (1.12).

Corollary B.4. *Let $f \in \mathcal{D}_{\boldsymbol{w},\kappa}(\mathfrak{L})$ satisfy*

$$f > 0, \qquad \frac{\mathfrak{L}f}{f} \in C_b(H). \tag{B.10}$$

Then, for any $\lambda \in \mathcal{P}(H)$, it follows

$$\int_H \frac{-\mathfrak{L}f}{f} \mathrm{d}\lambda = \lim_{\alpha \to +\infty} \int_H \frac{-\mathfrak{L}\mathcal{R}_\alpha f}{\mathcal{R}_\alpha f} \mathrm{d}\lambda, \tag{B.11}$$

and

$$\int_H -\frac{\mathfrak{L}\mathcal{R}_\alpha f}{\mathcal{R}_\alpha f} \mathrm{d}\lambda \leq \liminf_{N \to +\infty} \int_H -\frac{\mathfrak{L}\mathcal{R}_\alpha f_N}{\mathcal{R}_\alpha f_N} \mathrm{d}\lambda \tag{B.12}$$

for any $\alpha > 0$, where $f_N := (f \wedge N) \vee \frac{1}{N}$.

Proof. Let $f \in \mathcal{D}_{\boldsymbol{w},\kappa}(\mathfrak{L})$ satisfy (B.10). By Propositions B.2 and B.3, it follows

$$\lim_{\alpha \to +\infty} \frac{\mathfrak{L}\mathcal{R}_\alpha f(u)}{\mathcal{R}_\alpha f(u)} = \lim_{\alpha \to +\infty} \frac{\alpha \mathcal{R}_\alpha \mathfrak{L} f(u)}{\alpha \mathcal{R}_\alpha f(u)} = \frac{\mathfrak{L}f(u)}{f(u)}$$

for any $u \in H$. Moreover,

$$\begin{aligned} |\mathcal{R}_\alpha \mathfrak{L} f(u)| &\leq \int_0^{+\infty} e^{-\alpha t} |\mathfrak{P}_t \mathfrak{L} f(u)| \mathrm{d}t \\ &\leq \left\| \frac{\mathfrak{L}f}{f} \right\|_\infty \int_0^{+\infty} e^{-\alpha t} \mathfrak{P}_t f(u) \mathrm{d}t = \left\| \frac{\mathfrak{L}f}{f} \right\|_\infty \mathcal{R}_\alpha f(u), \end{aligned}$$

which implies

$$\left\| \frac{-\mathfrak{L}\mathcal{R}_\alpha f}{\mathcal{R}_\alpha f} \right\|_\infty \leq \left\| \frac{\mathfrak{L}f}{f} \right\|_\infty$$

for any $\alpha > 0$. Thus, an application of the dominated convergence theorem implies (B.11).

We turn to the proof of (B.12). To this end, notice that f_N converges to f pointwise on H and

$$\sup_N \|f_N\|_{C_{\boldsymbol{w},\kappa}} \leq 1 \vee \|f\|_{C_{\boldsymbol{w},\kappa}} < \infty.$$

By Propositions B.2 and B.3, this implies

$$\lim_{N \to +\infty} -\frac{\mathfrak{L}\mathcal{R}_\alpha f_N(u)}{\mathcal{R}_\alpha f_N(u)} = -\frac{\mathfrak{L}\mathcal{R}_\alpha f(u)}{\mathcal{R}_\alpha f(u)}$$

for any $u \in H$. Besides, from Proposition B.3, it follows that

$$-\frac{\mathfrak{L}\mathcal{R}_\alpha f_N}{\mathcal{R}_\alpha f_N} = -\frac{\alpha \mathcal{R}_\alpha f_N - f_N}{\mathcal{R}_\alpha f_N} \geq -\alpha.$$

Hence, by applying the Fatou lemma, one obtains (B.12). □

C Basic properties of the Donsker–Varadhan entropy

In this section, we gather some basic properties of the Donsker–Varadhan entropy used throughout the paper. While these results were originally established in [DV83] under seven technical assumptions, we show that their validity extends beyond the original setting and

adapts to the case of the stochastic NS system. Let $\mathbf{H}(\boldsymbol{\lambda})$ denote the Donsker–Varadhan entropy defined in (3.1). For $t > 0$, we introduce

$$\mathbf{H}(t, \boldsymbol{\lambda}) := \begin{cases} \displaystyle\int_{\mathfrak{D}^{\mathbb{R}}} \mathrm{Ent}\left(\pi^*_{[0,t]}\boldsymbol{\lambda}(u_{(-\infty,0]}, \cdot) \,\middle|\, \pi^*_{[0,t]}\mathbb{P}_{u(0)}\right) \boldsymbol{\lambda}(\mathrm{d}u), & \text{if } \boldsymbol{\lambda} \in \mathcal{P}_s(\mathfrak{D}^{\mathbb{R}}), \\ +\infty, & \text{otherwise.} \end{cases} \tag{C.1}$$

where for $r \in \mathbb{R}$ and $\mathcal{J} \subset \mathbb{R}$, π^*_r and $\pi^*_{\mathcal{J}}$ are the push-forward operators defined in (1.21) and (1.22), respectively. We introduce $\mathcal{B}(\mathscr{F}^0_t)$ as the space of all bounded $\mathscr{F}^0_t$-measurable functions $\Psi : \mathfrak{D}^{\mathbb{R}} \to \mathbb{R}$, and recall that each $\Psi \in \mathcal{B}(\mathscr{F}^0_t)$ has the following representation

$$\Psi(u_{(-\infty,\infty)}) = \Phi \circ \pi_{[0,t]}(u_{(-\infty,\infty)}), \qquad \forall u_{(-\infty,\infty)} \in \mathfrak{D}^{\mathbb{R}},$$

where $\Phi : \mathfrak{D}^{[0,t]} \to \mathbb{R}$ is some bounded Borel-measurable function, see Theorem 2.12.3 in [Bog07]. For simplicity, we do not distinguish the function Ψ and the corresponding representation Φ in what follows. Define

$$\bar{\mathbf{H}}(t, \boldsymbol{\lambda}) := \begin{cases} \displaystyle\sup_{\Psi \in \mathcal{B}(\mathscr{F}^0_t)} \int_{\mathfrak{D}^{\mathbb{R}}} \left(\Psi(u_{[0,t]}) - \log\left(\int_{\mathfrak{D}} e^{\Psi}\, \mathrm{d}\mathbb{P}_{u(0)}\right)\right) \boldsymbol{\lambda}(\mathrm{d}u), & \text{if } \boldsymbol{\lambda} \in \mathcal{P}_s(\mathfrak{D}^{\mathbb{R}}), \\ +\infty, & \text{otherwise.} \end{cases} \tag{C.2}$$

We need the following properties of $\mathbf{H}(\boldsymbol{\lambda})$, $\mathbf{H}(t, \boldsymbol{\lambda})$, and $\bar{\mathbf{H}}(t, \boldsymbol{\lambda})$.

Proposition C.1. *For any $\boldsymbol{\lambda} \in \mathcal{P}(\mathfrak{D}^{\mathbb{R}})$, it follows*

$$\mathbf{H}(t, \boldsymbol{\lambda}) = t\mathbf{H}(\boldsymbol{\lambda}), \qquad \bar{\mathbf{H}}(t, \boldsymbol{\lambda}) \le t\mathbf{H}(\boldsymbol{\lambda}), \qquad \forall t > 0, \tag{C.3}$$

and

$$\lim_{t \to +\infty} \frac{\bar{\mathbf{H}}(t, \boldsymbol{\lambda})}{t} = \mathbf{H}(\boldsymbol{\lambda}). \tag{C.4}$$

Furthermore, if $\boldsymbol{\lambda} \in \mathcal{P}_s(\mathfrak{D}^{\mathbb{R}})$, then

$$\bar{\mathbf{H}}(t, \boldsymbol{\lambda}) = \sup_{\Psi \in \mathcal{Y}_t} \left(\int_{\mathfrak{D}^{\mathbb{R}}} \left(\Psi(u_{[0,t]}) - \log \int_{\mathfrak{D}} e^{\Psi} \mathrm{d}\mathbb{P}_{u(0)}\right) \boldsymbol{\lambda}(\mathrm{d}u_{(-\infty,\infty)})\right) \tag{C.5}$$

$$= \sup_{\Psi \in \mathcal{Z}_t} \int_{\mathfrak{D}^{\mathbb{R}}} \Psi \mathrm{d}\boldsymbol{\lambda}, \tag{C.6}$$

where $\mathcal{Y}_t$ denotes the space of functions $\Psi : \mathfrak{D}^{\mathbb{R}} \to \mathbb{R}$ of the form

$$\Psi = V \circ \pi_{t_1, t_2, \ldots, t_n}$$

for some integer $n \ge 1$, partition $0 \le t_1 < t_2 < \ldots < t_n \le t$, and function $V \in C_b(H^n)$, and $\mathcal{Z}_t$ denotes the set of functions $\Psi \in \mathcal{Y}_t$ such that

$$\int_{\mathfrak{D}} e^{\Psi} \mathrm{d}\mathbb{P}_u \le 1, \qquad \forall u \in H. \tag{C.7}$$

Proof. The proof of (C.3) and (C.4) can be found in Theorems 3.1 and 3.6 in [DV83], while (C.5) can be established by using the definition (C.2) combined with Dynkin's monotone class argument. Thus, we omit the details and turn to (C.6). Let $\tilde{\mathbf{H}}(t, \boldsymbol{\lambda})$ denote the right-hand side of (C.6). Applying (C.5), it follows

$$\bar{\mathbf{H}}(t, \boldsymbol{\lambda}) \ge \tilde{\mathbf{H}}(t, \boldsymbol{\lambda}).$$

To prove the above inequality in the other direction, we fix any $\Psi \in \mathcal{Y}_t$ and write

$$\bar{\Psi}(u) := \log \int_{\mathfrak{D}} e^{\Psi} \mathrm{d}\mathbb{P}_u.$$

Notice that $\bar{\Psi} \in C_b(H)$ and $\Psi - \bar{\Psi} \circ \pi_0 \in \mathcal{Y}_t$. Moreover, one has

$$\int_{\mathfrak{D}} e^{\Psi - \bar{\Psi} \circ \pi_0} \mathrm{d}\mathbb{P}_u = e^{-\bar{\Psi}(u)} \int_{\mathfrak{D}} e^{\Psi} \mathrm{d}\mathbb{P}_u = 1, \qquad \forall u \in H.$$

Therefore, $\Psi - \bar{\Psi} \circ \pi_0 \in \mathcal{Z}_t$. This implies

$$\begin{aligned} \int_{\mathfrak{D}^{\mathbb{R}}} \Psi \mathrm{d}\boldsymbol{\lambda} &= \int_{\mathfrak{D}^{\mathbb{R}}} (\Psi - \bar{\Psi} \circ \pi_0) \mathrm{d}\boldsymbol{\lambda} + \int_{\mathfrak{D}^{\mathbb{R}}} \bar{\Psi} \circ \pi_0 \mathrm{d}\boldsymbol{\lambda} \\ &\leq \tilde{\mathbf{H}}(t, \boldsymbol{\lambda}) + \int_{\mathfrak{D}^{\mathbb{R}}} \bar{\Psi} \circ \pi_0 \mathrm{d}\boldsymbol{\lambda}, \end{aligned}$$

and thus

$$\int_{\mathfrak{D}^{\mathbb{R}}} \left(\Psi - \log \int_{\mathfrak{D}} e^{\Psi} \mathrm{d}\mathbb{P}_{u(0)} \right) \mathrm{d}\boldsymbol{\lambda} \leq \tilde{\mathbf{H}}(t, \boldsymbol{\lambda}).$$

Taking supremum in $\Psi \in \mathcal{Y}_t$ and using (C.5), one obtains (C.6). □

Proposition C.2. $\mathbf{H}(\boldsymbol{\lambda})$ *is lower semi-continuous in* $\boldsymbol{\lambda}$.

Proof. By Proposition 2.4.7 in [KS12], the map $u \mapsto \mathbb{P}_u$ is continuous. Combining this and Theorem 3.3 in [DV83], we have the lower semi-continuity of $\mathbf{H}(\boldsymbol{\lambda})$. □

The main result of this section is the following LD upper bound.

Proposition C.3. *Let* $F \subset \mathcal{P}_s(\mathfrak{D}^{\mathbb{R}})$ *be a closed set such that the family* $\{\pi_0^* \boldsymbol{\lambda} | \boldsymbol{\lambda} \in F\}$ *is tight in* $\mathcal{P}(H)$. *Then,*

$$\limsup_{t \to +\infty} \frac{1}{t} \log \left(\sup_{u \in H} \mathbb{P}_u \{ \boldsymbol{\zeta}_t^{\mathrm{Per}} \in F \} \right) \leq - \inf_{\boldsymbol{\lambda} \in F} \mathbf{H}(\boldsymbol{\lambda}). \tag{C.8}$$

The proof is based on the following three auxiliary lemmas.

Lemma C.4. *A sequence* $\{\boldsymbol{\lambda}_n\}_{n \geq 1} \subset \mathcal{P}_s(\mathfrak{D}^{\mathbb{R}})$ *is tight if and only if* $\{\pi_I^* \boldsymbol{\lambda}_n\}_{n \geq 1}$ *is tight in* $\mathcal{P}(\mathfrak{D}^I)$ *for some interval* $I \subset \mathbb{R}$.

Lemma C.5. *For any* $A \subset \mathcal{P}_s(\mathfrak{D}^{\mathbb{R}})$, *it follows*

$$\limsup_{t \to +\infty} \frac{1}{t} \log \left(\sup_{u \in H} \mathbb{P}_u \{ \boldsymbol{\zeta}_t^{\mathrm{Per}} \in A \} \right) \leq - \sup_{t > 0} \sup_{\Psi \in \mathcal{Z}_t} \inf_{\boldsymbol{\lambda} \in A} \frac{1}{t} \int_{\mathfrak{D}^{\mathbb{R}}} \Psi \mathrm{d}\boldsymbol{\lambda}. \tag{C.9}$$

Lemma C.6. *Let* $A \subset \mathcal{P}_s(\mathfrak{D}^{\mathbb{R}})$ *be a compact set. The following assertions hold.*

1. *If* $\inf_{\boldsymbol{\lambda} \in A} \mathbf{H}(\boldsymbol{\lambda}) < \infty$, *then for any* $\epsilon > 0$, *there are open sets* $A_1, A_2 \ldots, A_l$ *such that* $A \subset \bigcup_{i=1}^l A_i$ *and*

$$- \inf_{1 \leq i \leq l} \sup_{t > 0} \sup_{\Psi \in \mathcal{Z}_t} \inf_{\boldsymbol{\lambda} \in A_i} \frac{1}{t} \int_{\mathfrak{D}^{\mathbb{R}}} \Psi \mathrm{d}\boldsymbol{\lambda} \leq - \inf_{\boldsymbol{\lambda} \in A} \mathbf{H}(\boldsymbol{\lambda}) + \epsilon. \tag{C.10}$$

2. *If* $\inf_{\boldsymbol{\lambda} \in A} \mathbf{H}(\boldsymbol{\lambda}) = \infty$, *then for any* $L > 0$, *there are open sets* $A_1, A_2 \ldots, A_l$ *such that* $A \subset \bigcup_{i=1}^l A_i$ *and*

$$- \inf_{1 \leq i \leq l} \sup_{t > 0} \sup_{\Psi \in \mathcal{Z}_t} \inf_{\boldsymbol{\lambda} \in A_i} \frac{1}{t} \int_{\mathfrak{D}^{\mathbb{R}}} \Psi \mathrm{d}\boldsymbol{\lambda} \leq -L. \tag{C.11}$$

In both cases, it follows

$$\limsup_{t\to+\infty}\frac{1}{t}\log\left(\sup_{u\in H}\mathbb{P}_u\{\boldsymbol{\zeta}_t^{\mathrm{Per}}\in A\}\right)\le-\inf_{\boldsymbol{\lambda}\in A}\mathbf{H}(\boldsymbol{\lambda}). \tag{C.12}$$

Proof of Lemma C.4. It suffices to prove the tightness of $\{\boldsymbol{\lambda}_n\}_{n\ge1}$ by using the tightness of $\{\pi_I^*\boldsymbol{\lambda}_n\}_{n\ge1}$ for some interval $I\subset\mathbb{R}$. Without loss of generality, we take $I:=[0,1]$. We need the following result, which follows from Theorem 16.8 in [Bil99].

Lemma C.7. *A sequence $\{\boldsymbol{\lambda}_n\}_{n\ge1}\subset\mathcal{P}(\mathfrak{D}^{\mathbb{R}})$ is tight if and only if the following two conditions hold.*

1. *For each integer $m\ge1$,*

$$\lim_{a\to+\infty}\limsup_{n\to\infty}\boldsymbol{\lambda}_n\left\{\boldsymbol{u}\in\mathfrak{D}^{\mathbb{R}}\middle|\sup_{|t|\le m}\|\boldsymbol{u}(t)\|\ge a\right\}=0.$$

2. *For each integer $m\ge1$ and $\epsilon>0$,*

$$\lim_{\delta\to0^+}\limsup_{n\to\infty}\boldsymbol{\lambda}_n\left\{\boldsymbol{u}\in\mathfrak{D}^{\mathbb{R}}\middle|w'_m(\boldsymbol{u},\delta)\ge\epsilon\right\}=0,$$

where

$$w'_m(\boldsymbol{u},\delta):=\inf\max_{1\le i\le k}\sup_{s,r\in[t_{i-1},t_i)}\|\boldsymbol{u}(s)-\boldsymbol{u}(r)\|,$$

and the infimum is taken over all partitions $-m=t_0<t_1<\ldots<t_k=m$ such that $t_i-t_{i-1}>\delta$ for all $1<i<k$[7].

The tightness of $\{\boldsymbol{\lambda}_n\}_{n\ge1}$ is proved by verifying the above conditions. For simplicity, we confine ourselves to the verification of the second condition, as the other follows similarly. For integer $-m<j\le m$, introduce

$$w'_{[j-1,j]}(\boldsymbol{u},\delta):=\inf\max_{1\le i\le k_j}\sup_{s,r\in[t_{l-1}^j,t_l^j)}\|\boldsymbol{u}(s)-\boldsymbol{u}(r)\|,$$

where the infimum is taken over partitions $j-1=t_0^j<\ldots<t_{k_j}^j=j$ satisfying $t_l^j-t_{l-1}^j>\delta$ for all $1\le l\le k_j$. For any $\epsilon'>0$, we take the partition of $[j-1,j]$ such that

$$\max_{1\le i\le k_j}\sup_{s,r\in[t_{l-1}^j,t_l^j)}\|\boldsymbol{u}(s)-\boldsymbol{u}(r)\|\le w'_{[j-1,j]}(\boldsymbol{u},\delta)+\epsilon',\qquad\forall-m<j\le m.$$

By taking maximum in j and then sending ϵ' to 0, it follows that

$$w'_m(\boldsymbol{u},\delta)\le\max_{-m<j\le m}w'_{[j-1,j]}(\boldsymbol{u},\delta),$$

which, together with the translation-invariance of $\{\boldsymbol{\lambda}_n\}_{n\ge1}$, the tightness of $\{\pi^*_{[0,1]}\boldsymbol{\lambda}_n\}_{n\ge1}$, and Theorem 13.2 in [Bil99], implies

$$\begin{aligned}&\lim_{\delta\to0^+}\limsup_{n\to\infty}\boldsymbol{\lambda}_n\left\{\boldsymbol{u}\in\mathfrak{D}^{\mathbb{R}}\middle|w'_m(\boldsymbol{u},\delta)\ge\epsilon\right\}\\&\qquad\le2m\lim_{\delta\to0^+}\limsup_{n\to\infty}\boldsymbol{\lambda}_n\left\{\boldsymbol{u}\in\mathfrak{D}^{\mathbb{R}}\middle|w'_{[0,1]}(\boldsymbol{u},\delta)\ge\epsilon\right\}=0.\end{aligned}$$

This verifies the second condition in Lemma C.7 and thus completes the proof. □

[7] The endpoints $i=1,k$ are excluded, due to the definition of the Skorohod topology; see Theorem 16.4 in [Bil99] and the subsequent discussion.

Proof of Lemma C.5. Fix $t > 0$ and $\Psi \in \mathcal{B}(\mathscr{F}_t^0)$ satisfying $\mathbb{E}_u e^{\Psi} \leq 1$ for any $u \in H$. We claim that

$$\int_{\mathfrak{D}} \exp\left(\frac{1}{t}\int_0^s \Psi(\theta_r u_{[0,t]})\mathrm{d}r\right)\mathrm{d}\mathbb{P}_u \leq 1 \tag{C.13}$$

for any $s \geq 0$ and $u \in H$. Indeed, for any $s \in [0,t]$ and $u \in H$, by the Jensen inequality, and the Markov property, it follows

$$\begin{aligned}\int_{\mathfrak{D}} \exp\left(\frac{1}{t}\int_0^s \Psi(\theta_r u_{[0,t]})\mathrm{d}r\right)\mathrm{d}\mathbb{P}_u &\leq \frac{1}{t}\int_0^t \int_{\mathfrak{D}} \exp\left(\mathbb{I}_{\{r\leq s\}}\Psi(\theta_r u_{[0,t]})\right)\mathrm{d}\mathbb{P}_u\mathrm{d}r \\ &\leq \frac{1}{t}\int_0^t \left(\mathbb{I}_{\{r\leq s\}}\mathbb{E}_u \exp\left(\Psi(\theta_r u_{[0,t]})\right) + \mathbb{I}_{\{r>s\}}\right)\mathrm{d}r \\ &= \frac{1}{t}\int_0^t \left(\mathbb{I}_{\{r\leq s\}}\mathbb{E}_u\left(\mathbb{E}_{u(r)}\exp\left(\Psi(u_{[0,t]})\right)\right) + \mathbb{I}_{\{r>s\}}\right)\mathrm{d}r \leq 1.\end{aligned}$$

For $s > t$, define

$$\bar{\Psi}(r, u_{[0,t]}) := \sum_{k=0}^{[s/t]-1} \Psi(\theta_{r+kt}u_{[0,t]}) + \mathbb{I}_{\{0\leq r\leq s-[s/t]t\}}\Psi(\theta_{r+[s/t]t}u_{[0,t]}),$$

so one has

$$\int_0^s \Psi(\theta_r u_{[0,t]})\mathrm{d}r = \int_0^t \bar{\Psi}(r, u_{[0,t]})\mathrm{d}r.$$

Then, an application of the Jensen inequality gives

$$\int_{\mathfrak{D}} \exp\left(\frac{1}{t}\int_0^s \Psi(\theta_r u_{[0,t]})\mathrm{d}r\right)\mathrm{d}\mathbb{P}_u \leq \frac{1}{t}\int_0^t \int_{\mathfrak{D}} \exp\left(\bar{\Psi}(r, u_{[0,t]})\right)\mathrm{d}\mathbb{P}_u\mathrm{d}r,$$

where, by successively applying the Markov property, it follows

$$\begin{aligned}\int_{\mathfrak{D}} \exp\left(\bar{\Psi}(r, u_{[0,t]})\right)\mathrm{d}\mathbb{P}_u &= \mathbb{E}_u\left(\mathbb{E}_u\left(\exp\left(\bar{\Psi}(r, u_{[0,t]})\right)\middle|\mathscr{F}_{r+[s/t]t}\right)\right) \\ &\leq \mathbb{E}_u\left(\exp\left(\sum_{k=0}^{[s/t]-1}\Psi(\theta_{r+kt}u_{[0,t]})\right)\right) \leq \ldots \leq 1.\end{aligned}$$

This implies (C.13) as claimed. To conclude the proof, notice that

$$\left|\int_0^s \Psi(\theta_r u_{\mathrm{Per},s})\mathrm{d}r - \int_0^s \Psi(\theta_r u_{[0,t]})\mathrm{d}r\right| \leq 2t\sup_{\mathfrak{D}^{\mathbb{R}}}|\Psi|.$$

Combining this with (C.13), one has

$$\int_{\mathfrak{D}} \exp\left(\frac{s}{t}\int_{\mathfrak{D}^{\mathbb{R}}}\Psi\mathrm{d}\boldsymbol{\zeta}_s^{\mathrm{Per}}\right)\mathrm{d}\mathbb{P}_u = \int_{\mathfrak{D}} \exp\left(\frac{1}{t}\int_0^s \Psi(\theta_r u_{\mathrm{Per},s})\mathrm{d}r\right)\mathrm{d}\mathbb{P}_u \leq \exp\left(2\sup_{\mathfrak{D}^{\mathbb{R}}}|\Psi|\right)$$

for any $s > 0$ and $u \in H$. Hence, for any $A \subset \mathcal{P}_s(\mathfrak{D}^{\mathbb{R}})$, it follows

$$\begin{aligned}\mathbb{P}_u\{\boldsymbol{\zeta}_s^{\mathrm{Per}} \in A\} &\leq \exp\left(-\inf_{\boldsymbol{\lambda}\in A}\frac{s}{t}\int_{\mathfrak{D}^{\mathbb{R}}}\Psi\mathrm{d}\boldsymbol{\lambda}\right)\mathbb{E}_u\exp\left(\frac{s}{t}\int_{\mathfrak{D}^{\mathbb{R}}}\Psi\mathrm{d}\boldsymbol{\zeta}_s^{\mathrm{Per}}\right) \\ &\leq \exp\left(-\inf_{\boldsymbol{\lambda}\in A}\frac{s}{t}\int_{\mathfrak{D}^{\mathbb{R}}}\Psi\mathrm{d}\boldsymbol{\lambda} + 2\sup_{\mathfrak{D}^{\mathbb{R}}}|\Psi|\right),\end{aligned}$$

which implies

$$\limsup_{s\to+\infty}\frac{1}{s}\log\left(\sup_{u\in H}\mathbb{P}_u\{\boldsymbol{\zeta}_s^{\mathrm{Per}}\in A\}\right)\leq-\inf_{\boldsymbol{\lambda}\in A}\frac{1}{t}\int_{\mathfrak{D}^{\mathbb{R}}}\Psi\mathrm{d}\boldsymbol{\lambda}. \tag{C.14}$$

In particular, this estimate holds for any $t>0$ and $\Psi\in\mathcal{Z}_t$. By taking infimum in t,Ψ, one obtains (C.9). □

Proof of Lemma C.6. First, we prove (C.10). Let $\boldsymbol{\lambda}_0\in A$. Applying (C.4) and (C.6), for any $\epsilon>0$, there is a time $t_0>0$ and a function $\Psi_0\in\mathcal{Z}_{t_0}$ such that

$$\inf_{\boldsymbol{\lambda}\in A}\mathbf{H}(\boldsymbol{\lambda})\leq\frac{1}{t_0}\int_{\mathfrak{D}^{\mathbb{R}}}\Psi_0\mathrm{d}\boldsymbol{\lambda}_0+\frac{1}{2}\epsilon.$$

Introduce the following continuous functional

$$\boldsymbol{\lambda}\mapsto\frac{1}{t_0}\int_{\mathfrak{D}^{\mathbb{R}}}\Psi_0\mathrm{d}\boldsymbol{\lambda},\qquad\mathcal{P}_s(\mathfrak{D}^{\mathbb{R}})\to\mathbb{R}.$$

Then, there is a neighborhood A_0 of $\boldsymbol{\lambda}_0$ such that

$$\inf_{\boldsymbol{\lambda}\in A}\mathbf{H}(\boldsymbol{\lambda})\leq\inf_{\boldsymbol{\lambda}\in A_0}\frac{1}{t_0}\int_{\mathfrak{D}^{\mathbb{R}}}\Psi_0\mathrm{d}\boldsymbol{\lambda}+\epsilon\leq\sup_{t>0}\sup_{\Psi\in\mathcal{Z}_t}\inf_{\boldsymbol{\lambda}\in A_0}\frac{1}{t}\int_{\mathfrak{D}^{\mathbb{R}}}\Psi\mathrm{d}\boldsymbol{\lambda}+\epsilon.$$

All these neighborhoods form an open cover of the compact set A. Hence, there are open sets $A_1,A_2,\ldots,A_l$ such that $A\subset\bigcup_{i=1}^l A_i$ and

$$\inf_{\boldsymbol{\lambda}\in A}\mathbf{H}(\boldsymbol{\lambda})\leq\sup_{t>0}\sup_{\Psi\in\mathcal{Z}_t}\inf_{\boldsymbol{\lambda}\in A_i}\frac{1}{t}\int_{\mathfrak{D}^{\mathbb{R}}}\Psi\mathrm{d}\boldsymbol{\lambda}+\epsilon$$

for any $i=1,2,\ldots,l$. This leads to (C.10) as desired.

Since (C.11) follows from the same argument, we omit the details and turn to the proof of (C.12). Suppose that $\inf_{\boldsymbol{\lambda}\in A}\mathbf{H}(\boldsymbol{\lambda})<\infty$. Fix $\epsilon>0$ and choose $\{A_i\}_{i=1}^l$ as the open cover of A satisfying (C.10). Then, by Lemma C.5, one has

$$\begin{aligned}\limsup_{t\to+\infty}\frac{1}{t}\log\left(\sup_{u\in H}\mathbb{P}_u\{\boldsymbol{\zeta}_t^{\mathrm{Per}}\in A\}\right)&\leq\sup_{1\leq i\leq l}\limsup_{t\to+\infty}\frac{1}{t}\log\left(\sup_{u\in H}\mathbb{P}_u\left\{\boldsymbol{\zeta}_t^{\mathrm{Per}}\in A_i\right\}\right)\\&\leq-\inf_{1\leq i\leq l}\sup_{t>0}\sup_{\Psi\in\mathcal{Z}_t}\inf_{\boldsymbol{\lambda}\in A_i}\frac{1}{t}\int_{\mathfrak{D}^{\mathbb{R}}}\Psi\mathrm{d}\boldsymbol{\lambda}\\&\leq-\inf_{\boldsymbol{\lambda}\in A}\mathbf{H}(\boldsymbol{\lambda})+\epsilon,\end{aligned}$$

which implies (C.12) by taking $\epsilon\to0^+$. The case $\inf_{\boldsymbol{\lambda}\in A}\mathbf{H}(\boldsymbol{\lambda})=\infty$ follows in the same way, and its proof is thus omitted. □

Now we are in a position to prove Proposition C.3.

Proof of Proposition C.3. Step 1. Set $F_{\mathcal{M}}:=\{\pi_0^*\boldsymbol{\lambda}|\boldsymbol{\lambda}\in F\}$ and take any sequences $\epsilon_n,\tilde{\epsilon}_n$ such that $\epsilon_n,\tilde{\epsilon}_n\to0$, as $n\to\infty$. By the tightness of $F_{\mathcal{M}}$, for any $n\geq1$, there is a compact set $K_n\subset H$ such that

$$\mu(K_n)\geq1-\epsilon_n \tag{C.15}$$

for any $\mu\in F_{\mathcal{M}}$. As $\{\mathbb{P}_u\}_{u\in K_n}$ is tight, one can find a compact set $\mathcal{K}_n\subset\mathfrak{D}^{[0,1]}$ such that

$$\mathbb{P}_u(\mathcal{K}_n)\geq1-\tilde{\epsilon}_n \tag{C.16}$$

for any $u \in K_n$. Repeating the argument of the derivation of (3.7), it follows

$$\mathbb{E}_u \left(\exp \left(c \mathbb{I}_{K_n}(u(0)) \mathbb{I}_{\mathcal{K}_n^c}(u_{[0,1]}) \right) \right) \leq 1 + (e^c - 1)\tilde{\epsilon}_n$$

for any $c > 0$ and $u \in H$. Applying (C.13) with

$$\Psi(u_{[0,1]}) := c \mathbb{I}_{K_n}(u(0)) \mathbb{I}_{\mathcal{K}_n^c}(u_{[0,1]}) - \log(1 + (e^c - 1)\tilde{\epsilon}_n),$$

one gets

$$\int_{\mathfrak{D}} \exp \left(c \int_0^t \mathbb{I}_{K_n}(u(s)) \mathbb{I}_{\mathcal{K}_n^c}(u_{[s,s+1]}) \mathrm{d}s \right) \mathrm{d}\mathbb{P}_u \leq \exp \left(t \log \left(1 + (e^c - 1)\tilde{\epsilon}_n \right) \right) \tag{C.17}$$

for any $t \geq 0$ and $u \in H$. Notice that

$$\begin{aligned}
\int_0^t \mathbb{I}_{K_n}(u(s)) \mathbb{I}_{\mathcal{K}_n^c}(u_{[s,s+1]}) \mathrm{d}s &\geq \int_0^t \mathbb{I}_{\mathcal{K}_n^c}(u_{[s,s+1]}) \mathrm{d}s - \int_0^t \mathbb{I}_{K_n^c}(u(s)) \mathrm{d}s \\
&\geq \int_0^t \mathbb{I}_{\mathcal{K}_n^c} \circ \pi_{[0,1]}(\theta_s u_{\mathrm{Per},t}) \mathrm{d}s - \int_0^t \mathbb{I}_{K_n^c} \circ \pi_0(\theta_s u_{\mathrm{Per},t}) \mathrm{d}s - 1 \\
&= t \pi^*_{[0,1]} \boldsymbol{\zeta}_t^{\mathrm{Per}}(\mathcal{K}_n^c) - t \pi_0^* \boldsymbol{\zeta}_t^{\mathrm{Per}}(K_n^c) - 1.
\end{aligned}$$

This, together with (C.17), implies

$$\mathbb{E}_u \exp \left(ct \left(\pi^*_{[0,1]} \boldsymbol{\zeta}_t^{\mathrm{Per}}(\mathcal{K}_n^c) - \pi_0^* \boldsymbol{\zeta}_t^{\mathrm{Per}}(K_n^c) - \frac{1}{t} \right) \right) \leq \exp \left(t \log \left(1 + (e^c - 1)\tilde{\epsilon}_n \right) \right).$$

Moreover, utilizing (C.15), it follows

$$\pi_0^* \boldsymbol{\zeta}_t^{\mathrm{Per}}(K_n^c) \leq \epsilon_n, \qquad \forall n \geq 1$$

on the event $\{\boldsymbol{\zeta}_t^{\mathrm{Per}} \in F\}$. Therefore,

$$\begin{aligned}
&\exp(t \log \left(1 + (e^c - 1)\tilde{\epsilon}_n \right)) \\
&\quad \geq \mathbb{E}_u \left(\mathbb{I}_{\{\boldsymbol{\zeta}_t^{\mathrm{Per}} \in F\}} \exp \left(ct \left(\pi^*_{[0,1]} \boldsymbol{\zeta}_t^{\mathrm{Per}}(\mathcal{K}_n^c) - \epsilon_n - \frac{1}{t} \right) \right) \right) \\
&\quad \geq e^{c\epsilon_n t} \mathbb{P}_u \left\{ \boldsymbol{\zeta}_t^{\mathrm{Per}} \in F \bigcap \left\{ \boldsymbol{\lambda} \middle| \pi^*_{[0,1]} \boldsymbol{\lambda}(\mathcal{K}_n^c) > 2\epsilon_n + \frac{1}{t} \right\} \right\}.
\end{aligned} \tag{C.18}$$

Choose

$$\epsilon_n = \frac{1}{n}, \qquad \tilde{\epsilon}_n = e^{-an^2}, \qquad c = an^2$$

in (C.18), where $a > 0$. Then,

$$\mathbb{P}_u \left\{ \boldsymbol{\zeta}_t^{\mathrm{Per}} \in F \bigcap \left\{ \boldsymbol{\lambda} \middle| \pi^*_{[0,1]} \boldsymbol{\lambda}(\mathcal{K}_n^c) > 2\epsilon_n + \frac{1}{t} \right\} \right\} \leq e^{t \log 2 - ant},$$

which implies

$$\mathbb{P}_u \left\{ \boldsymbol{\zeta}_t^{\mathrm{Per}} \in F \bigcap F_t^c \right\} \leq e^{t \log 2} \frac{e^{-at}}{1 - e^{-at}},$$

where

$$F_t := \bigcap_{n=1}^{\infty} \left\{ \boldsymbol{\lambda} \in \mathcal{P}_s(\mathfrak{D}^{\mathbb{R}}) \middle| \pi^*_{[0,1]} \boldsymbol{\lambda}(\mathcal{K}_n^c) \leq \frac{2}{n} + \frac{1}{t} \right\}.$$

This yields

$$\limsup_{t\to+\infty}\frac{1}{t}\log\left(\sup_{u\in H}\mathbb{P}_u\left\{\boldsymbol{\zeta}_t^{\mathrm{Per}}\in F\bigcap F_t^c\right\}\right)\leq\log 2-a. \tag{C.19}$$

Step 2. Write

$$F_\infty:=\bigcap_{t>0}F_t.$$

By the Portmanteau theorem, F_∞ is a closed set. Let $G\subset\mathcal{P}_s(\mathfrak{D}^{\mathbb{R}})$ be an open set such that

$$F\bigcap F_\infty\subset G.$$

We claim that there is $t_0>0$ such that $F\bigcap F_{t_0}\subset G$. Indeed, by contradiction, assume that for any $k\geq 1$,

$$(F\bigcap F_k)\bigcap G^c\neq\emptyset.$$

For each $k\geq 1$, choose $\boldsymbol{\lambda}_k\in(F\bigcap F_k)\bigcap G^c$. Then, for any $k,n\geq 1$,

$$\pi^*_{[0,1]}\boldsymbol{\lambda}_k(\mathcal{K}_n^c)\leq\frac{2}{n}+\frac{1}{k}.$$

This, together with Lemma C.4, implies the tightness of $\{\boldsymbol{\lambda}_k\}_{k=1}^\infty$. Therefore, one gets a contradiction, since any limit point of $\{\boldsymbol{\lambda}_k\}_{k=1}^\infty$ must belong to

$$G^c\bigcap(F\bigcap F_\infty)=\emptyset.$$

Step 3. To conclude the proof, note that by the Prokhorov theorem, $F\bigcap F_\infty$ is a compact set. Suppose that

$$\inf_{\boldsymbol{\lambda}\in F\bigcap F_\infty}\mathbf{H}(\boldsymbol{\lambda})<\infty.$$

Then, by applying Lemma C.6, for any $\epsilon>0$, there is an open cover $\{A_i\}_{i=1}^l$ of $F\bigcap F_\infty$ satisfying (C.10). As shown in the previous step, one can find $t_0>0$ such that

$$F\bigcap F_{t_0}\subset\bigcup_{i=1}^l A_i.$$

Therefore,

$$\begin{aligned}\mathbb{P}_u\left\{\boldsymbol{\zeta}_t^{\mathrm{Per}}\in F\right\}&\leq\mathbb{P}_u\left\{\boldsymbol{\zeta}_t^{\mathrm{Per}}\in F\bigcap F_{t_0}\right\}+\mathbb{P}_u\left\{\boldsymbol{\zeta}_t^{\mathrm{Per}}\in F\bigcap F_{t_0}^c\right\}\\&\leq\sum_{i=1}^l\mathbb{P}_u\left\{\boldsymbol{\zeta}_t^{\mathrm{Per}}\in A_i\right\}+\mathbb{P}_u\left\{\boldsymbol{\zeta}_t^{\mathrm{Per}}\in F\bigcap F_t^c\right\}\end{aligned}$$

for any $t\geq t_0$ and $u\in H$. This, together with (C.9), (C.10), and (C.19), implies

$$\begin{aligned}&\limsup_{t\to+\infty}\frac{1}{t}\log\left(\sup_{u\in H}\mathbb{P}_u\left\{\boldsymbol{\zeta}_t^{\mathrm{Per}}\in F\right\}\right)\\&\qquad\leq\max\left\{-\inf_{1\leq i\leq l}\sup_{t>0}\sup_{\Psi\in\mathcal{Z}_t}\inf_{\boldsymbol{\lambda}\in A_i}\frac{1}{t}\int_{\mathfrak{D}^{\mathbb{R}}}\Psi\mathrm{d}\boldsymbol{\lambda},\log 2-a\right\}\\&\qquad\leq\max\left\{-\inf_{\boldsymbol{\lambda}\in F}\mathbf{H}(\boldsymbol{\lambda})+\epsilon,\log 2-a\right\}.\end{aligned}$$

As F is independent of a, by first choosing a sufficiently large and then taking $\epsilon\to 0^+$, (C.8) holds. The case $\inf_{\boldsymbol{\lambda}\in F\bigcap F_\infty}\mathbf{H}(\boldsymbol{\lambda})=\infty$ follows from the above argument combined with (C.11), and the details are thus omitted. □

D An improved version of Kifer's criterion

We provide an improvement of the Kifer-type criterion introduced in Section 3 in [JNPS18] by weakening the exponential growth condition (3.4) therein to the exponential tightness along any subsequence.

To formulate the result, we recall the following basic settings. Let $\mathfrak{X}$ be a Polish space, and let $\mathcal{P}(\mathfrak{X})$ denote the space of all Borel probability measures on $\mathfrak{X}$ endowed with the topology of weak convergence. Consider a family of random probability measures $\{\zeta_\theta\}_{\theta\in\Theta}$ on $\mathfrak{X}$ that are defined on some probability spaces $(\Omega_\theta, \mathscr{F}_\theta, \mathbb{P}_\theta)$, where $\Theta := (\Theta, \prec)$ is a directed set possessing a cofinal sequence. In what follows, the dependence of the probability spaces on θ shall be omitted, which will not lead to any confusion. Let $r(\theta)$ be a positive function satisfying $\lim_{\theta\in\Theta} r(\theta) = +\infty$. Define the pressure function

$$Q(V) := \lim_{\theta\in\Theta} \frac{1}{r(\theta)} \log \int_\Omega \exp(r(\theta)\langle V, \zeta_\theta\rangle) \mathrm{d}\mathbb{P}, \qquad V \in C_b(\mathfrak{X}), \tag{D.1}$$

if the above limit exists, and introduce the Legendre transform of Q

$$I(\lambda) := \sup_{V\in C_b(\mathfrak{X})} (\langle V, \lambda\rangle - Q(V)), \tag{D.2}$$

where $\lambda \in \mathcal{P}(\mathfrak{X})$. A measure $\sigma_V \in \mathcal{P}(\mathfrak{X})$ satisfying the relation

$$Q(V) = \langle V, \sigma_V\rangle - I(\sigma_V)$$

is called an equilibrium state for V. The basic properties of Q, I can be found in [JNPS18].

Proposition D.1. *Suppose that the following two conditions hold for $\{\zeta_\theta\}_{\theta\in\Theta}$.*

1. ***Exponential tightness along any subsequence.*** *For any subsequence $\{\theta_n\}_{n\geq 1}$ such that $r(\theta_n) \to +\infty$, as $n \to +\infty$, $\{\zeta_{\theta_n}\}_{n\geq 1}$ is exponentially tight, that is, for any $l > 0$, there is a compact set $\mathcal{K}_l \subset \mathcal{P}(\mathfrak{X})$ such that*

$$\limsup_{n\to+\infty} \frac{1}{r(\theta_n)} \log \mathbb{P}\{\zeta_{\theta_n} \in \mathcal{K}_l^c\} \leq -l. \tag{D.3}$$

2. ***Existence of pressure.*** *The limit (D.1) exists for any $V \in C_b(\mathfrak{X})$.*

Then, the relation (D.2) defines a good rate function, and the following upper bound holds: for any closed set $F \subset \mathcal{P}(\mathfrak{X})$,

$$\limsup_{\theta\in\Theta} \frac{1}{r(\theta)} \log \mathbb{P}\{\zeta_\theta \in F\} \leq -\inf_{\lambda\in F} I(\lambda). \tag{D.4}$$

Furthermore, the following lower bound holds: for any open set $G \subset \mathcal{P}(\mathfrak{X})$,

$$\liminf_{\theta\in\Theta} \frac{1}{r(\theta)} \log \mathbb{P}\{\zeta_\theta \in G\} \geq -\inf_{\lambda\in G} I(\lambda), \tag{D.5}$$

provided that the following condition is also satisfied for $\{\zeta_\theta\}_{\theta\in\Theta}$.

3. ***Uniqueness of equilibrium.*** *There exists a vector space $\mathcal{V} \subset C_b(\mathfrak{X})$ such that the restrictions of the functions in $\mathcal{V}$ on any compact subset $K \subset \mathfrak{X}$ form a dense subspace of $C(K)$, and that, for each $V \in \mathcal{V}$, there is a unique[8] equilibrium state.*

Remark D.2. In the case of level-2 LDP, the exponential growth condition (3.4) in [JNPS18] can be verified for the white-forced NS system by using (A.3). However, in the level-3 case, such a condition, as reformulated in (1.18), does not necessarily hold. As discussed in Section 1.2, it fails to hold in the case of a bounded domain, due to the lack of exponential

[8]Notice that if I is a good rate function, then the set of equilibrium states for any $V \in C_b(\mathfrak{X})$ is non-empty.

moment estimates for the H^1-norm. Moreover, we emphasize that even in the torus case, where exponential moment estimates are available for all H^s-norms, (1.18) may not be satisfied. Indeed, due to the Aubin–Lions theorem, a natural choice for $\mathbf{\Phi}$ is

$$\mathbf{\Phi}(u_{[0,T]}) := \|u_{[0,T]}\|^2_{C([0,T];H^1)} + \|u_{[0,T]}\|^2_{C^{\frac{1}{4}}([0,T];H^{-1})}.$$

By the Itô formula, one has

$$\frac{1}{2}\mathrm{d}\|u(t)\|^2_{H^1} = \left(-\|Lu\|^2 + \langle h, Lu\rangle + \frac{1}{2}\mathcal{B}_1\right)\mathrm{d}t + \sum_j \alpha_j b_j \langle u, e_j\rangle \mathrm{d}\beta_j.$$

Then, there are constants $C, c > 0$ such that

$$\|u(t)\|^2_{H^1} - C(t-s) \le \|u(s)\|^2_{H^1} + M_s(t) - \frac{1}{2}c\langle M_s\rangle(t),$$

where $M_s(t) := 2\sum_j \alpha_j b_j \int_s^t \langle u, e_j\rangle \mathrm{d}\beta_j$. Hence,

$$\begin{aligned}\int_0^t \|u(\cdot)\|^2_{C([s,s+T];H^1)}\mathrm{d}s \le CTt + \int_0^t \|u(s)\|^2_{H^1} ds \\ + \int_0^t \sup_{r\in[s,s+T]}\left(M_s(r) - \frac{1}{2}c\langle M_s\rangle(r)\right)\mathrm{d}s.\end{aligned}$$

It is not clear whether the third term admits an exponential moment estimate, even though the quantity inside the integral has finite exponential moments. Furthermore, as the second and third terms are dependent, the right-hand side may not have finite exponential moments, even if each term individually does. This illustrates that (1.18) does not immediately follow using merely the Itô formula and energy estimates. Therefore, to obtain a level-3 LDP, the Kifer-type criterion in [JNPS18] needs to be improved.

Proof of Proposition D.1. Fix an arbitrary subsequence $\{\theta_n\}_{n\ge1}$ satisfying $r(\theta_n) \to +\infty$ as $n \to +\infty$. It is known that if the LDP holds for $\{\zeta_{\theta_n}\}_{n\ge1}$ with a rate function I independent of the underlying subsequence $\{\theta_n\}_{n\ge1}$, then the whole family $\{\zeta_\theta\}$ satisfies the LDP with the same rate function I. Therefore, we fix such a subsequence $\{\zeta_{\theta_n}\}_{n\ge1}$.
Step 1: Covering property. Compared to the proof of Theorem 3.3 in [JNPS18], the main difficulty in our situation is to establish the following covering property for the level sets of I: for any $a \ge 0$, there is a sequence of compact sets $\{\mathfrak{M}_{a,l}\}_{l\ge1} \subset \mathcal{P}(\mathfrak{X})$ such that

$$L_a := \{\lambda \in \mathcal{P}(\mathfrak{X}) | I(\lambda) \le a\} \subset \textstyle\bigcap_{l\ge1}\mathfrak{M}_{a,l}, \tag{D.6}$$

and

$$\limsup_{n\to+\infty} \frac{1}{r(\theta_n)} \log \mathbb{P}\{\zeta_{\theta_n} \in \mathfrak{M}^c_{a,l}\} \le -\frac{l}{2} \tag{D.7}$$

for any integer $l \ge 1$. This covering property is established by using an estimate derived in [CLXZ24]. To formulate the estimate, for each integer $l \ge 1$, let $\mathcal{K}_l$ be the compact set satisfying (D.3). As we are considering a sequence of random probability measures, we can assume, up to enlarging the compact set $\mathcal{K}_l$, that

$$\mathbb{P}\{\zeta_{\theta_n} \in \mathcal{K}^c_l\} \le e^{-\frac{l}{2}r(\theta_n)} \tag{D.8}$$

for all $n \ge 1$. Since $\mathcal{K}_l$ is compact, for each integer $m \ge 1$, there is a compact set $K_{l,m}$ such that

$$\sigma(K^c_{l,m}) \le \frac{1}{m}$$

for any $\sigma \in \mathcal{K}_l$. Moreover, in view of Corollary 3.4 in [CLXZ24], it follows

$$I(\lambda) \geq l\left(\lambda(K_{l,m}^c) - \frac{1}{m}\right) \tag{D.9}$$

for any $\lambda \in \mathcal{P}(\mathfrak{X})$ and any $l, m \geq 1$. Define

$$\mathfrak{M}_{a,l} := \bigcap_{j\geq l}\bigcap_{m\geq 1}\left\{\lambda \in \mathcal{P}(\mathfrak{X})\middle|\lambda(K_{j,m}^c) \leq \frac{a}{j} + \frac{1}{m}\right\}.$$

By the Portmanteau theorem and the Prokhorov theorem, for any $a \geq 0$ and any $l \geq 1$, $\mathfrak{M}_{a,l}$ is a compact subset of $\mathcal{P}(\mathfrak{X})$. Furthermore, by the definition of $\mathfrak{M}_{a,l}$ and (D.9), one has

$$L_a \subset \mathfrak{M}_{a,1} = \bigcap_{l\geq 1}\mathfrak{M}_{a,l},$$

which implies (D.6). As for (D.7), note that for any $l \geq 1$,

$$\bigcap_{j\geq l}\mathcal{K}_j \subset \mathfrak{M}_{a,l}.$$

This, together with (D.8), implies

$$\mathbb{P}\{\zeta_{\theta_n} \in \mathfrak{M}_{a,l}^c\} \leq \sum_{j\geq l}\mathbb{P}\{\zeta_{\theta_n} \in \mathcal{K}_j^c\} \leq \sum_{j\geq l} e^{-\frac{j}{2}r(\theta_n)} = \frac{e^{-\frac{l}{2}r(\theta_n)}}{1 - e^{-\frac{r(\theta_n)}{2}}}$$

and therefore (D.7) as desired.

Step 2: Goodness of I and LD upper bound. Since I is lower semi-continuous, for any $a \geq 0$, L_a is a closed subset of $\mathcal{P}(\mathfrak{X})$. Combining this with (D.6), one gets the compactness of L_a and therefore the goodness of the rate function I. The LD upper bound (D.4) follows in the same way as in [JNPS18], and the proof is omitted.

Step 3: LD lower bound. The LD lower bound (D.5) is established by using the covering property (D.6) combined with the estimate (D.7) and the following lemma established in [JNPS18], see Proposition 3.4 therein.

Lemma D.3. *Let $V_1, V_2, \ldots, V_m \in \mathcal{V}$ and define*

$$\begin{aligned} f_m :&\mathcal{P}(\mathfrak{X}) \to \mathbb{R}^m \\ &\lambda \mapsto (\langle V_1, \lambda\rangle, \langle V_2, \lambda\rangle, \ldots, \langle V_m, \lambda\rangle) \end{aligned} \tag{D.10}$$

and $\zeta_{\theta_n}^m := f_m(\zeta_{\theta_n})$. Then, under the conditions of Proposition D.1, for any closed set $\mathcal{F} \subset \mathbb{R}^m$,

$$\limsup_{n\to+\infty}\frac{1}{r(\theta_n)}\log\mathbb{P}\{\zeta_{\theta_n}^m \in \mathcal{F}\} \leq -\inf_{\alpha\in\mathcal{F}} I_m(\alpha),$$

and for any open set $\mathcal{W} \subset \mathbb{R}^m$,

$$\liminf_{n\to+\infty}\frac{1}{r(\theta_n)}\log\mathbb{P}\{\zeta_{\theta_n}^m \in \mathcal{W}\} \geq -\inf_{\alpha\in\mathcal{W}} I_m(\alpha),$$

where $I_m(\alpha) := \inf_{\lambda\in f_m^{-1}(\alpha)} I(\lambda)$.

Let $G \subset \mathcal{P}(\mathfrak{X})$ be an open set. Without loss of generality, assume that $\inf_{\lambda\in G} I(\lambda) < \infty$. Then, for any $\epsilon > 0$, there is a measure $\lambda_\epsilon \in G$ such that

$$I(\lambda_\epsilon) \leq \inf_{\lambda\in G} I(\lambda) + \epsilon. \tag{D.11}$$

Set

$$a = \inf_{\lambda\in G} I(\lambda) + \epsilon, \qquad l = \lfloor 2(a+2)\rfloor + 1, \tag{D.12}$$

so one has $\lambda_\epsilon \in L_a \subset \mathfrak{M}_{a,l}$. Moreover, let $\{V_k\}_{k\geq 1} \subset \mathcal{V}$ be a sequence of functions such that $\|V_k\|_\infty = 1$ for any $k \geq 1$ and

$$d(\lambda_1, \lambda_2) := \sum_{k=1}^{\infty} 2^{-k} |\langle V_k, \lambda_1\rangle - \langle V_k, \lambda_2\rangle|$$

metrizes the topology of weak convergence on $\mathfrak{M}_{a,l}$[9]. As G is open, there is an integer $m \geq 1$ and $\delta > 0$ such that if $\lambda \in \mathfrak{M}_{a,l}$ satisfies

$$\sum_{k=1}^{m} 2^{-k} |\langle V_k, \lambda\rangle - \langle V_k, \lambda_\epsilon\rangle| < \delta, \tag{D.13}$$

then $\lambda \in G$. Endow $\mathbb{R}^m$ with the norm

$$\|x\|_m := \sum_{k=1}^{m} 2^{-k} |x_k|, \qquad x = (x_1, x_2, \ldots, x_m),$$

and define the map f_m by (D.10) with the functions $V_1, V_2, \ldots, V_m$ given in (D.13). Let $x_\epsilon := f_m(\lambda_\epsilon)$. Under the above notations, it follows

$$f_m^{-1}(B_{\mathbb{R}^m}(x_\epsilon, \delta)) \bigcap \mathfrak{M}_{a,l} \subset G.$$

Therefore,

$$\begin{aligned}\mathbb{P}\{\zeta_{\theta_n} \in G\} &\geq \mathbb{P}\{\zeta_{\theta_n} \in G \bigcap \mathfrak{M}_{a,l}\} \geq \mathbb{P}\{\zeta_{\theta_n} \in f_m^{-1}(B_{\mathbb{R}^m}(x_\epsilon, \delta)) \bigcap \mathfrak{M}_{a,l}\} \\ &\geq \mathbb{P}\{\zeta_{\theta_n}^m \in B_{\mathbb{R}^m}(x_\epsilon, \delta)\} - \mathbb{P}\{\zeta_{\theta_n} \in \mathfrak{M}_{a,l}^c\}.\end{aligned}$$

Applying Lemma D.3, the relations (D.7), (D.11), and (D.12), there is an integer $n_0 \geq 1$ such that if $n \geq n_0$, then

$$\begin{aligned}\mathbb{P}\{\zeta_{\theta_n}^m \in B_{\mathbb{R}^m}(x_\epsilon, \delta)\} &\geq \exp\left(r(\theta_n)\left(-\inf_{\alpha \in B_{\mathbb{R}^m}(x_\epsilon, \delta)} I_m(\alpha) - 1\right)\right) \\ &\geq \exp\left(r(\theta_n)\left(-I(\lambda_\epsilon) - 1\right)\right) \\ &\geq \exp\left(r(\theta_n)\left(-\frac{l}{2} + 1\right)\right) \geq \sqrt{e}\mathbb{P}\{\zeta_{\theta_n} \in \mathfrak{M}_{a,l}^c\},\end{aligned}$$

which implies

$$\mathbb{P}\{\zeta_{\theta_n} \in G\} \geq \left(1 - \frac{1}{\sqrt{e}}\right) \mathbb{P}\{\zeta_{\theta_n}^m \in B_{\mathbb{R}^m}(x_\epsilon, \delta)\}.$$

This, together with Lemma D.3 and (D.11) again, leads to

$$\begin{aligned}\liminf_{n\to+\infty} \frac{1}{r(\theta_n)} \log \mathbb{P}\{\zeta_{\theta_n} \in G\} &\geq \liminf_{n\to+\infty} \frac{1}{r(\theta_n)} \log \mathbb{P}\{\zeta_{\theta_n}^m \in B_{\mathbb{R}^m}(x_\epsilon, \delta)\} \\ &\geq -\inf_{\alpha \in B_{\mathbb{R}^m}(x_\epsilon, \delta)} I_m(\alpha) \geq -I(\lambda_\epsilon) \geq -\inf_{\lambda \in G} I(\lambda) - \epsilon.\end{aligned}$$

As ϵ is arbitrary, the LD lower bound (D.5) holds. This completes the proof. □

[9]For any given compact set $\mathcal{K} \subset \mathcal{P}(\mathfrak{X})$, the existence of such functions is proved in [JNPS18].

E Large-time asymptotics of generalized Markov semigroups: improved abstract criterion

We present a new version of the abstract criteria given in Theorem A.4 in [MN18] and Theorem 4.1 in [JNPS18]. In these works, sufficient conditions are provided, which ensure a multiplicative ergodic theorem (MET) for generalized Markov semigroups. This MET ensures the existence of the pressure function and the uniqueness of equilibrium required in Kifer's criterion. However, as discussed in Section 1.2, these criteria do not apply to the weakly dissipative case considered in the present paper. The difficulty is overcome by using a stronger irreducibility property ensured by the parabolic smoothing effect, cf. (A.14) in [MN18] and (E.5).

To formulate the result, we start with some definitions. Let $\mathfrak{X}$ be a separable Banach space endowed with the norm $\|\cdot\|$, let $\mathcal{M}^+(\mathfrak{X})$ be the set of the non-negative Borel measures on $\mathfrak{X}$, and let $\{P_t(u,\cdot), u\in\mathfrak{X}, t\geq 0\}$ be a generalized Markov family of transition kernels, see Definition A.3 in [MN18]. We define two associated semigroups by the following relations

$$\mathfrak{P}_t : C_b(\mathfrak{X}) \to C_b(\mathfrak{X}), \qquad \mathfrak{P}_t f(u) := \int_{\mathfrak{X}} f(v) P_t(u, \mathrm{d}v),$$

$$\mathfrak{P}_t^* : \mathcal{M}^+(\mathfrak{X}) \to \mathcal{M}^+(\mathfrak{X}), \qquad \mathfrak{P}_t^* \mu(A) := \int_{\mathfrak{X}} P_t(u, A)\mu(\mathrm{d}u).$$

We assume the following structure for the phase space $\mathfrak{X}$: there is an increasing sequence of compact subsets $\{\mathfrak{X}_R\}_{R=1}^{\infty}$ such that

$$\mathfrak{X}_\infty := \textstyle\bigcup_{R=1}^{\infty} \mathfrak{X}_R$$

is dense in $\mathfrak{X}$ and for each $R\geq 1$,

$$\mathfrak{X}_R \subset \overline{B_{\mathfrak{X}}(R)}. \tag{E.1}$$

Define the following weight function

$$\mathfrak{w}(u) := \mathcal{G}(\|u\|) \tag{E.2}$$

where $\mathcal{G} : [0,+\infty) \to [1,+\infty)$ is a continuous increasing function satisfying

$$\lim_{y\to+\infty} \mathcal{G}(y) = +\infty.$$

We denote by $C_{\mathfrak{w}}(\mathfrak{X})$ (respectively, $L^\infty_{\mathfrak{w}}(\mathfrak{X})$) the weighted space of continuous (measurable) functions $f : \mathfrak{X} \to \mathbb{R}$ such that the norm

$$\|f\|_{L^\infty_{\mathfrak{w}}} := \sup_{u\in\mathfrak{X}} \frac{|f(u)|}{\mathfrak{w}(u)} < \infty.$$

Let $\mathcal{P}_{\mathfrak{w}}(\mathfrak{X})$ be the set of measures $\mu \in \mathcal{P}(\mathfrak{X})$ such that $\langle \mathfrak{w}, \mu\rangle < \infty$. We say a family $\mathcal{C} \subset C_b(\mathfrak{X})$ is determining, if any two measures $\mu, \nu \in \mathcal{M}^+(\mathfrak{X})$, satisfying

$$\langle f, \mu\rangle = \langle f, \nu\rangle$$

for any $f \in \mathcal{C}$, must coincide.

We have the following improved version of Theorem 4.1 in [JNPS18].

Proposition E.1. *Let $\{P_k(u,\cdot)|u\in\mathfrak{X}, k\in\mathbb{N}\}$ be a generalized Markov family of transition kernels satisfying the following conditions.*

1. ***Concentration property.*** *There is an integer $l_0 \geq 1$ such that $P_{l_0}(u,\cdot)$ is concentrated on $\mathfrak{X}_\infty$ for any $u\in\mathfrak{X}$.*

2. ***Growth condition.*** *There is a weight function $\mathfrak{w}$ satisfying (E.2) and an integer $R_0 \geq 1$ such that*

$$\sup_{k\geq 0} \frac{\|\mathfrak{P}_k \mathfrak{w}\|_{L^\infty_{\mathfrak{w}}}}{\|\mathfrak{P}_k \mathbf{1}\|_{R_0}} < \infty,$$

where $\mathbf{1}$ is the function on $\mathfrak{X}$ identically equal to 1, and $\|\cdot\|_R$ denotes the L^∞-norm on $\mathfrak{X}_R$.

3. ***Uniform irreducibility.*** *For any integers $\rho, R \geq 1$ and $r > 0$, there is an integer $l = l(\rho, R, r) \geq 1$ and a positive number $p = p(\rho, R, r)$ such that*

$$P_l(u, B_{\mathfrak{X}_\rho}(\hat{u}, r)) \geq p, \tag{E.3}$$

where $u \in \overline{B_{\mathfrak{X}}(R)}$, $\hat{u} \in \mathfrak{X}_\rho$, and $B_{\mathfrak{X}_\rho}(\hat{u}, r) := \mathfrak{X}_\rho \bigcap B_{\mathfrak{X}}(\hat{u}, r)$.

4. ***Uniform Feller property.*** *There is a determining family $\mathcal{C} \subset C_b(\mathfrak{X})$ and an integer $R_0 \geq 1$ such that $\mathbf{1} \in \mathcal{C}$ and the sequence $\{\|\mathfrak{P}_k \mathbf{1}\|_R^{-1} \mathfrak{P}_k f\}_{k\geq 0}$ is uniformly equicontinuous on $\mathfrak{X}_R$ for any $f \in \mathcal{C}$ and $R \geq R_0$.*

5. ***Uniform tail probability estimate.*** *There is an integer $m \geq 1$ such that for any integer $\mathcal{R} \geq 1$,*

$$\sup_{u\in\overline{B_{\mathfrak{X}}(\mathcal{R})}} \int_{\mathfrak{X}^c_R} \mathfrak{w}(v) P_m(u, \mathrm{d}v) \to 0, \tag{E.4}$$

as $R \to +\infty$.

Then, the following properties hold.

1. *There is at most one measure $\mu \in \mathcal{P}_{\mathfrak{w}}(\mathfrak{X})$ satisfying the relation*

$$\mathfrak{P}_1^* \mu = c\mu$$

for some number $c \geq 0$, as well as the following tail probability estimate: for each positive integer ρ and $r > 0$,

$$p^{-1}(\rho, R, r)\mathcal{G}(R) \int_{\overline{B_{\mathfrak{X}}(R)}^c} \mathfrak{w} \mathrm{d}\mu \to 0, \tag{E.5}$$

as $R \to +\infty$. Here, $\mathcal{G}$ is the function in (E.2), and $p(\rho, R, r)$ is given in (E.3).

2. *If such a measure μ exists, then the corresponding eigenvalue c is positive, the support of μ coincides with $\mathfrak{X}$, and there is a positive function $h \in C_{\mathfrak{w}}(\mathfrak{X})$ such that*

$$\langle h, \mu \rangle = 1, \qquad \mathfrak{P}_1 h = ch.$$

Moreover, for any $f \in \mathcal{C}$ and integer $R \geq 1$, it follows

$$c^{-k} \mathfrak{P}_k f \to \langle f, \mu \rangle h \tag{E.6}$$

in $C_b(\overline{B_{\mathfrak{X}}(R)}) \bigcap L^1(\mathfrak{X}, \mu)$, as $k \to +\infty$.

Sketch of the proof. The proof of Theorem 4.1 in [JNPS18] remains applicable to this case, with the only adjustment required in Step 6. More precisely, by utilizing the stronger uniform irreducibility condition, one can derive the estimate (4.23) in [JNPS18] with $\mathfrak{X}_R$ replaced by $\overline{B_{\mathfrak{X}}(R)}$. This, combined with the new condition (E.5), leads to the same contradiction. □

The following result is an improved version of Theorem A.4 in [MN18].

Proposition E.2. *Let $\{P_t(u, \cdot) | u \in \mathfrak{X}, t \geq 0\}$ be a generalized Markov family of transition kernels satisfying the following conditions.*

1. ***Concentration property.*** *There is a number $l_0 > 0$ such that $P_t(u, \cdot)$ is concentrated on $\mathfrak{X}_\infty$ for any $u \in \mathfrak{X}$ and $t \geq l_0$.*
2. ***Growth condition.*** *There is a weight function $\mathfrak{w}$ satisfying (E.2) and an integer $R_0 \geq 1$ such that*

$$\sup_{t \geq 0} \frac{\|\mathfrak{P}_t \mathfrak{w}\|_{L^\infty_\mathfrak{w}}}{\|\mathfrak{P}_t \mathbf{1}\|_{R_0}} < \infty, \tag{E.7}$$

$$\sup_{t \in [0,1]} \|\mathfrak{P}_t \mathbf{1}\|_{L^\infty(\mathfrak{X})} < \infty, \tag{E.8}$$

where $\mathbf{1}$ is the function on $\mathfrak{X}$ identically equal to 1, and $\|\cdot\|_R$ denotes the L^∞-norm on $\mathfrak{X}_R$.
3. ***Time-continuity.*** *For any function $f \in C_\mathfrak{w}(\mathfrak{X})$ and $u \in \mathfrak{X}$, the function $t \mapsto \mathfrak{P}_t f(u)$ is continuous from $[0, \infty)$ to $\mathbb{R}$.*
4. ***Dissipativity condition.*** *There exist constants $c_1, c_2 > 0$ such that for any integer $R \geq 1$, there is an integer $\mathcal{R} = \mathcal{R}(R) \geq 1$ satisfying*

$$P_t(u, \overline{B_\mathfrak{X}(\mathcal{R})}) \geq c_1 e^{-c_2 t} \tag{E.9}$$

for any $t \geq 0$ and $u \in \overline{B_\mathfrak{X}(R)}$.
5. ***Uniform irreducibility.*** *For any integers $\rho, R \geq 1$ and $r > 0$, there are positive numbers $l = l(\rho, R, r)$ and $p = p(\rho, R, r)$ such that*

$$P_l(u, B_{\mathfrak{X}_\rho}(\hat{u}, r)) \geq p, \tag{E.10}$$

where $u \in \overline{B_\mathfrak{X}(R)}$, $\hat{u} \in \mathfrak{X}_\rho$, and $B_{\mathfrak{X}_\rho}(\hat{u}, r) := \mathfrak{X}_\rho \bigcap B_\mathfrak{X}(\hat{u}, r)$.
6. ***Uniform Feller property.*** *There is a determining family $\mathcal{C} \subset C_b(\mathfrak{X})$ and an integer $R_0 \geq 1$ such that $\mathbf{1} \in \mathcal{C}$ and the sequence $\{\|\mathfrak{P}_t \mathbf{1}\|_R^{-1} \mathfrak{P}_t f\}_{t \geq 0}$ is uniformly equicontinuous on $\mathfrak{X}_R$ for any $f \in \mathcal{C}$ and $R \geq R_0$.*
7. ***Uniform tail probability estimate.*** *There is a number $l_0 > 0$ such that for any integer $\mathcal{R} \geq 1$ and $t \geq l_0$,*

$$\sup_{u \in \overline{B_\mathfrak{X}(\mathcal{R})}} \int_{\mathfrak{X}_R^c} \mathfrak{w}(v) P_t(u, \mathrm{d}v) \to 0,$$

as $R \to +\infty$.

Then, the following properties hold.

1. *For any $t > 0$, there is at most one measure $\mu_t \in \mathcal{P}_\mathfrak{w}(\mathfrak{X})$ satisfying*

$$\mathfrak{P}_t^* \mu_t = c_t \mu_t \tag{E.11}$$

and the tail probability estimate: for each positive integer ρ and $r > 0$,

$$p^{-1}(\rho, \mathcal{R}(R), r) \mathcal{G}(R) \int_{\overline{B_\mathfrak{X}(R)}^c} \mathfrak{w} \mathrm{d}\mu_t \to 0, \tag{E.12}$$

as $R \to +\infty$. Here, $\mathcal{G}$ is the function in (E.2), $\mathcal{R}(R)$ is given in (E.9), and $p(\rho, \mathcal{R}(R), r)$ is the lower bound in (E.10) with R replaced by $\mathcal{R}(R)$.
2. *If such a measure μ_t exists for all $t > 0$, then it is independent of t (we thus set $\mu := \mu_t$), the corresponding eigenvalue c_t is of the form $c_t = c^t, c > 0$, the support of μ coincides with $\mathfrak{X}$, and there is a positive function $h \in C_\mathfrak{w}(\mathfrak{X})$ such that*

$$\langle h, \mu \rangle = 1, \qquad \mathfrak{P}_t h = c^t h. \tag{E.13}$$

Moreover, for any $f \in \mathcal{C}$ and integer $R \geq 1$, it follows

$$c^{-t}\mathfrak{P}_t f \to \langle f, \mu\rangle h \tag{E.14}$$

in $C_b(\overline{B_{\mathfrak{X}}(R)}) \bigcap L^1(\mathfrak{X}, \mu)$, as $t \to +\infty$.

Sketch of the proof. Let $t > 0$ be an arbitrary number, and consider the transition kernels $\{\tilde{P}_k := P_{kt}, k \in \mathbb{N}\}$. The conditions of Proposition E.1 are satisfied[10] for $\tilde{P}_k(u, \cdot)$. Thus, the first statement holds.

Suppose that the measure μ_t exists for all $t > 0$, then by using (E.11), (E.12), and the Kolmogorov–Chapman relation, one has

$$\mu_t = \mu_1 =: \mu, \qquad c_t = c_1^t =: c^t$$

for any t in the set $\mathbb{Q}_+$ of positive rational numbers. Using the time-continuity property and (E.8), it follows

$$\mathfrak{P}_t^* \mu = c^t \mu$$

for any $t > 0$. Moreover, it is clear that $\operatorname{supp} \mu = \mathfrak{X}$, and there is a positive function $h_t \in C_{\mathfrak{w}}(\mathfrak{X})$ such that $\langle h_t, \mu\rangle = 1$ and

$$\mathfrak{P}_t h_t = c^t h_t$$

and

$$c^{-tk}\mathfrak{P}_{tk} f \to \langle f, \mu\rangle h_t \tag{E.15}$$

in $C_b(\overline{B_{\mathfrak{X}}(R)}) \bigcap L^1(\mathfrak{X}, \mu)$, as $k \to +\infty$ for any integer $R \geq 1$ and $f \in \mathcal{C}$. Taking $f = 1$ in (E.15), it follows that $h_t = h_1 =: h$ for all $t \in \mathbb{Q}_+$, which implies (E.13) by using the time-continuity property. The derivation of (E.14) follows in a similar way as in [MN18] and is omitted. □

Data availability

Data sharing is not applicable to this article, as no datasets were generated or analysed during the current study.

Conflict of interest

The author declares no conflict of interest.

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